



MMM YFG TRUNK STUD INSERTION MACHINE.

Operator / Maintenance Manual

Manufactured by:
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019



YFG TRUNK STUD INSERTION MACHINE.

Marelli Motherson Auto System Pvt Ltd.
Gat No 148/150, Chakan, Pune-410501(Near
Dwarka School Village Ambethan Taluka Khed)



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YFG TRUNK STUD INSERTION MACHINE.

Marelli Motherson Auto System Pvt Ltd.
Gat No 148/150, Chakan, Pune-410501(Near
Dwarka School Village Ambethan Taluka Khed)



SYSTEM INFORMATION:

Name : YFG TRUNK STUD INSERTION MACHINE.

Customer : Marelli Motherson Auto System Pvt Ltd.
Gat No 148/150, Chakan,
Pune-410501(Near Dwarka School Village
Ambethan Taluka Khed)

Manufacturing Date : Jan,2021

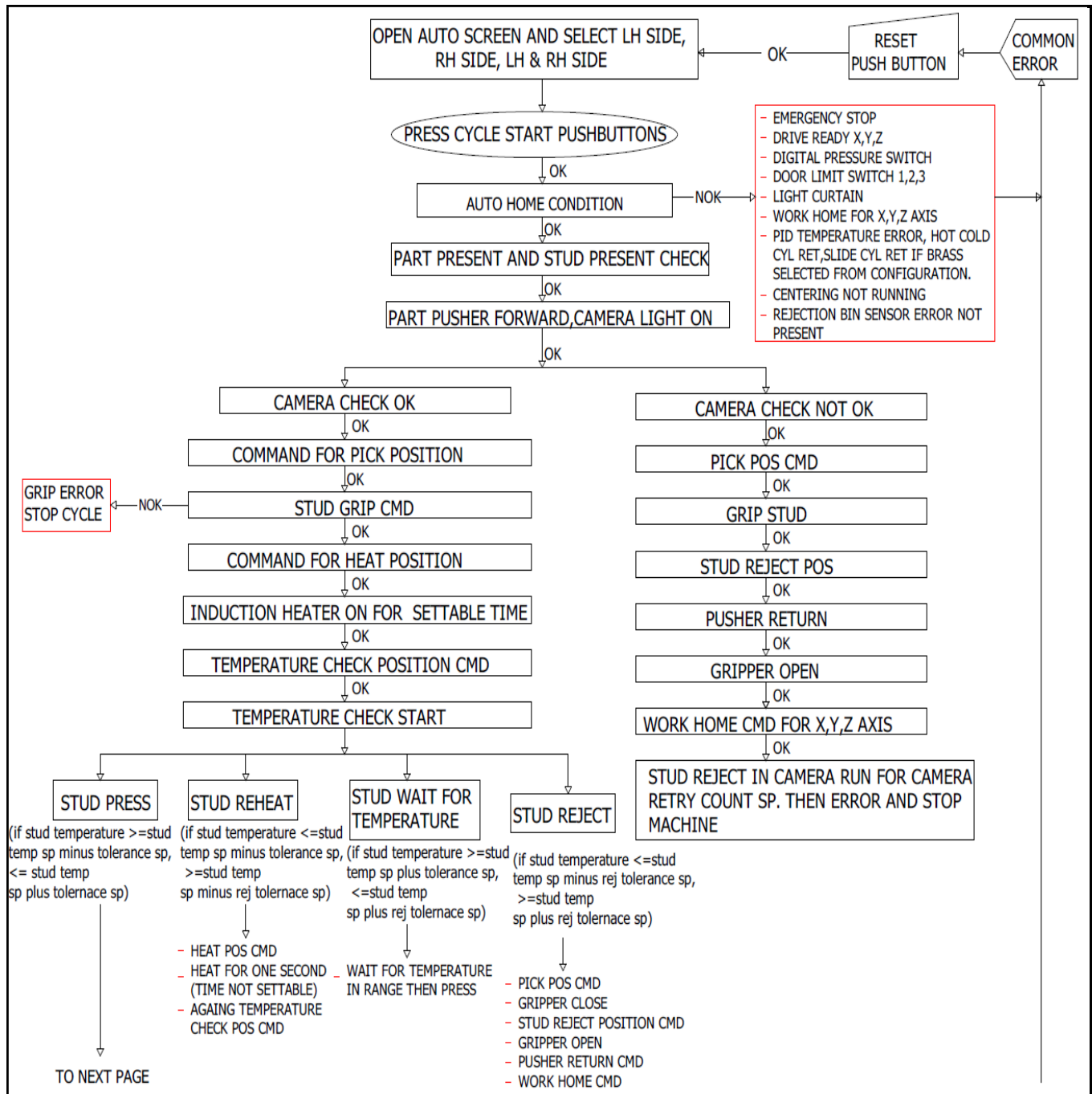
Supply Voltage : 230 VAC, 50/60 Hz

Power Consumption: 6.8kW

Manufacturer : **Paveway Automation Pvt. Ltd.**
Sanghvi Steel Complex,
128/2, Telco Road, Chinchwad,
Pune- 411019.
tel: 020-27450840

Email ID : applications@pavewayautomation.com

PROCESS SEQUENCE:

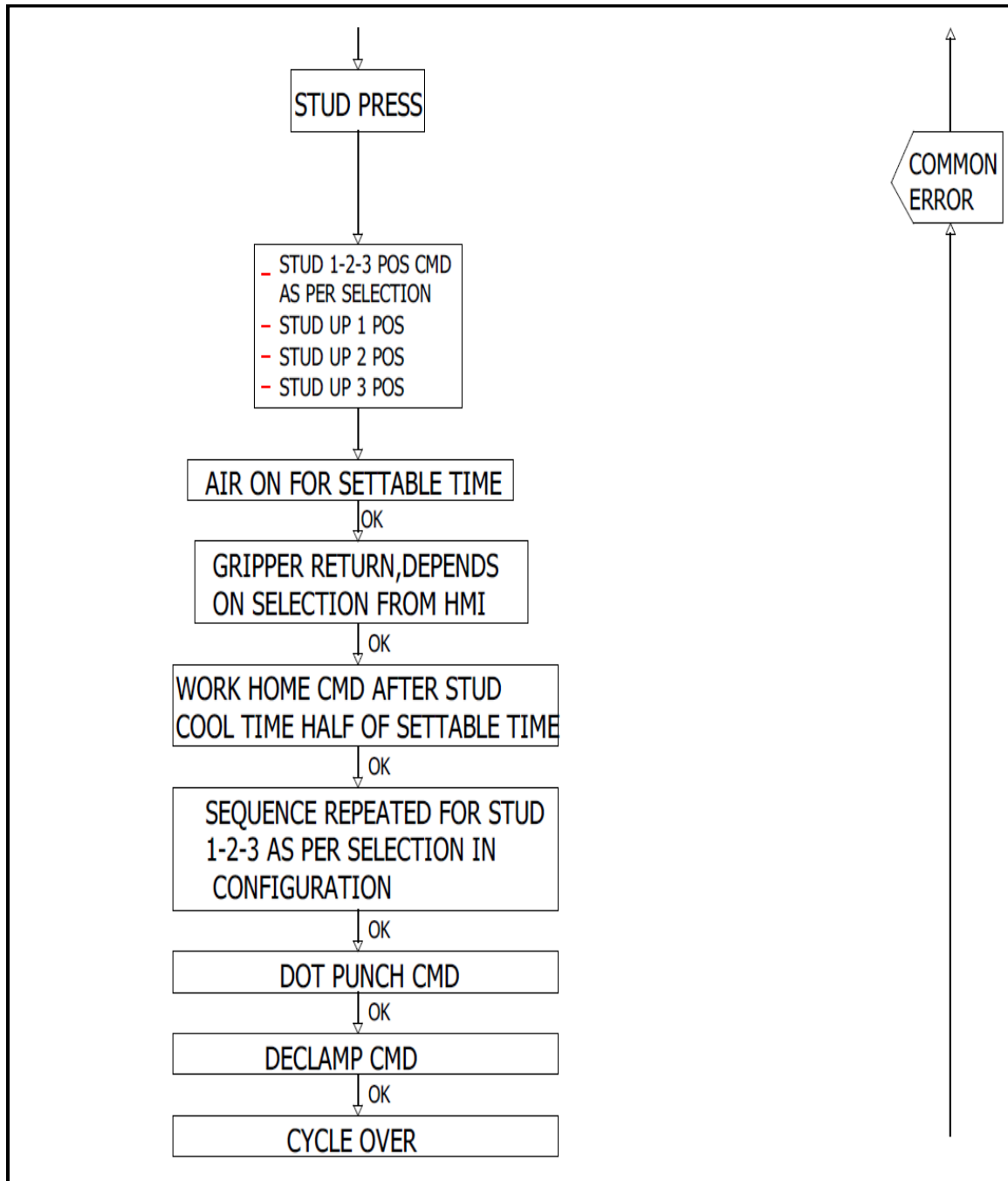


I

II

I

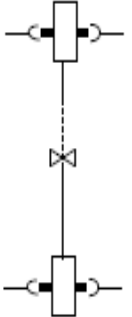
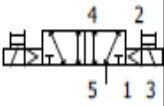
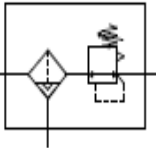
II



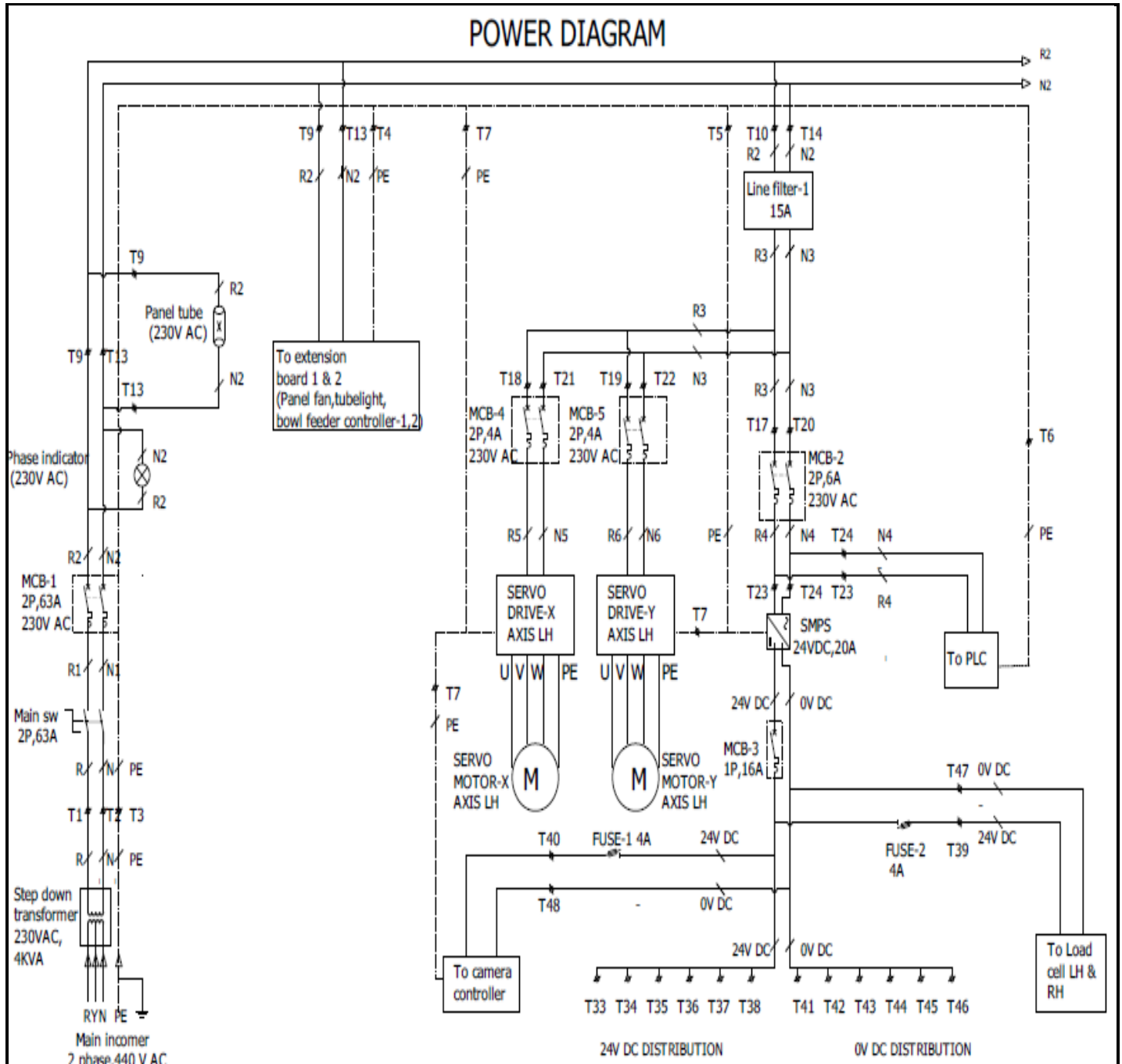
WIRING DIAGRAMS:

SYMBOL DESCRIPTION

SYMBOL DESCRIPTION

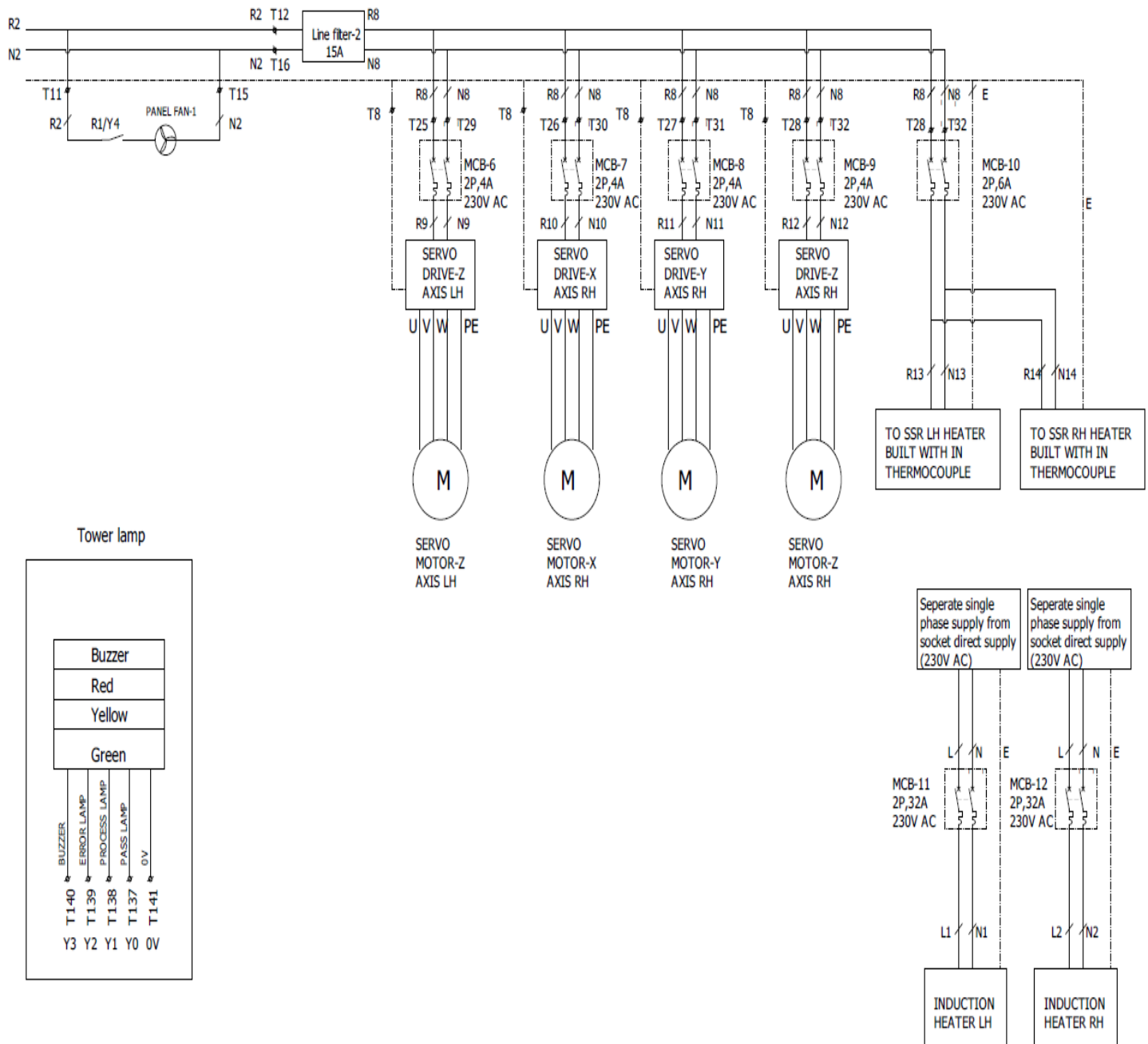
Symbol	Symbol description	Symbol	Symbol description
	Earthing		Buzzer
	2 Pole MCB		Double acting solenoid valve
	2 Pole on/off switch		
	Lamp		
	Tubelight		
	Fan		
	SMPS		
	1 pole MCB		Double acting solenoid valve (pneumatic)
	Push button		FR unit
	Relay coil		
	Emergency stop button		
	Proximity sensor		Airoff(Pneumatic)

POWER DIAGRAM

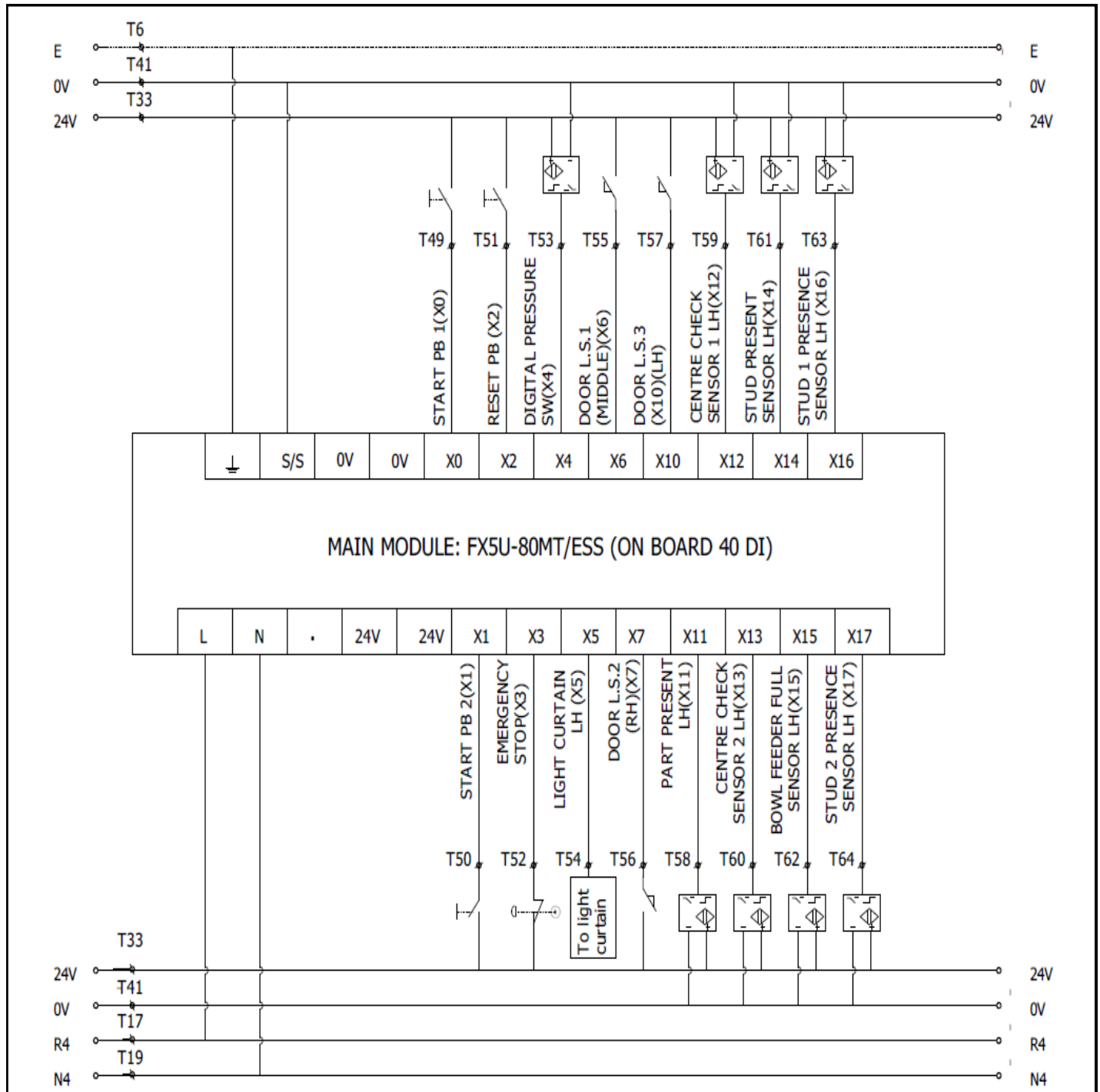


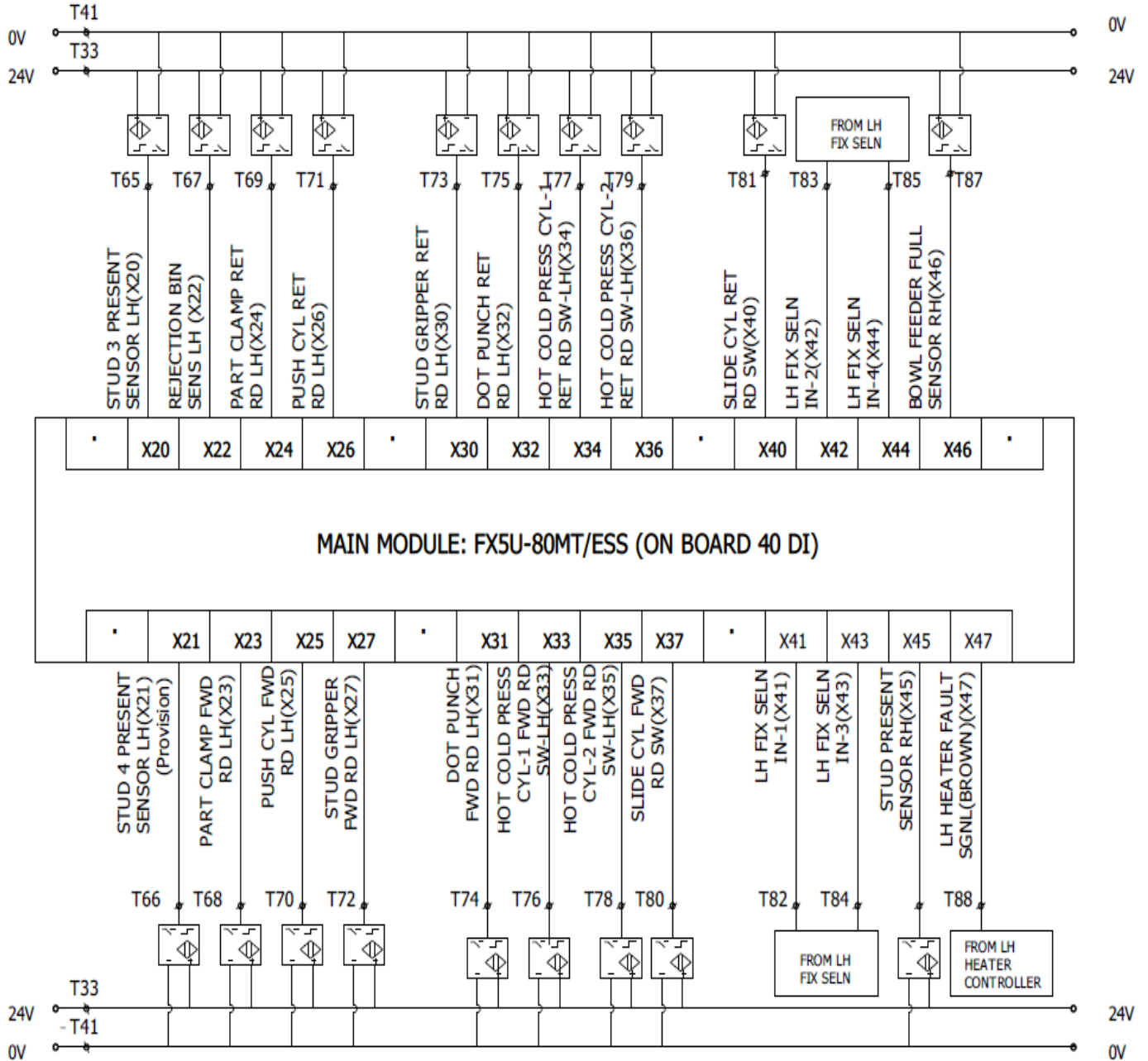
POWER DIGRAM-SERVO DRIVE

POWER DIAGRAM- SERVO DRIVE

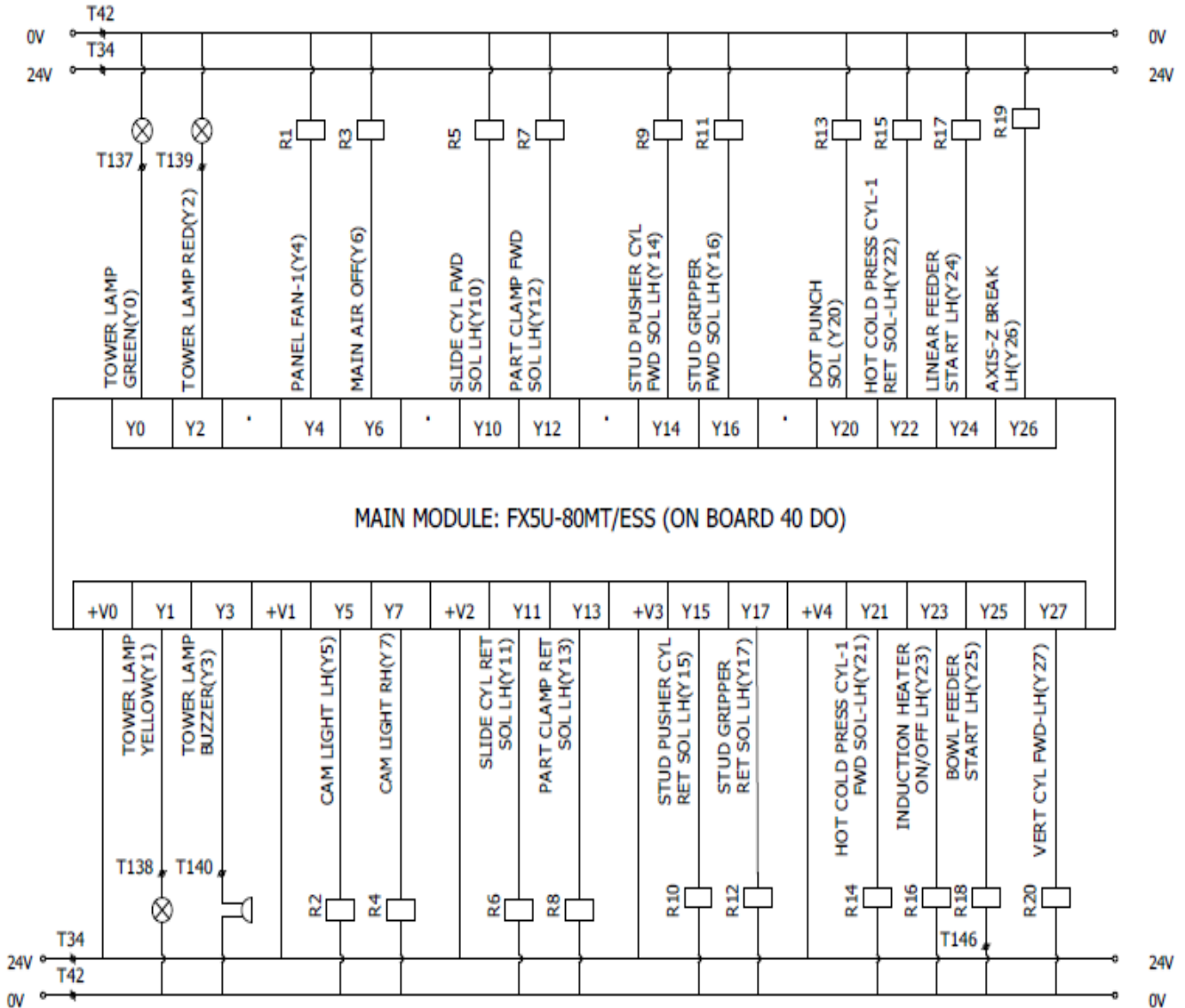


MAIN MODULE: FX5U-80MT/ESS(40DI)

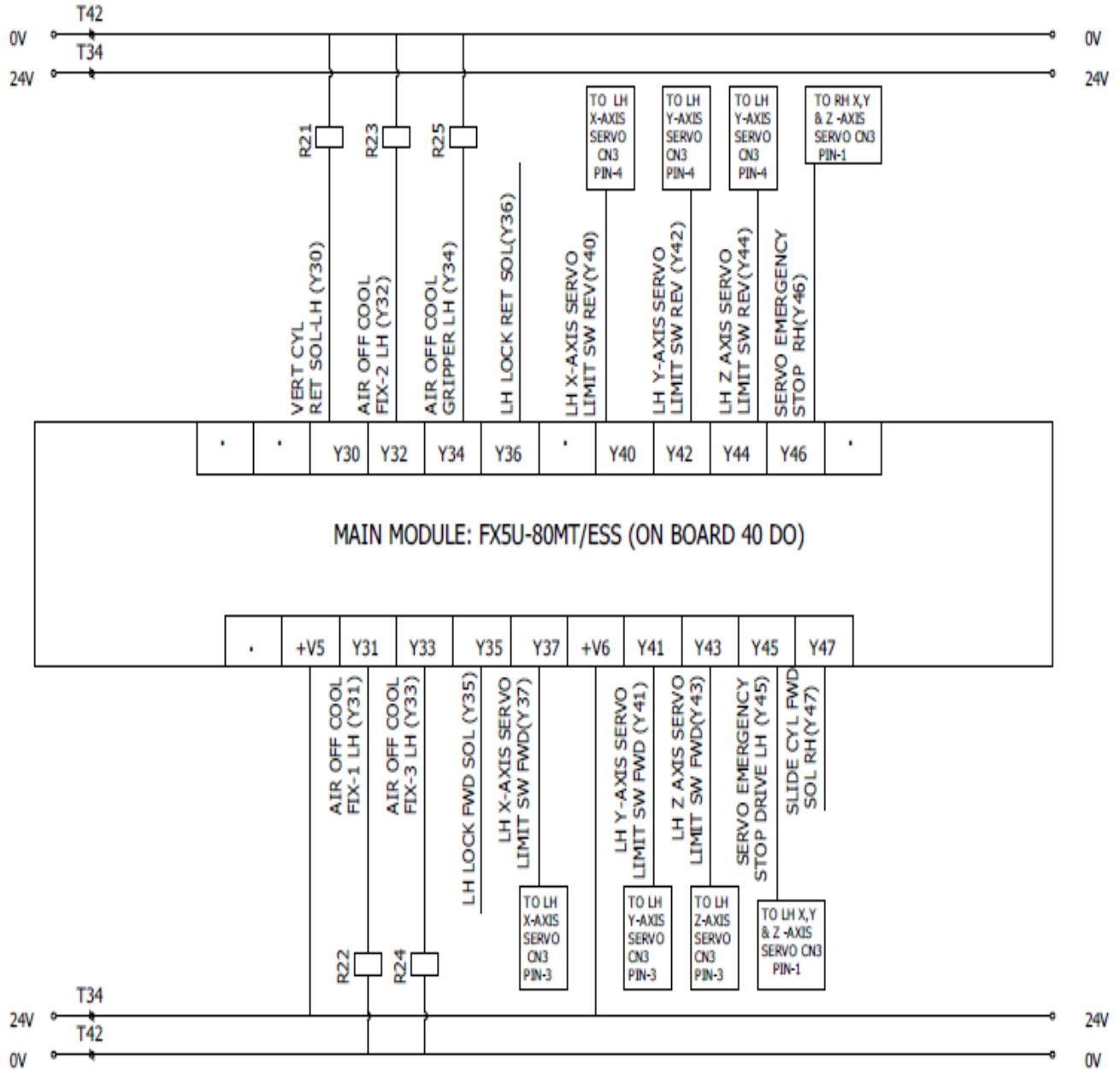




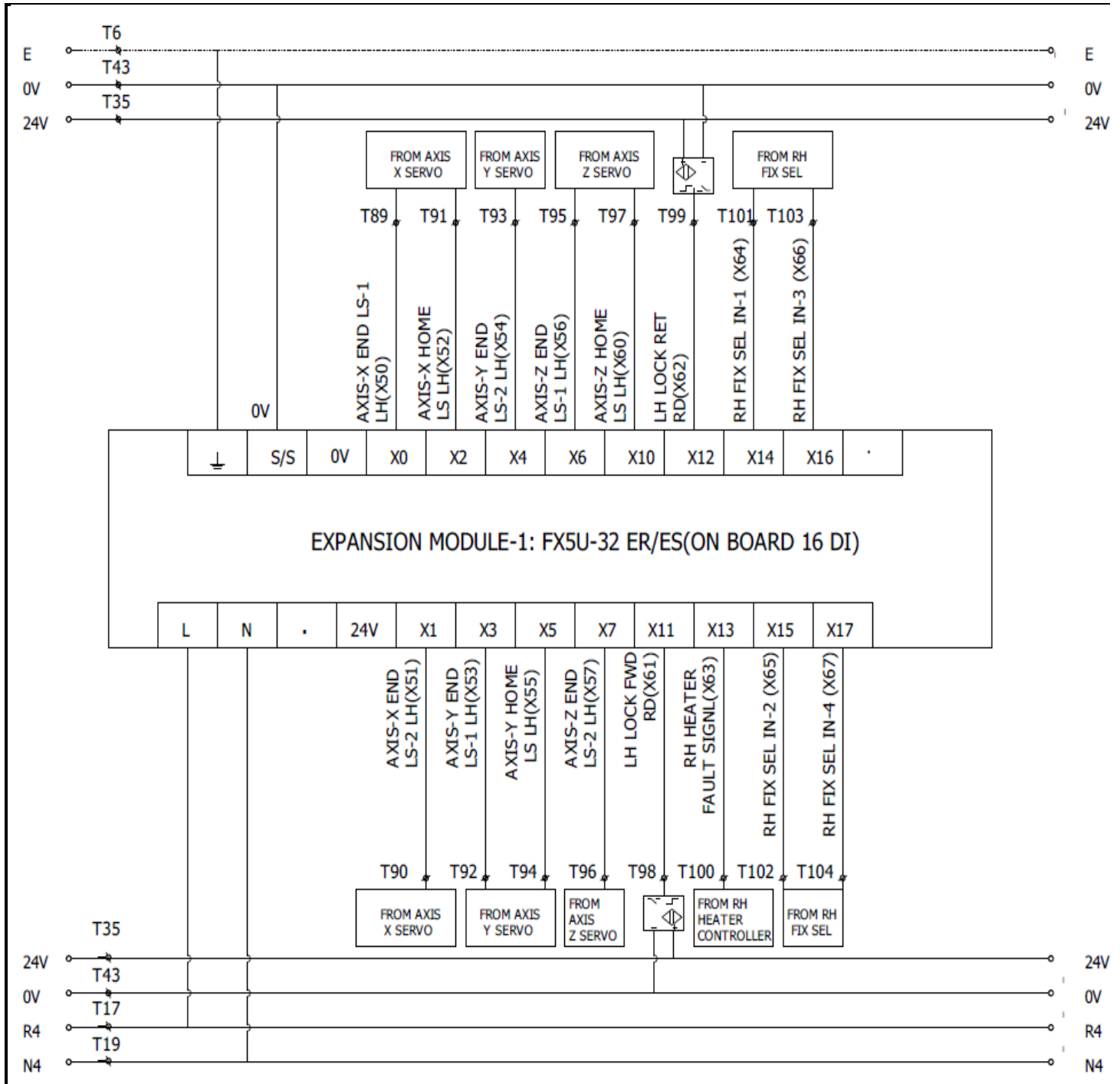
MAIN MODULE:FX5U-80MT/ESS(40DO)



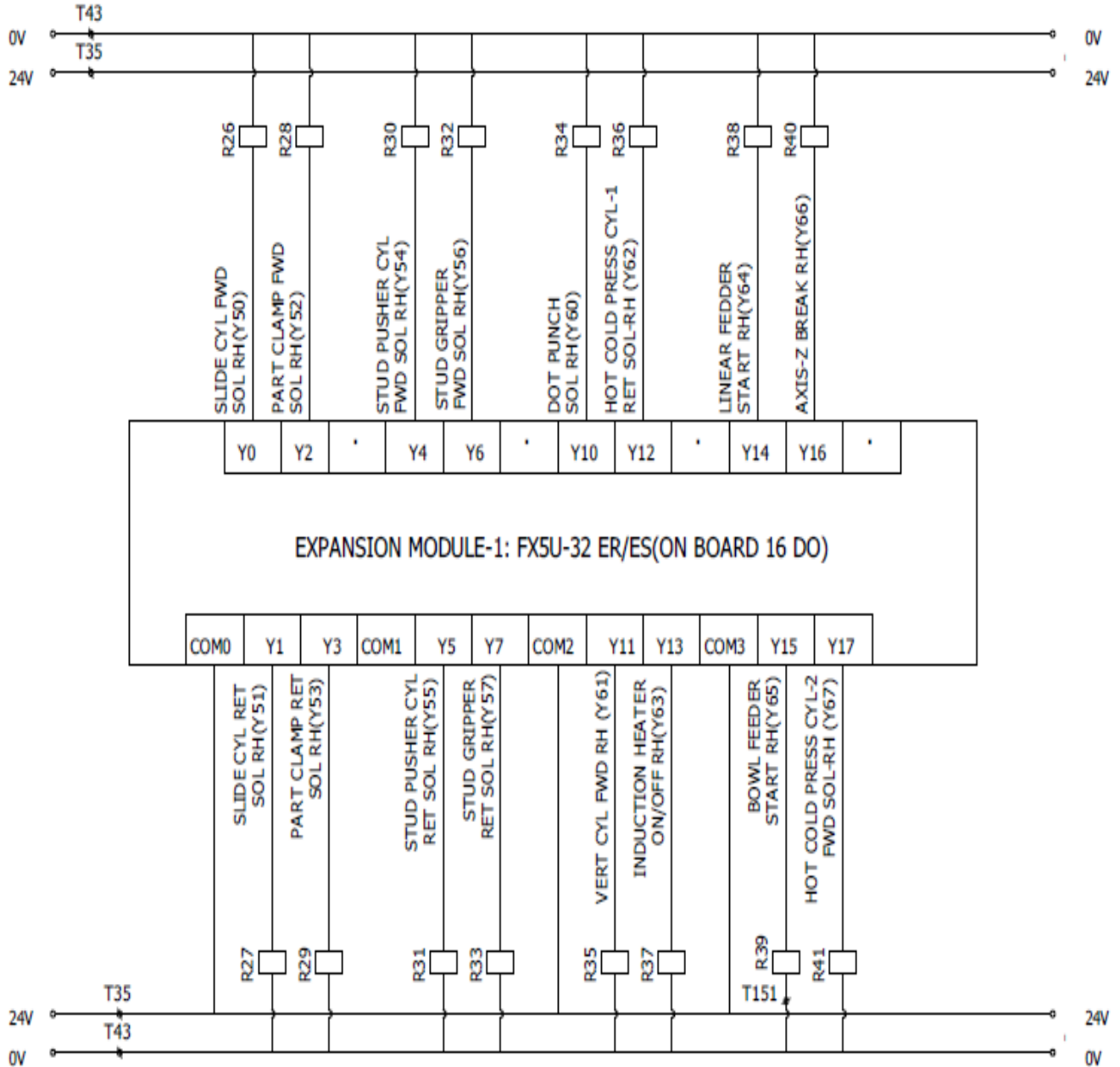
MAIN MODULE:FX5U-80MT/ESS(40DO)



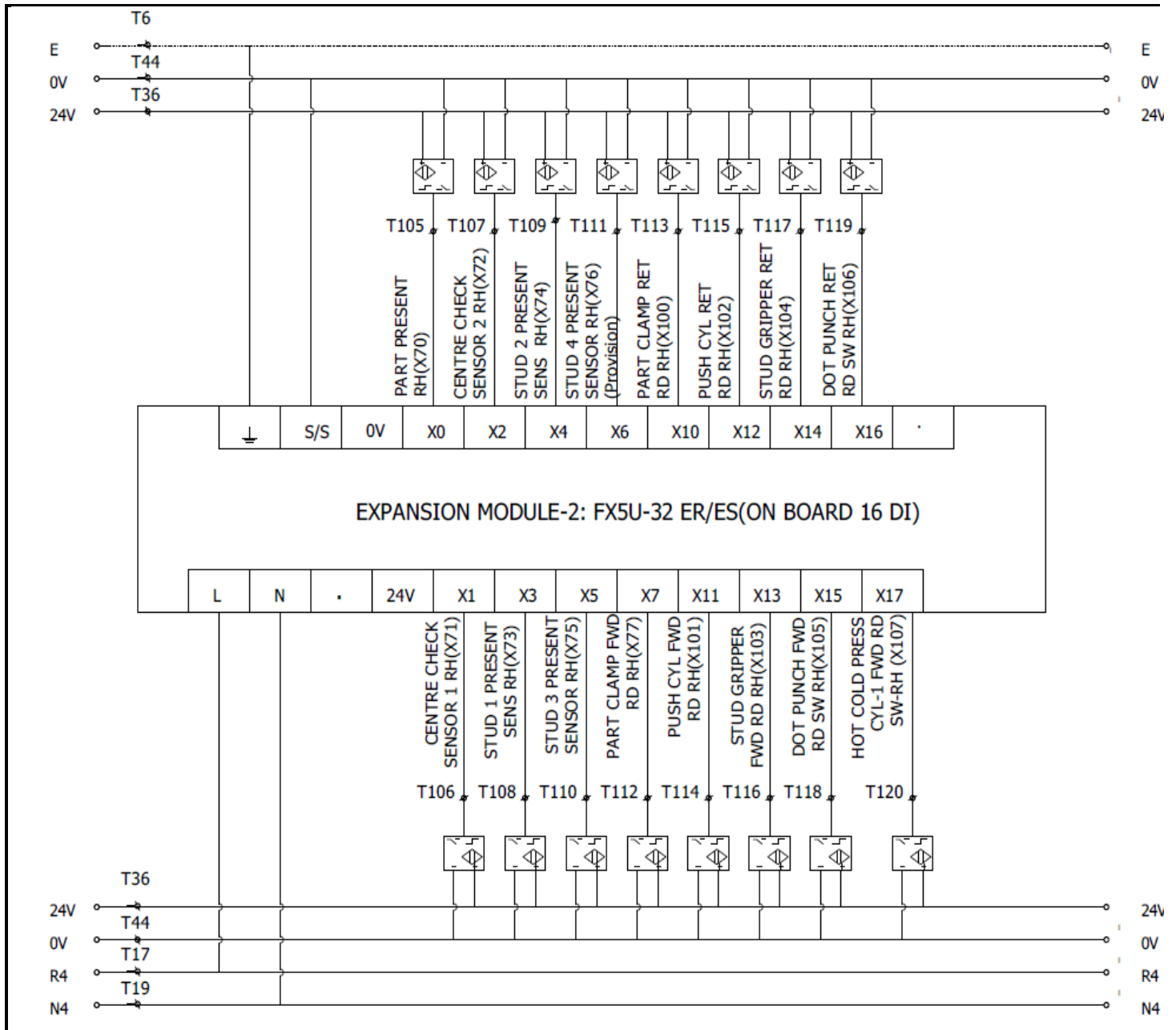
EXPANSION MODULE-1: FX5U-32ER/ES (16DI)



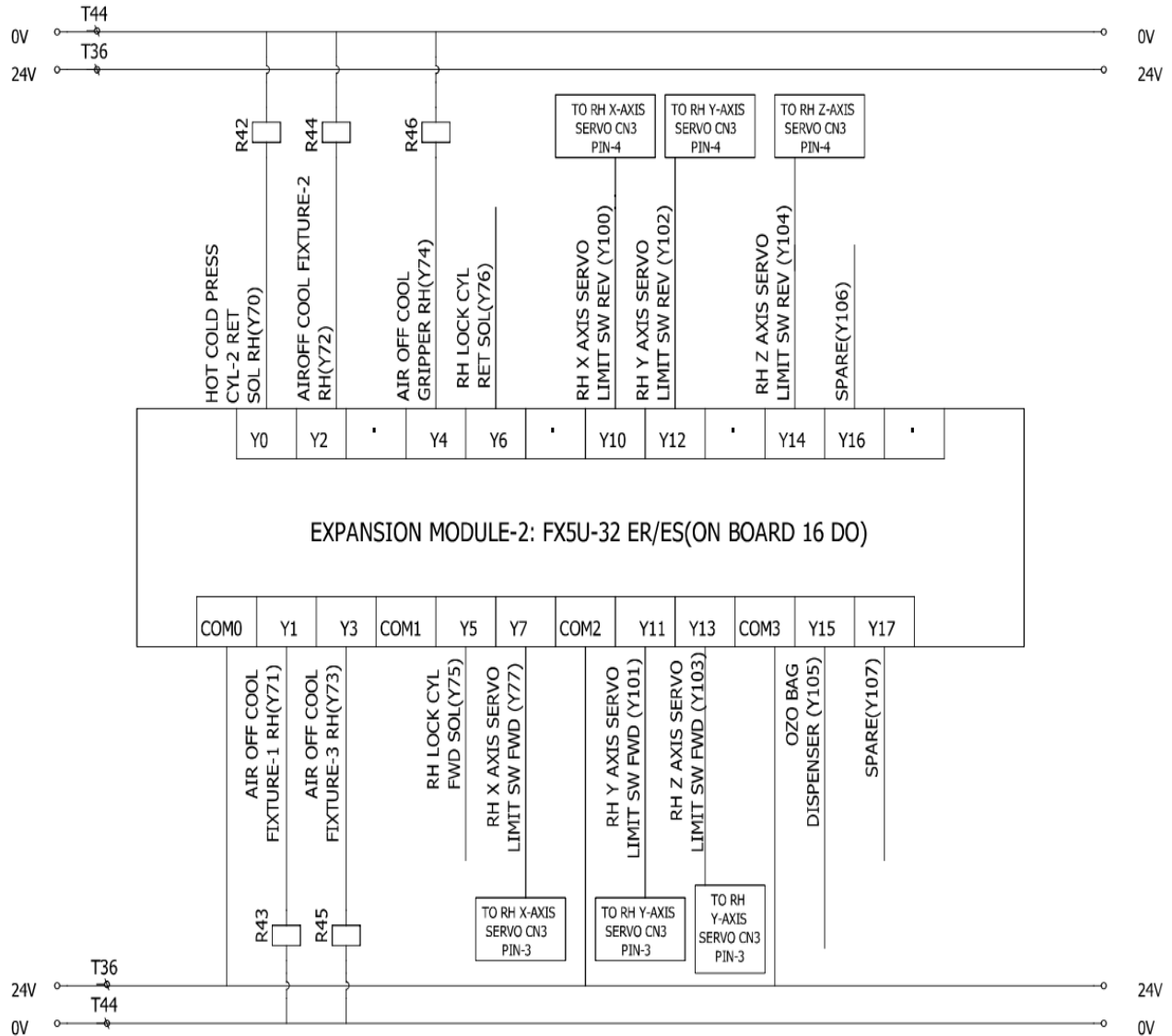
EXPANSION MODULE-1: FX5U-32ER/ES (16DO)



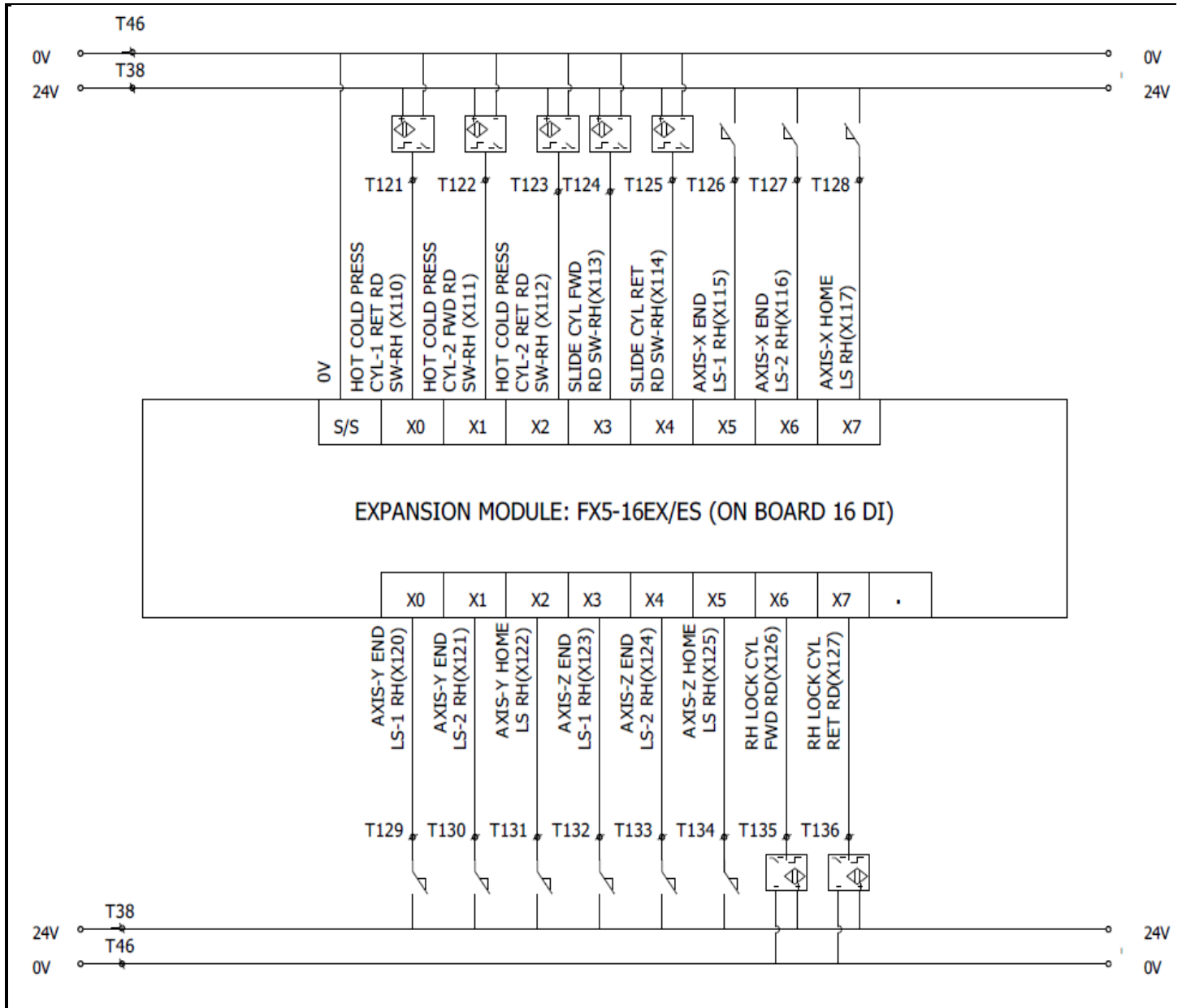
EXPANSION MODULE-2: FX5U-32ER/ES (16DI)



EXPANSION MODULE-2: FX5U-32ER/ES (16DO)

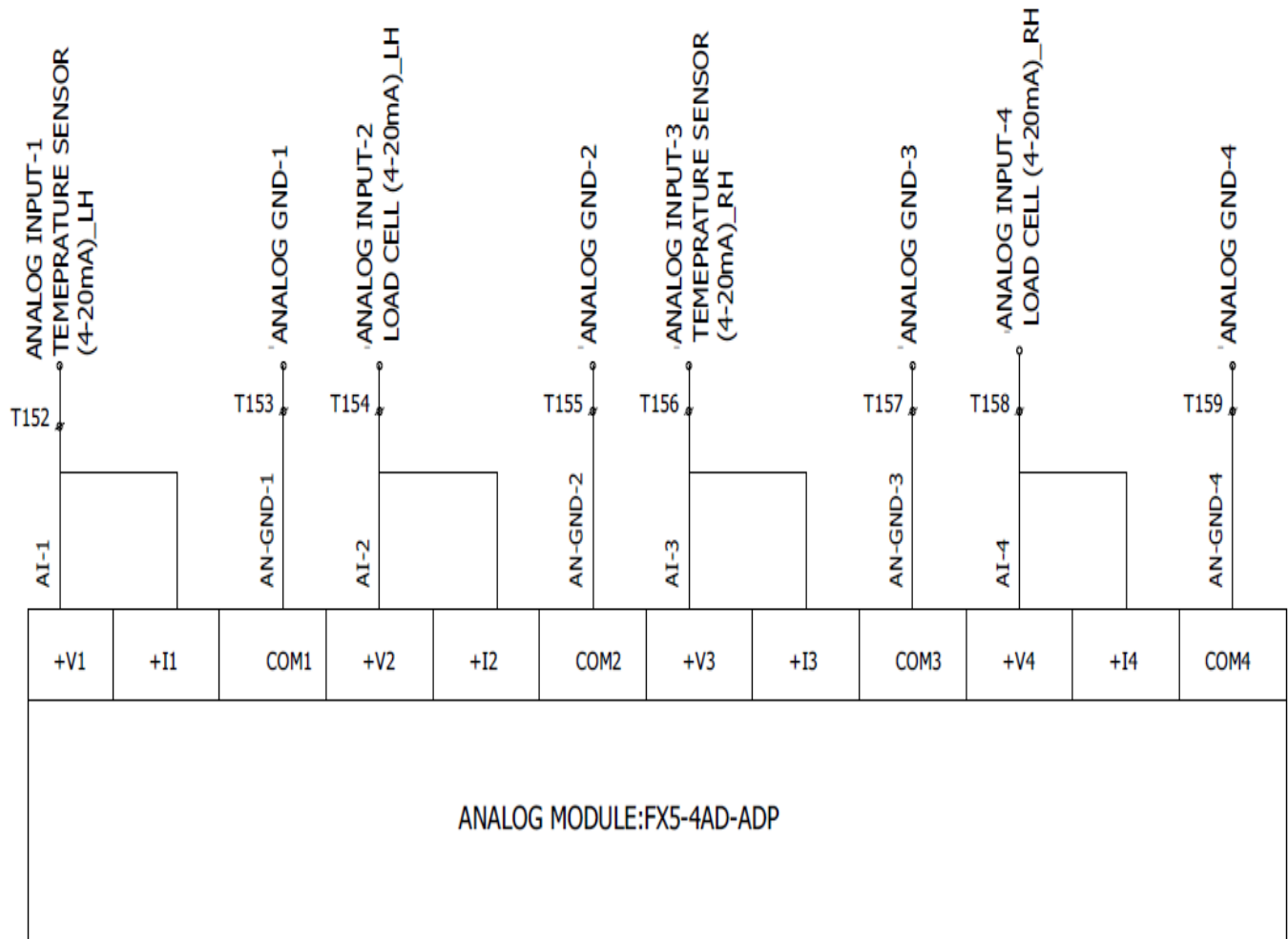


EXPANSION MODULE-3: FX5U-16EX/ES (16DI)



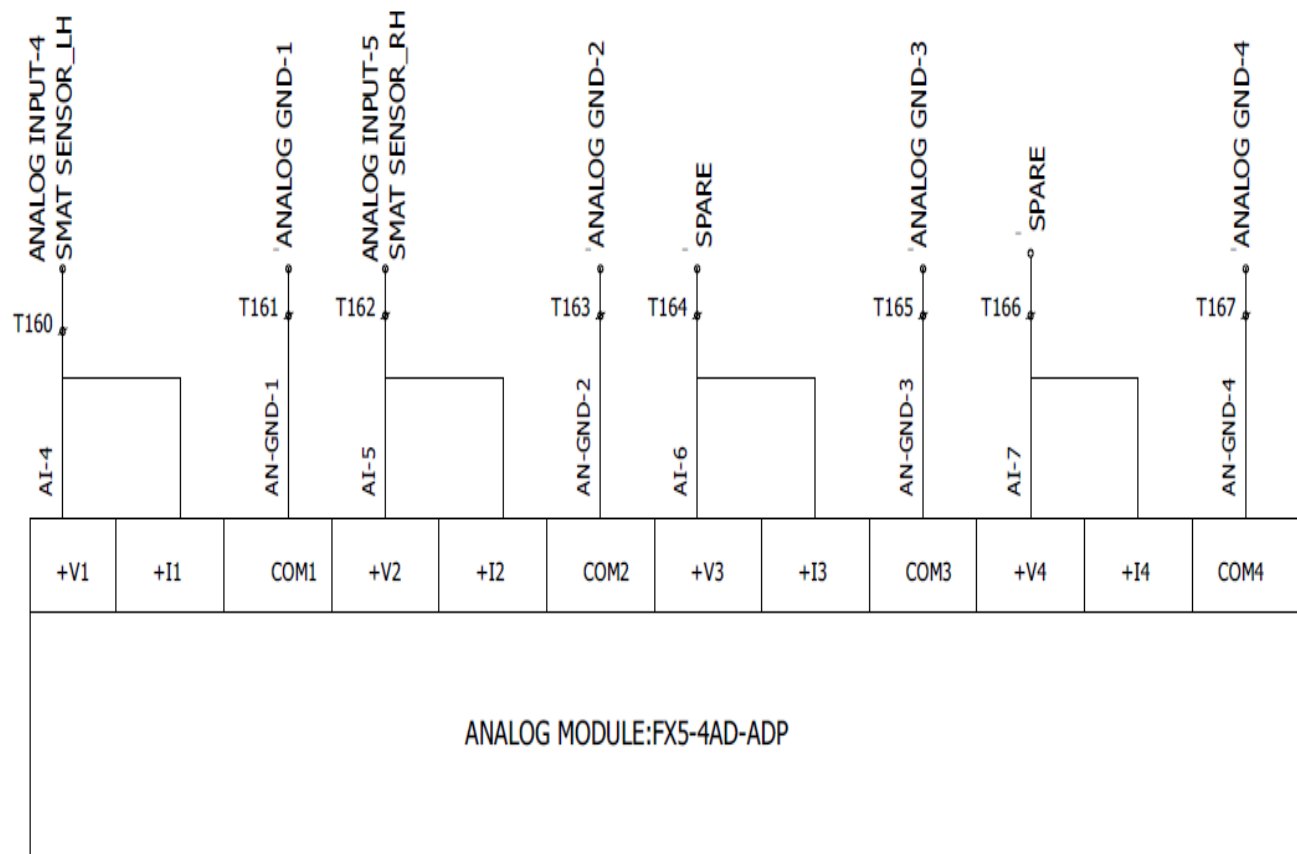
ANALOG MODULE: FX5-4AD-ADP-1

ANALOG MODULE INPUT-1



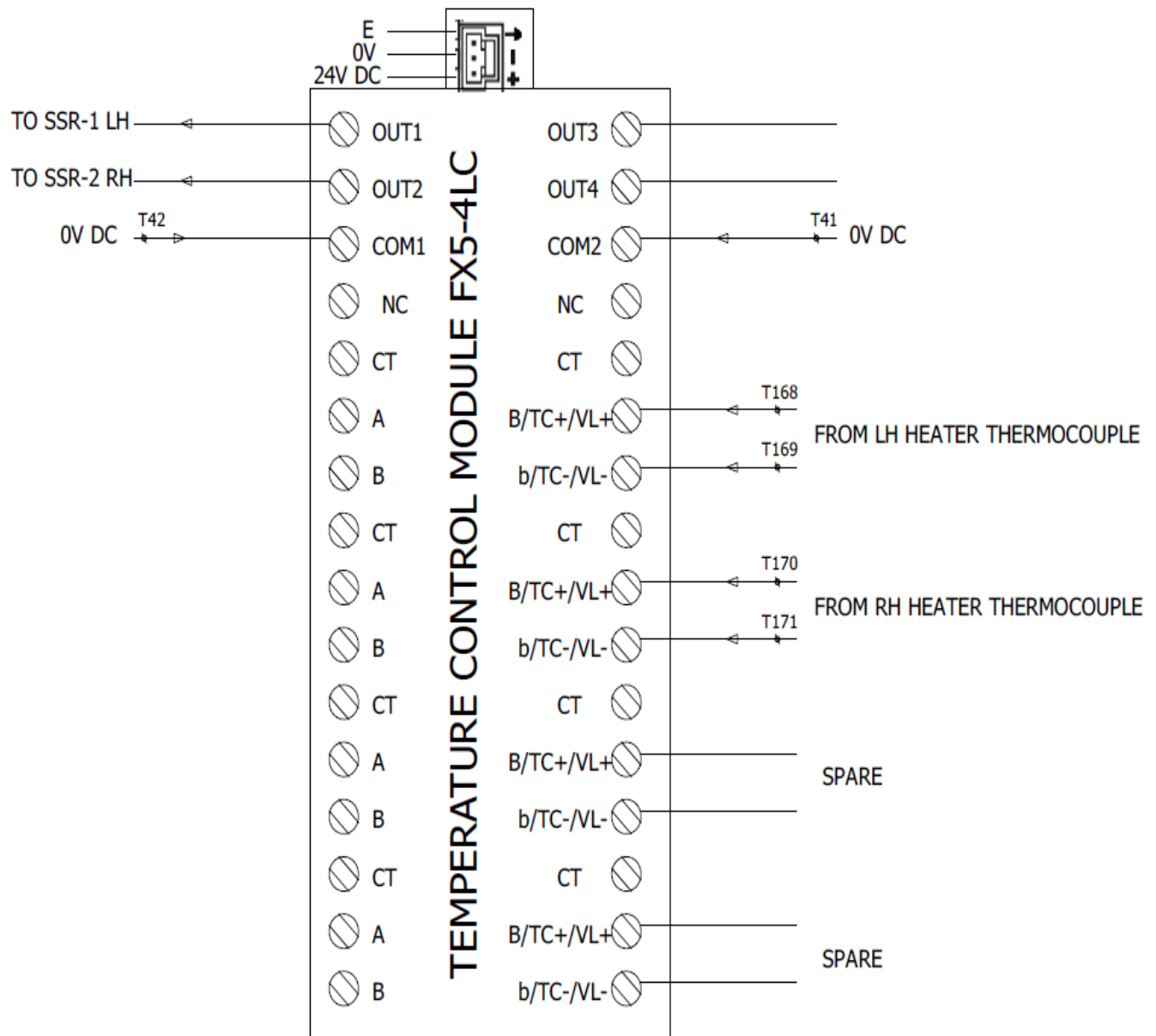
ANALOG MODULE: FX5-4AD-ADP-2

ANALOG MODULE INPUT-2



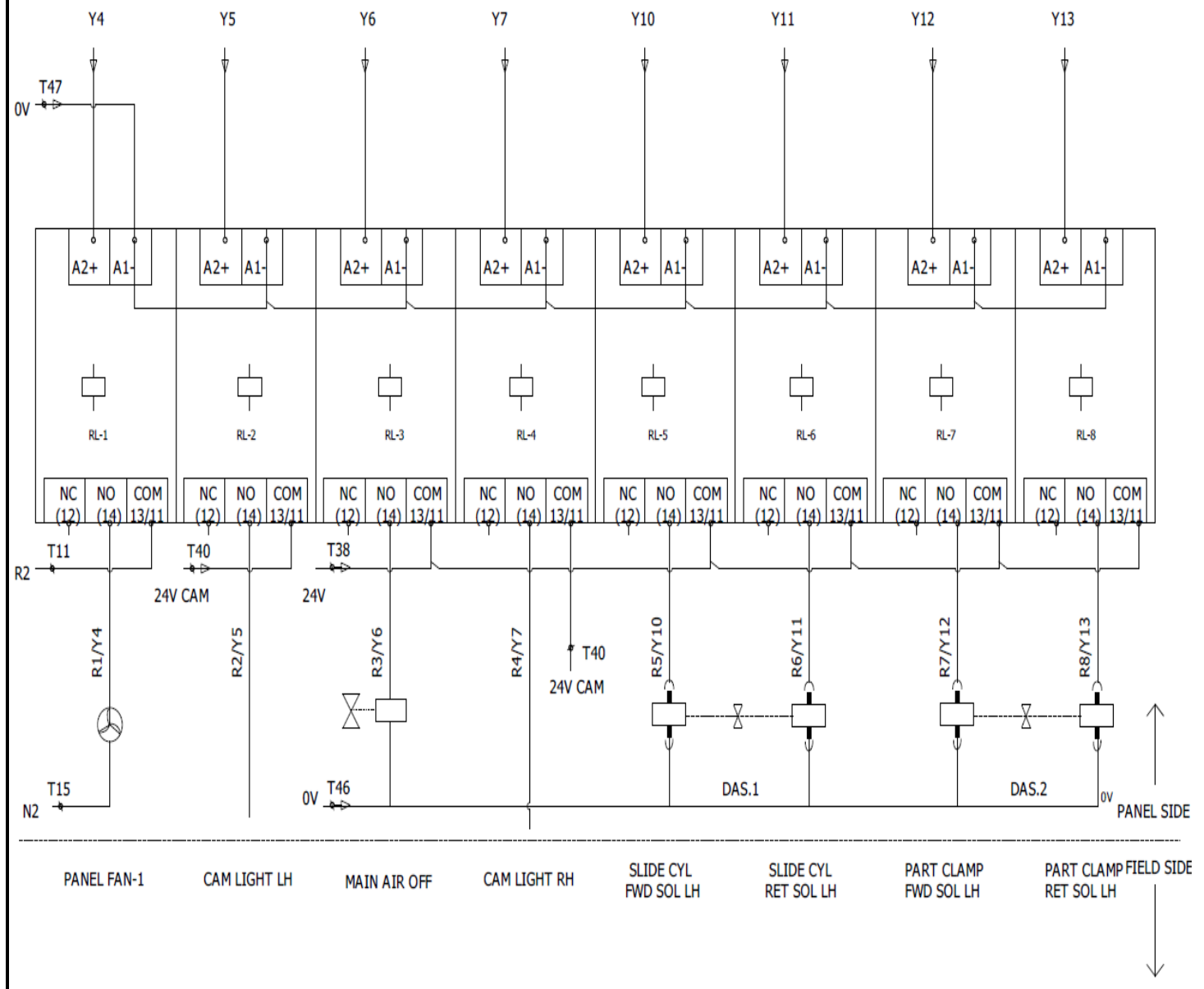
TEMEPERATURE CONTROL MODULE FX5-4LC

TEMPERATURE CONTROL MODULE FX5-4LC



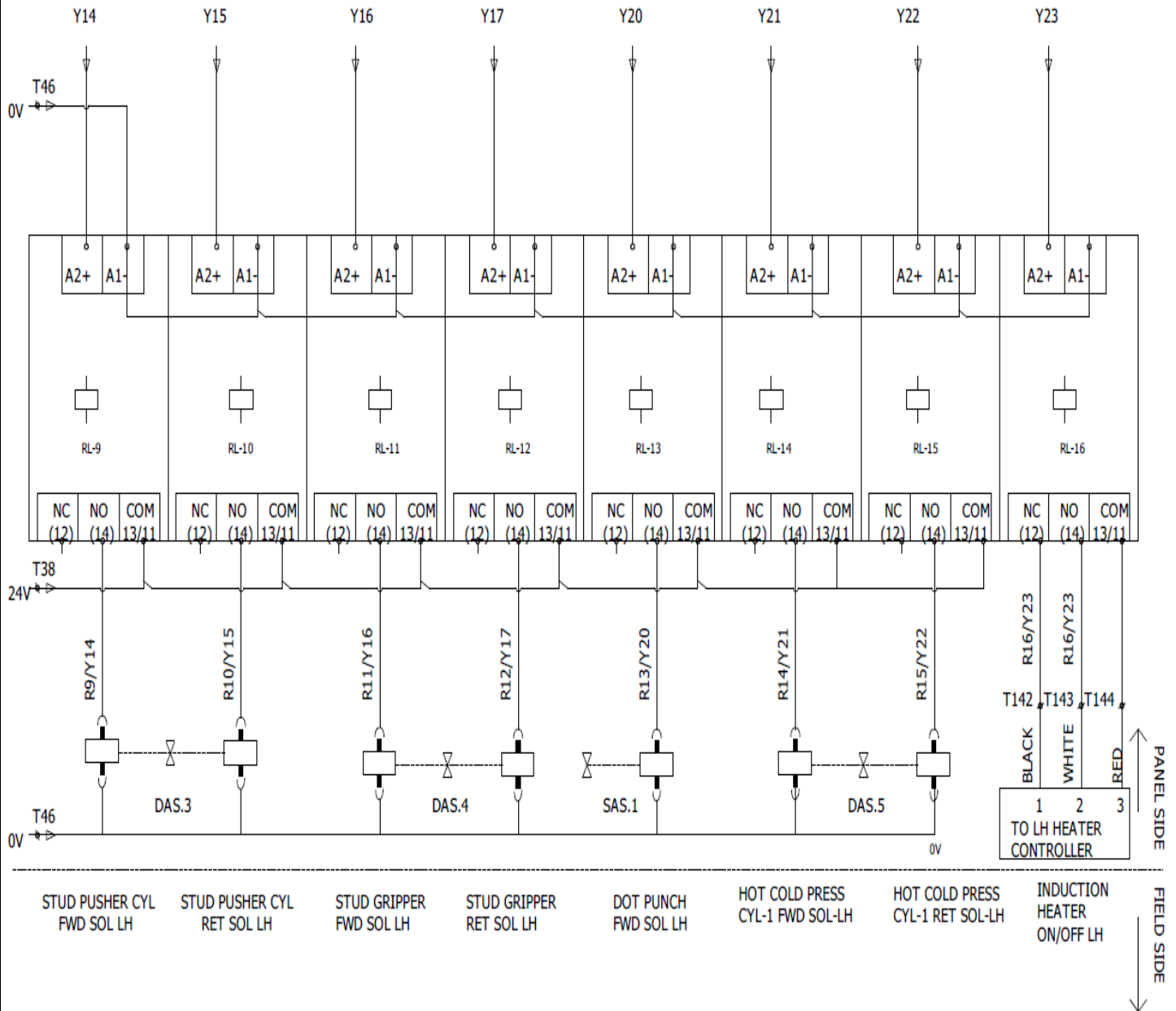
RELAY DIAGRAM 1

RELAY DIAGRAM-1



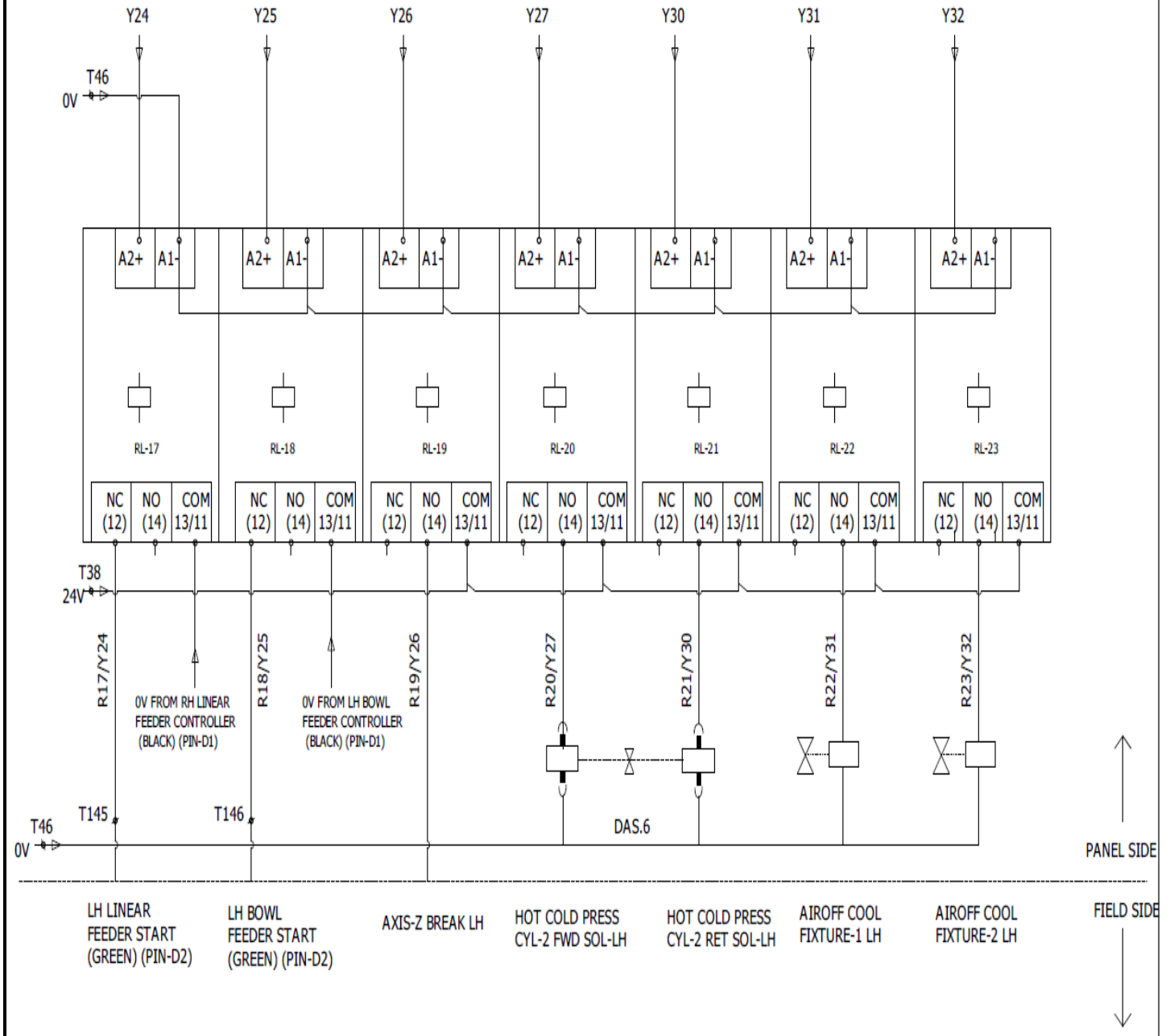
RELAY DIAGRAM 2

RELAY DIAGRAM-2



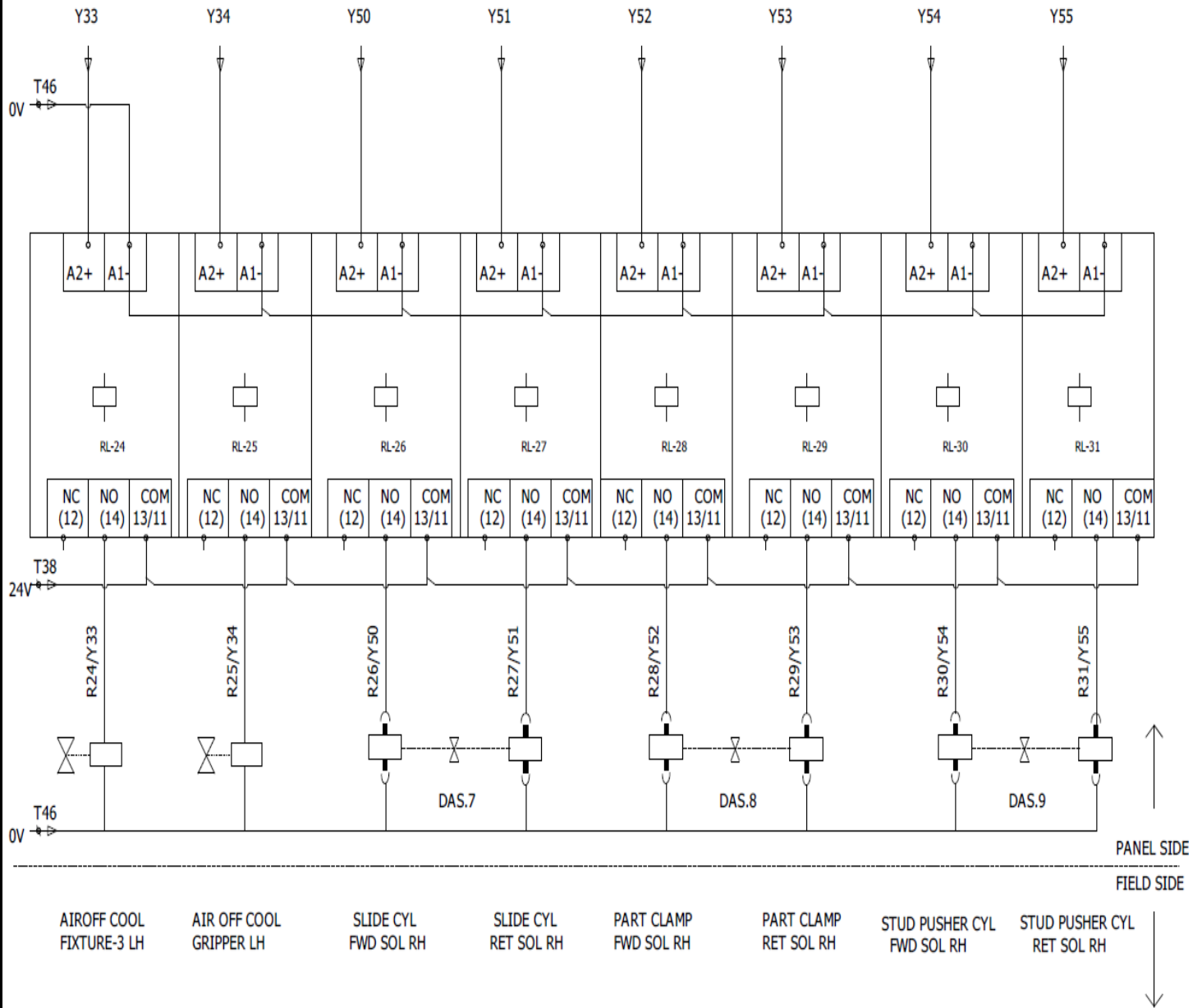
RELAY DIAGRAM 3

RELAY DIAGRAM-3



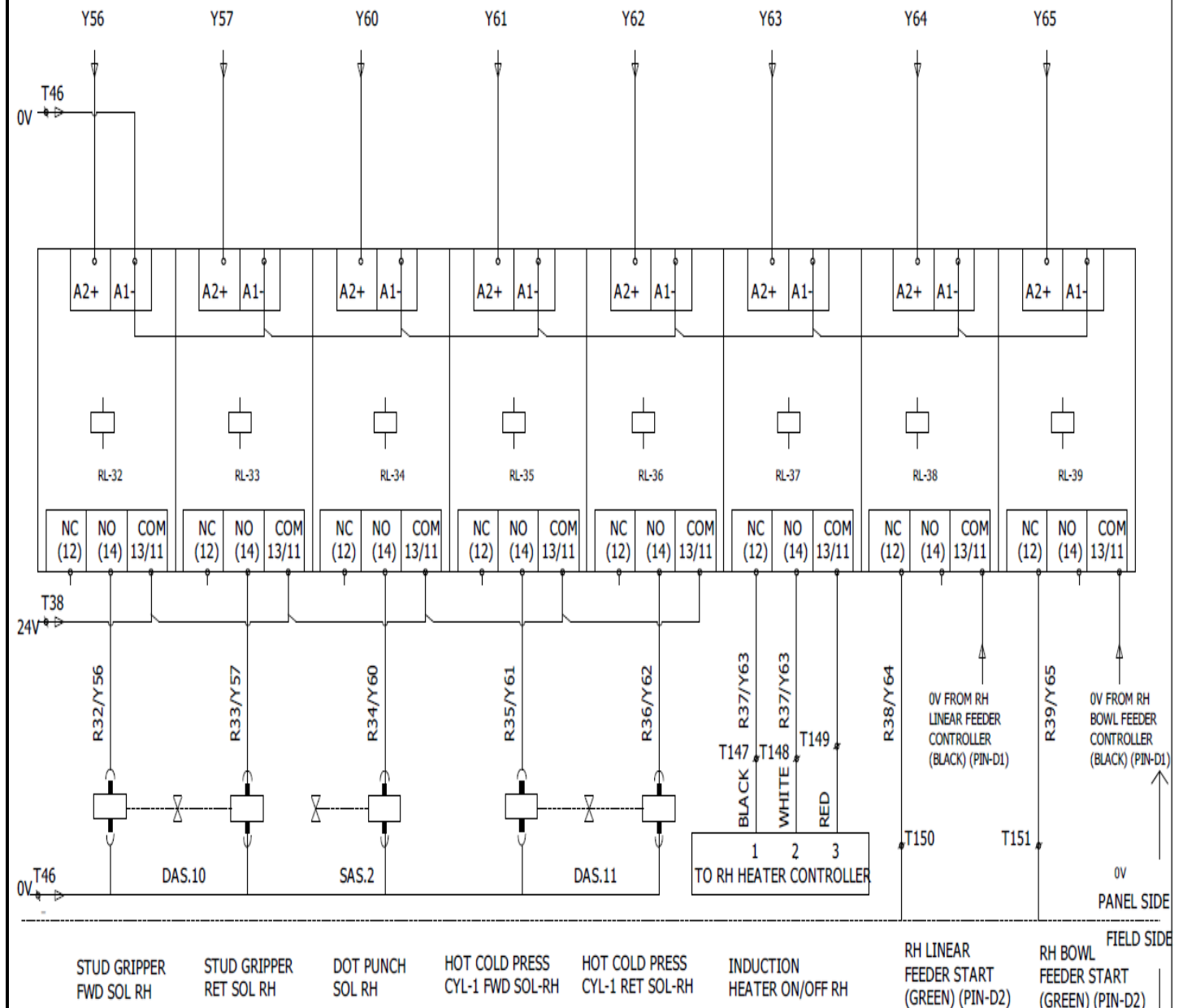
RELAY DIAGRAM-4

RELAY DIAGRAM-4



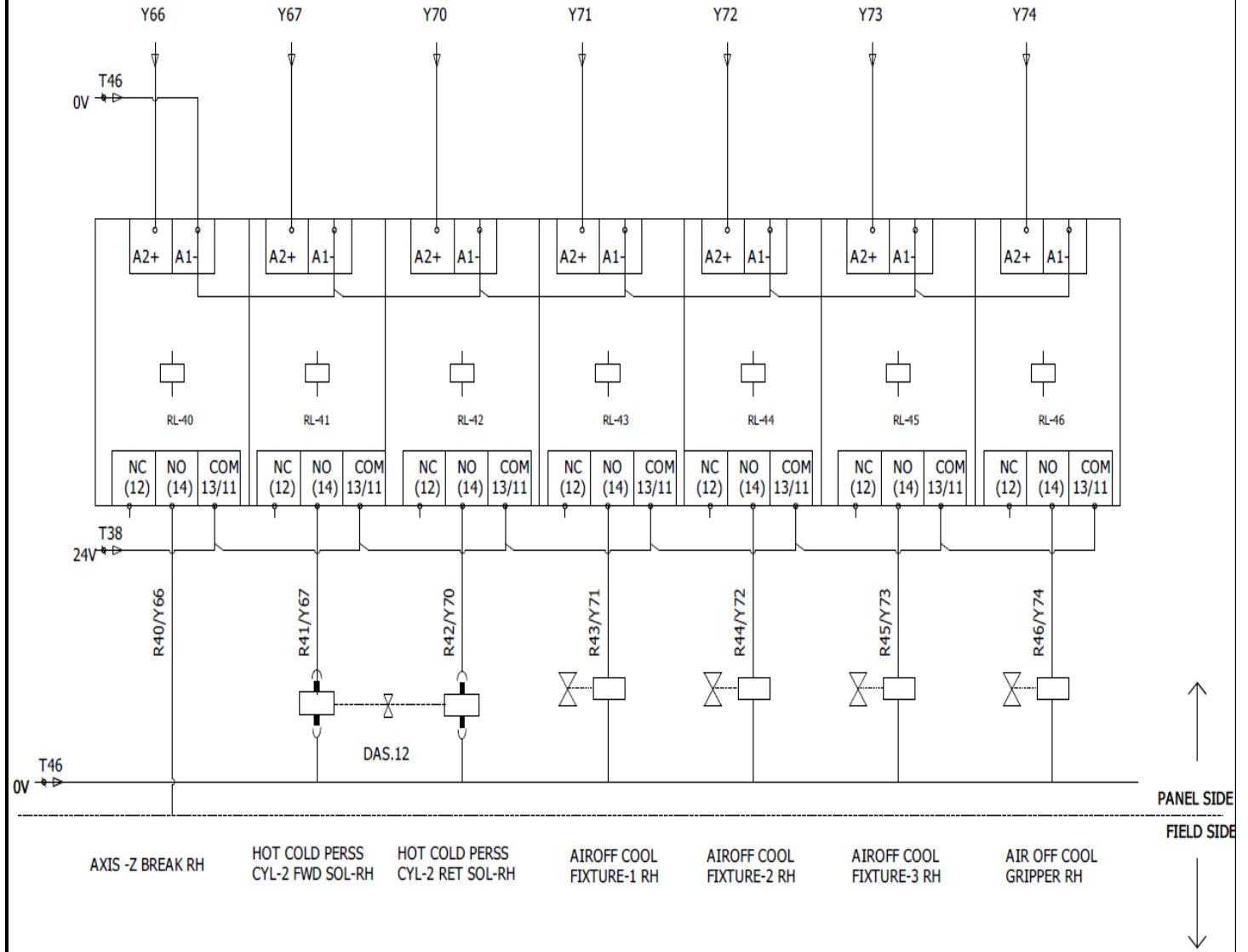
RELAY DIAGRAM-5

RELAY DIAGRAM-5



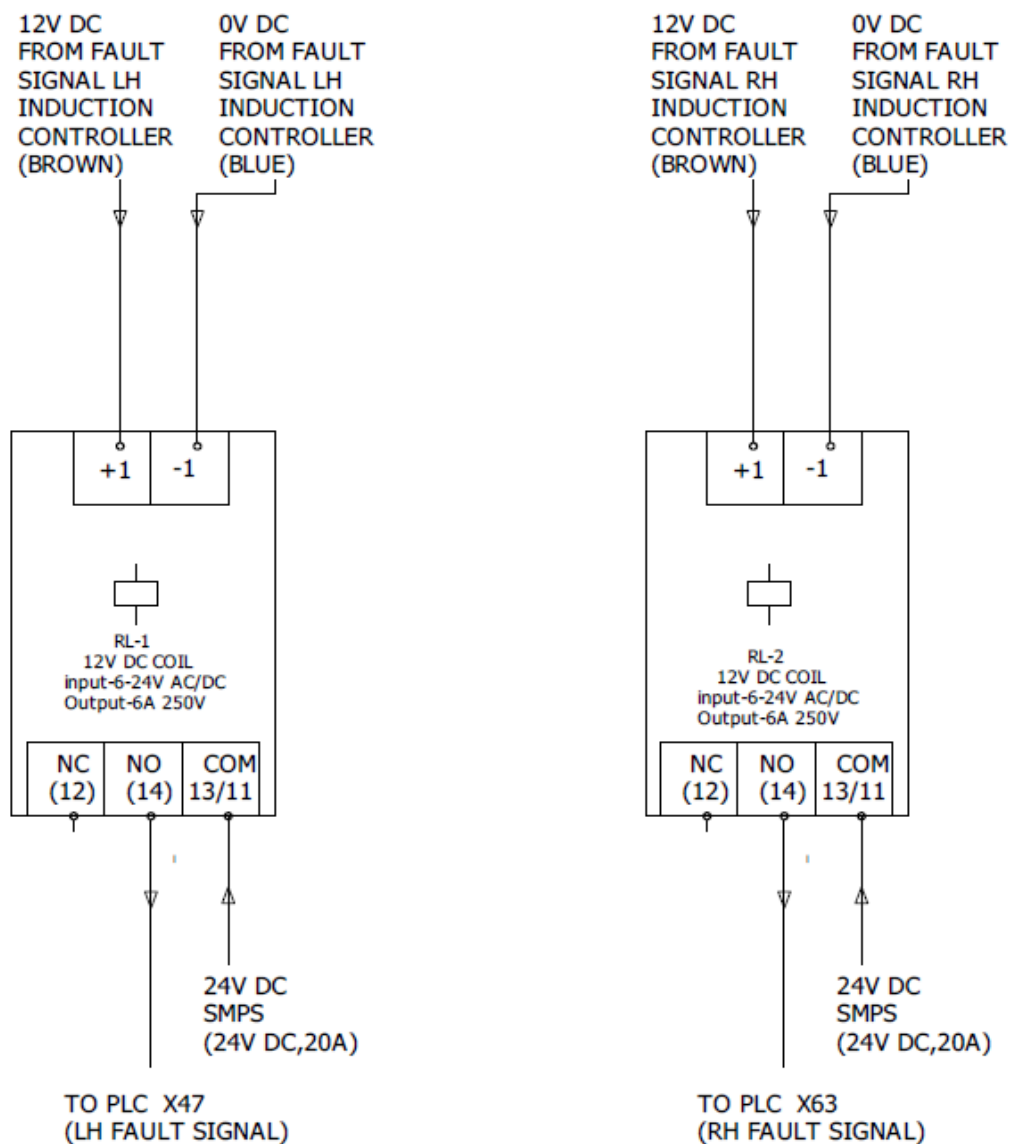
RELAY DIAGRAM-6

RELAY DIAGRAM-6

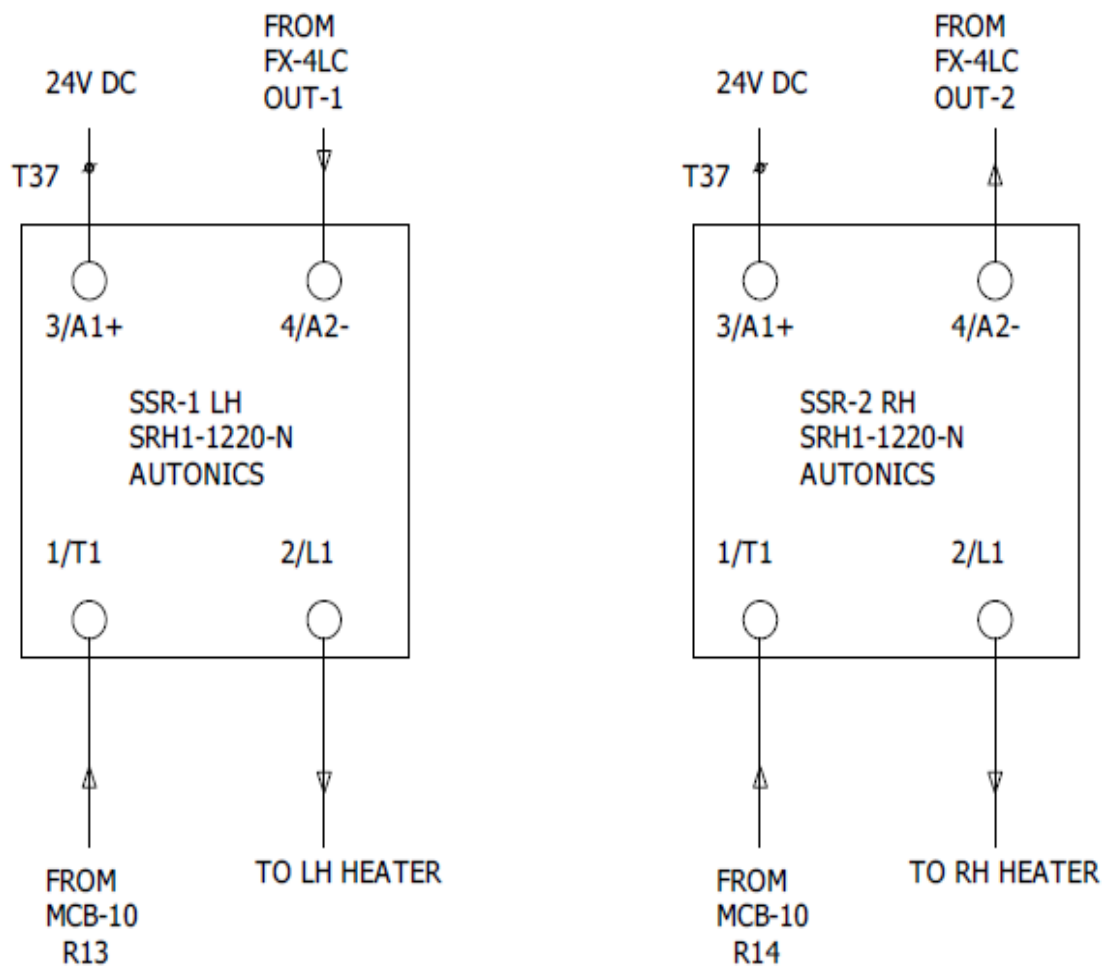


RELAY DIAGRAM-7

RELAY DIAGRAM-7(INDUCTION HEATER FAULT SIGNAL)



SSR CONNECTIONS-LH & RH



PANEL CONNECTOR DETAILS 1

PANEL CONNECTOR DETAILS-1

FIELD SIDE

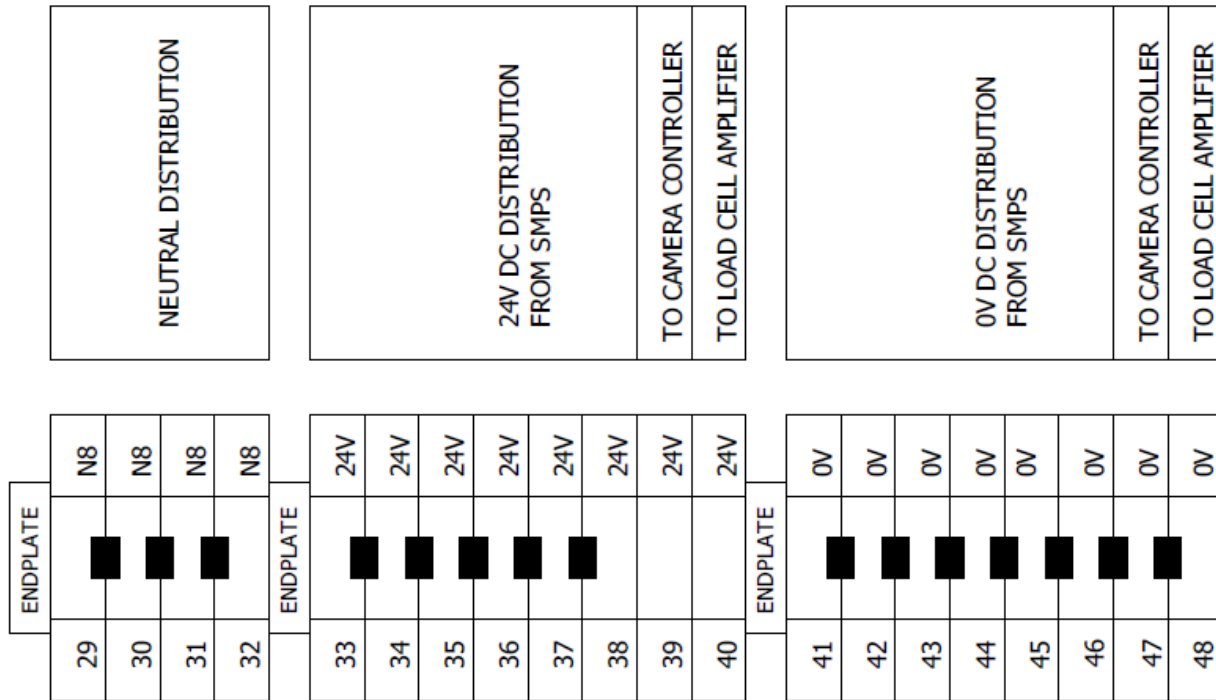


PANEL SIDE

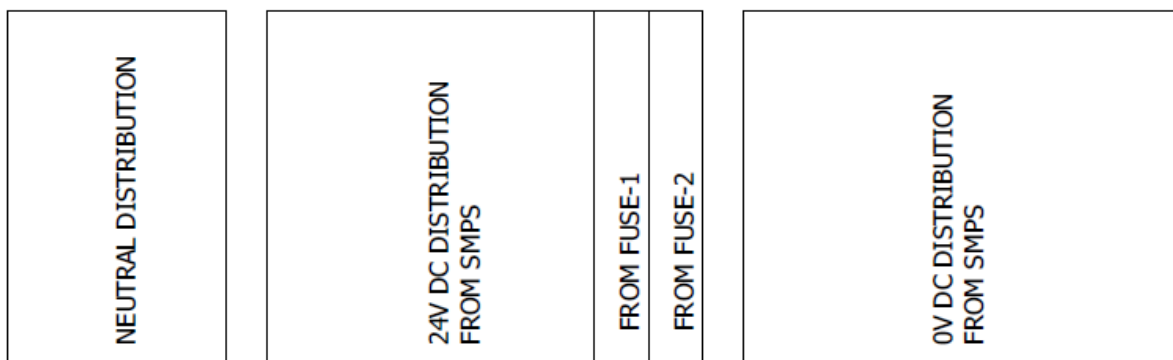
PANEL CONNECTOR DETAILS 2

PANEL CONNECTOR DETAILS-2

FIELD SIDE



PANEL SIDE



PANEL CONNECTOR DETAILS 3

PANEL CONNECTOR DETAILS-3

FIELD SIDE

ENDPLATE					
		49	X0	49	
START PB 2(X1)	X1	50			START PB 1(X0)
	X3	52		51	RESET PB (X2)
	X5	54		53	DIGITAL PRESSURE SW(X4)
LIGHT CURTAIN(X5)	X7	56		55	DOOR L.S.1(X6)
DOOR L.S.2(X7)	X11	58		57	DOOR L.S.3(X10)
PART PRESENT SENSOR LH(X11)	X13	60		59	CENTRE CHECK SENSOR 1 LH(X12)
BOWL FEEDER FULL SENSOR LH(X15)	X15	62		61	STUD PRESENT SENSOR LH(X14)
STUD 2 PRESENT SENS LH(X17)	X17	64		63	STUD 1 PRESENT SENSOR LH(X16)
		ENDPLATE		65	STUD 3 PRESENT SENSOR LH(X20)
STUD 4 PRESENT SENS LH(Provl.)(X21)	X21	66		65	REJECTION BIN SENS (X22)
PART CLAMP FWD RD LH(X23)	X23	68		67	PART CLAMP RET RD LH(X24)
PUSH CYL FWD RD LH(X25)	X25	70		69	PUSH CYL RET RD LH(X26)
STUD GRIPPER FWD RD LH(X27)	X27	72		71	STUD GRIPPER RET RD LH(X30)
DOT PUNCH FWD RD LH(X31)	X31	74		73	DOT PUNCH RET RD LH(X32)
HOT COLD PRESS CYL-1 FWD RD SW-LH(X33)	X33	76		75	HOT COLD PRESS CYL-1 RET RD SW-LH(X34)
HOT COLD PRESS CYL-2 FWD RD SW-LH(X35)	X35	78		77	HOT COLD PRESS CYL-2 RET RD SW-LH(X36)
SLIDE CYL FWD RD SW LH(X37)	X37	80		79	

PANEL SIDE

PANEL CONNECTOR DETAILS-4

PANEL CONNECTOR DETAILS-4

FIELD SIDE

ENDPLATE		SLIDE CYL RET RD SW-LH(X40)	
81	X40	81	X41
83	X42	82	X43
85	X44	84	X45
87	X46	86	X47
89	X50	88	X51
91	X52	90	X53
93	X54	92	X55
95	X56	94	X57
ENDPLATE		AXIS-Z HOME LS LH(X60)	
97	X60	97	X61
99	X62	98	X63
101	X64	100	X65
103	X66	102	X67
105	X70	104	X71
107	X72	106	X73
109	X74	108	X75
111	X76	110	X77

LH FIX SELN IN-1(X41)	82	X41	82	LH FIX SELN IN-2(42)
LH FIX SELN IN-3(X43)	84	X43	84	LH FIX SELN IN-4(44)
STUD PRESENT SENSOR RH(X45)	86	X45	86	BOWL FEEDER FULL SENSOR RH(X46)
SPARE(X47)	88	X47	88	AXIS-X END LS-1 LH(X50)
AXIS-X END LS-2 LH(X51)	90	X51	90	AXIS-X HOME LS LH(X52)
AXIS-Y END LS-1 LH(X53)	92	X53	92	AXIS-Y END LS-2 LH(X54)
AXIS-Y HOME LS LH(X55)	94	X55	94	AXIS-Z END LS-1 LH(X56)
AXIS-Z END LS-2 LH(X57)	96	X57	96	
SPARE(X61)	98	X61	98	SPARE(X62)
SPARE(X63)	100	X63	100	RH FIX SELN IN-1(X64)
RH FIX SELN IN-2(X65)	102	X65	102	RH FIX SELN IN-3(X66)
RH FIX SELN IN-4(X67)	104	X67	104	PART PRESENT SENSOR RH(X70)
CENTRE CHECK SENSOR 1 RH(X71)	106	X71	106	CENTRE CHECK SENSOR 2 RH(X72)
STUD 1 PRESENT SENSOR RH(X73)	108	X73	108	STUD 2 PRESENT SENS RH(X74)
STUD 3 PRESENT SENSOR RH(X75)	110	X75	110	STUD 4 PRESENT SENS RH(X76) (Provision)
PART CLAMP FWD RD RH(X77)	112	X77	112	

PANEL SIDE

PANEL CONNECTOR DETAILS-5

PANEL CONNECTOR DETAILS-5

FIELD SIDE

PANEL SIDE	ENDPLATE		113	X100	113	X101	114	PUSH CYL FWD RD RH(X101)
			115	X102	115	X103	116	STUD GRIPPER FWD RD RH(X103)
			117	X104	117	X105	118	DOT PUNCH FWD RD RH(X105)
			119	X106	119	X107	120	HOT COLD PRESS CYL-1 FWD RD SW-RH(X107)
			121	X110	121	X111	122	HOT COLD PRESS CYL-2 FWD RD SW-RH(X111)
			123	X112	123	X113	124	SLIDE CYL FWD RD SW RH(X113)
			125	X114	125	X115	126	AXIS-X END LS-1 RH(X115)
			127	X116	127	X117	128	AXIS-X HOME LS RH(X117)
	ENDPLATE		129	X120	129	X121	130	AXIS-Y END LS-2 RH(X121)
			131	X122	131	X123	132	AXIS-Z END LS-1 RH(X123)
FIELD SIDE			133	X124	133	X125	134	AXIS-Z HOME LS RH(X125)
			135	X126	135	X127	136	RH LOCK CYL RET RD(X127)
	ENDPLATE		137		Y0			
			138		Y1			
			139		Y2			
			140		Y3			
			141		OV			
	ENDPLATE		142		R16/ Y23 NO			
			143		R16/ Y23 NC			
			144		R16/ Y23 P			
TO TOWER LAMP			145		R17/ Y24 NC			
			146		R18/ Y25 NC			
			147		R37/ Y63 NO			
			148		R37/ Y63 NC			
			149		R37/ Y63 P			
			150		R38/ Y64 NC			
			151		R39/ Y65 NC			
TO INDUCTION HEATER LH								
TO BOWL FEEDER LH								
TO INDUCTION HEATER RH								
TO BOWL FEEDER RH								

PANEL SIDE

PANEL CONNECTOR DETAILS-6

PANEL CONNECTOR DETAILS-6

FIELD SIDE

TEMEPRATURE SENSOR LH
ANALOG GND-1
LOAD CELL (4-20mA) LH
ANALOG GND-2
TEMEPRATURE SENSOR RH
ANALOG GND-3
LOAD CELL (4-20mA) RH
ANALOG GND-4

SMAT SENSOR LH
ANALOG GND-1
SMAT SENSOR RH
ANALOG GND-2
SPARE
ANALOG GND-3
SPARE
ANALOG GND-4

FROM THERMOCOUPLE-LH
FROM THERMOCOUPLE-RH

0V DC TO LH SIDE SERVO (PIN NO-5)
0V DC TO RH SIDE SERVO (PIN NO-5)
TO LH SIDE X,Y,Z AXIS SERVO
TO RH SIDE X,Y,Z AXIS SERVO
TO LH SIDE X,Y,Z AXIS SERVO
EMG STOP (PIN NO-1)
TO RH SIDE X,Y,Z AXIS SERVO
EMG STOP (PIN NO-1)

AI-1	AI-1
AN-GND1	AN-GND1
AI-2	AI-2
AN-GND2	AN-GND2
AI-3	AI-3
AN-GND3	AN-GND3
AI-4	AI-4
AN-GND4	AN-GND4
152	ENDPLATE
153	160
154	161
155	162
156	163
157	164
158	165
159	166
	167

AI-5	AI-5
AN-GND1	AN-GND1
AI-6	AI-6
AN-GND2	AN-GND2
AI-7	AI-7
AN-GND3	AN-GND3
AI-8	AI-8
AN-GND4	AN-GND4
160	ENDPLATE
161	168
162	169
163	170
164	171
165	
166	
167	

CH-1	CH-1
CH-1	CH-1
CH-2	CH-2
CH-2	CH-2
168	ENDPLATE
169	172
170	173
171	174
	175
	176
	177

0VDC	0VDC
0VDC	0VDC
X2	X2
X2	X2
Y45	Y45
Y46	Y46
172	ENDPLATE
173	172
174	173
175	174
176	175
177	176

TEMEPRATURE SENSOR LH
ANALOG GND-1
LOAD CELL (4-20mA) LH
ANALOG GND-2
TEMEPRATURE SENSOR RH
ANALOG GND-3
LOAD CELL (4-20mA) RH
ANALOG GND-4

SMAT SENSOR LH
ANALOG GND-1
SMAT SENSOR RH
ANALOG GND-2
SPARE
ANALOG GND-3
SPARE
ANALOG GND-4

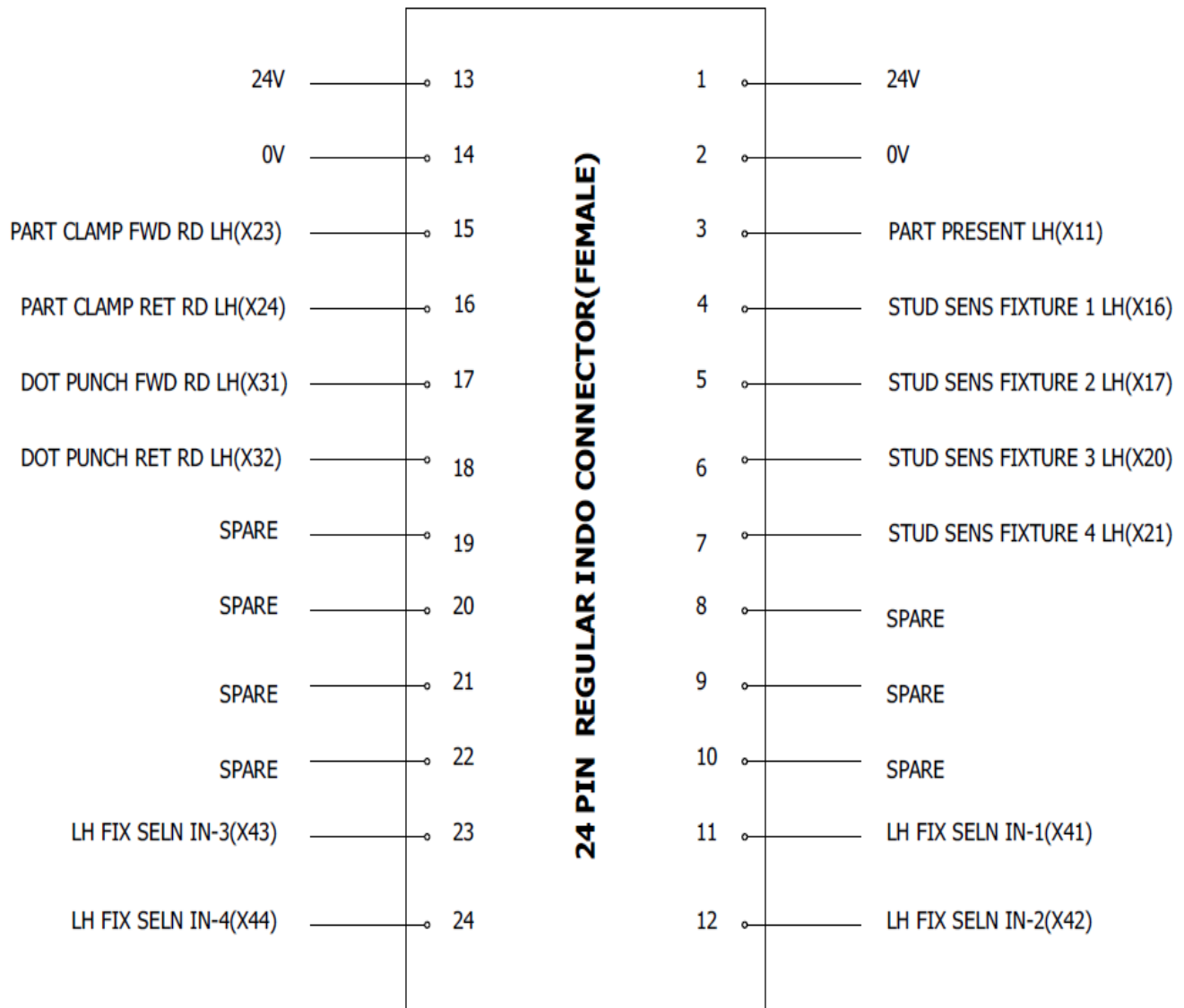
TO CHANNEL-1 B/TC+/VL+
TO CHANNEL-1 b/TC-/VL-
TO CHANNEL-2 B/TC+/VL+
TO CHANNEL-2 b/TC-/VL-

0V DC DISTRI.FROM SMPS
0V DC DISTRI.FROM SMPS
RESET PB (X2) PLC
RESET PB (X2) PLC
EMERGENCY STOP-LH (Y45)
EMERGENCY STOP-RH (Y46)

PANEL SIDE

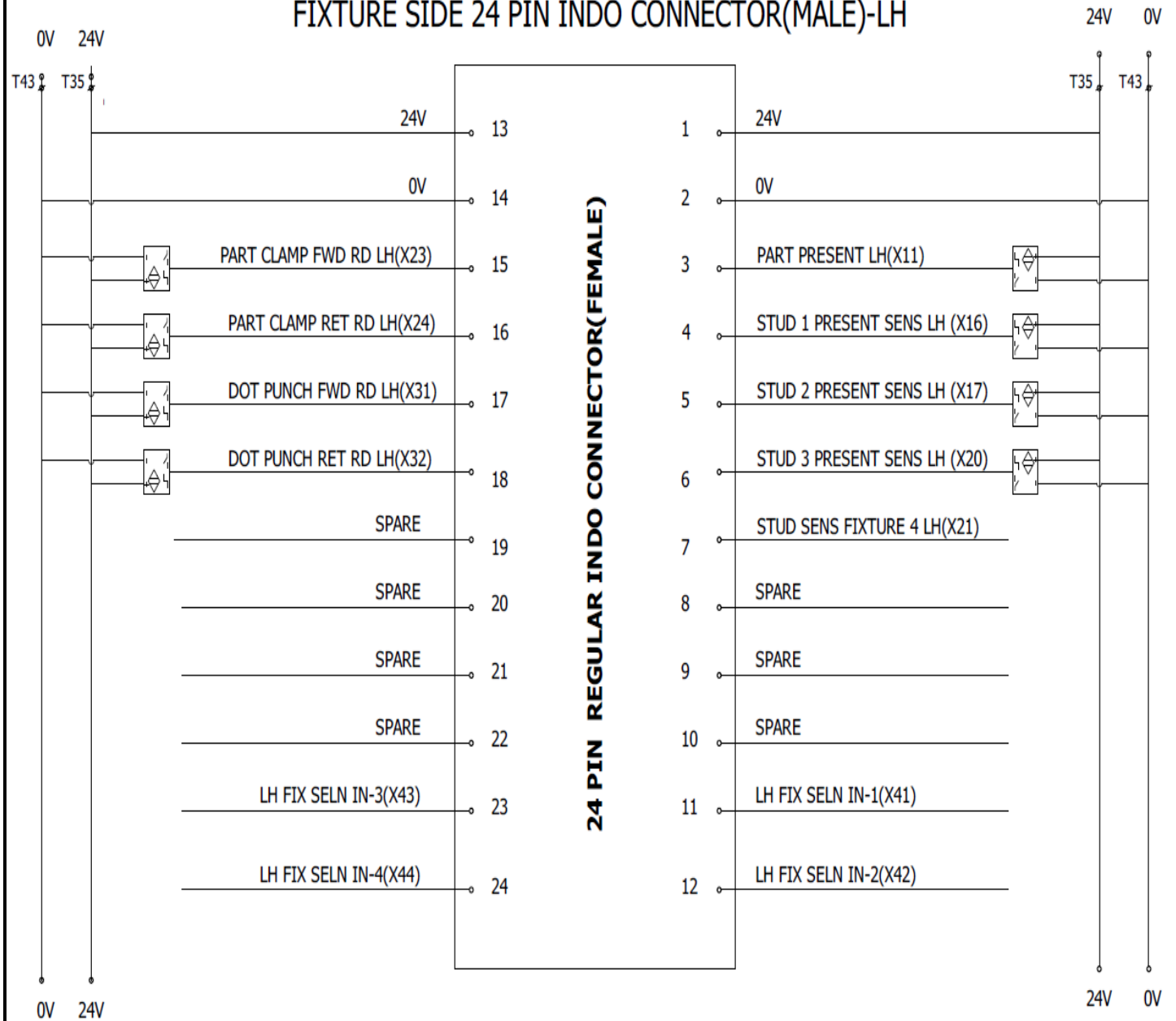
MACHINE SIDE 24 PIN INDO CONNECTOR - LH

MACHINE SIDE 24 PIN INDO CONNECTOR(FEMALE)-LH



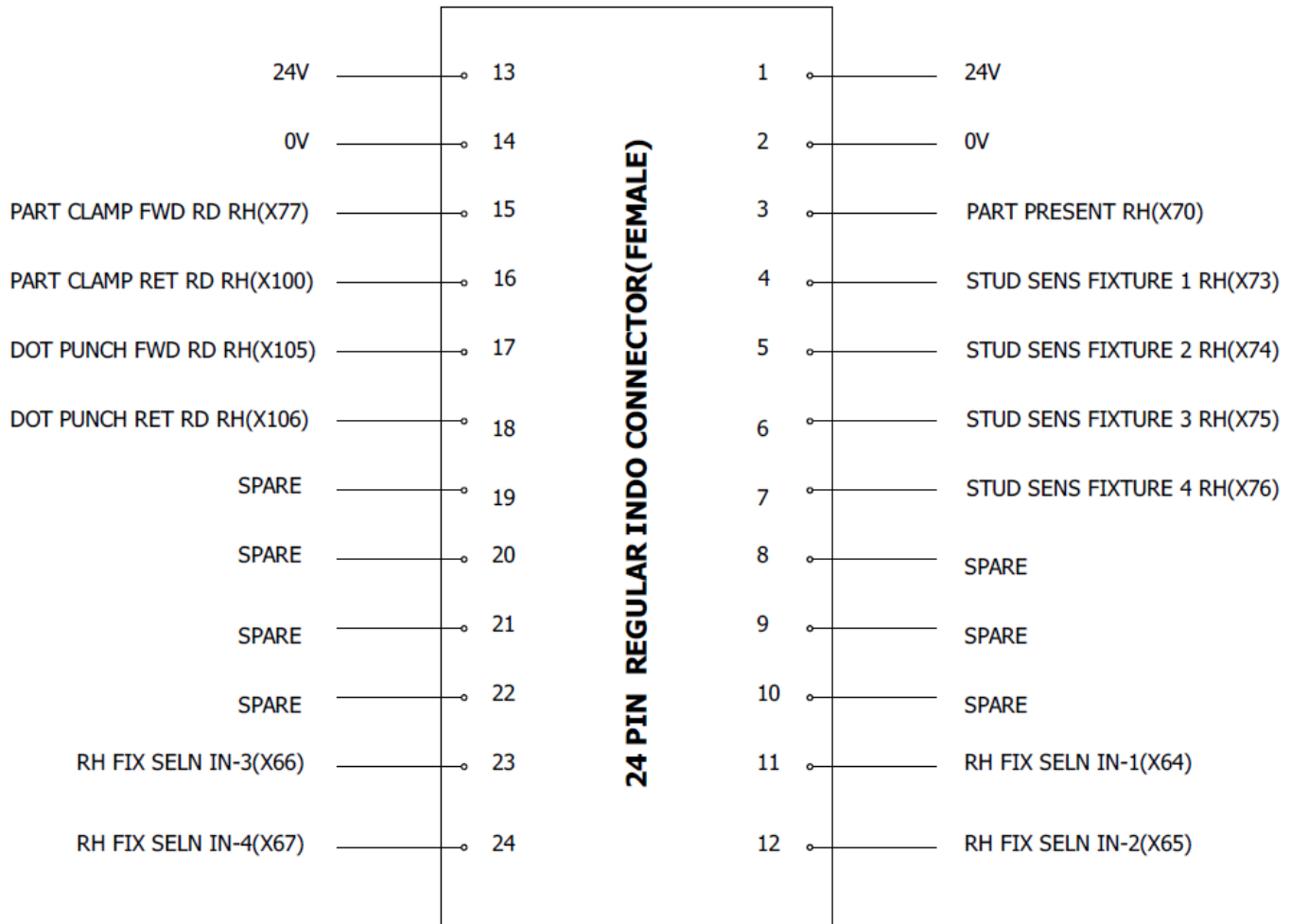
FIXTURE SIDE 24 PIN INDO CONNECTOR - LH

FIXTURE SIDE 24 PIN INDO CONNECTOR(MALE)-LH



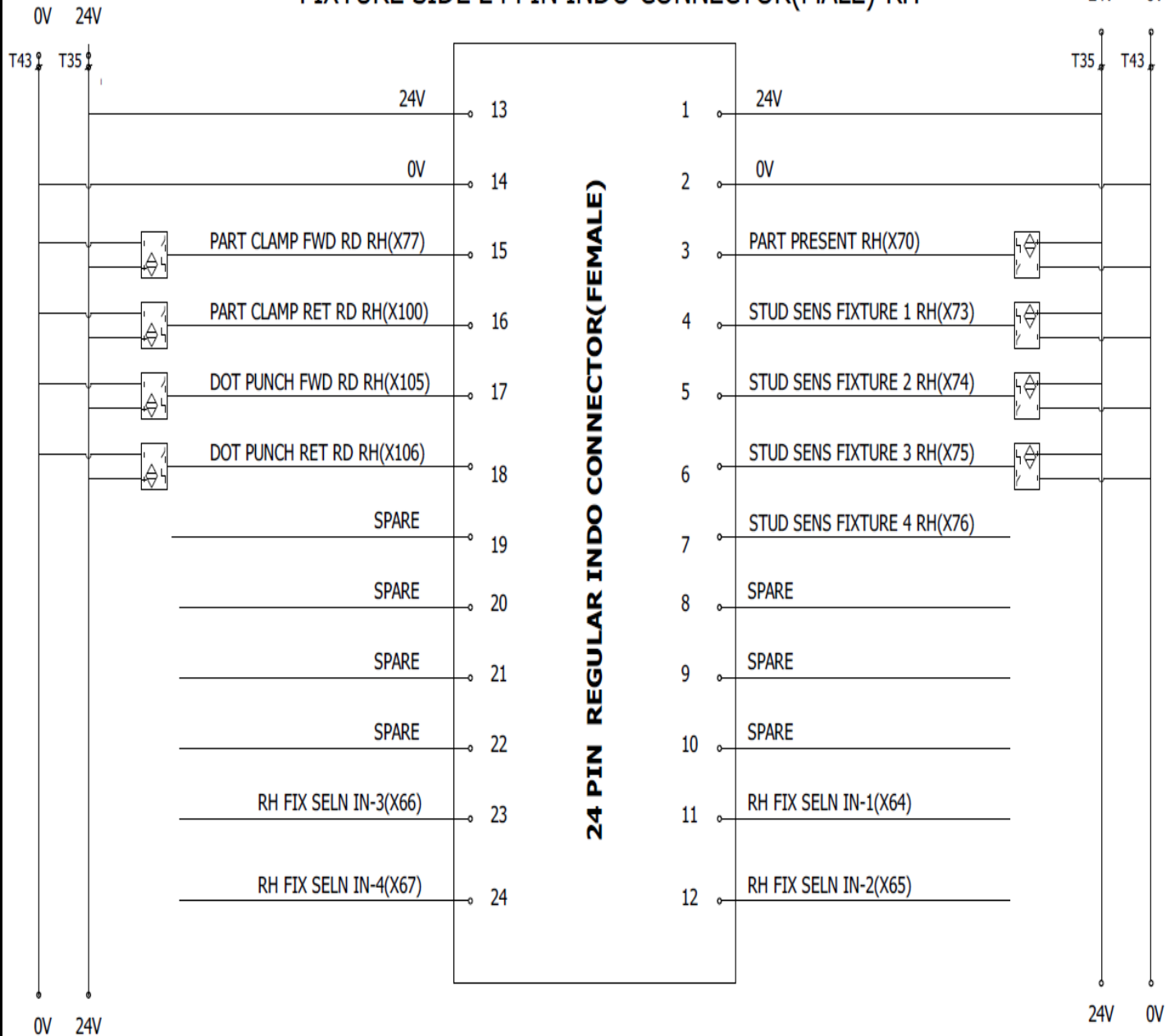
MACHINE SIDE 24 PIN INDO CONNECTOR - RH

MACHINE SIDE 24 PIN INDO CONNECTOR(FEMALE)-RH



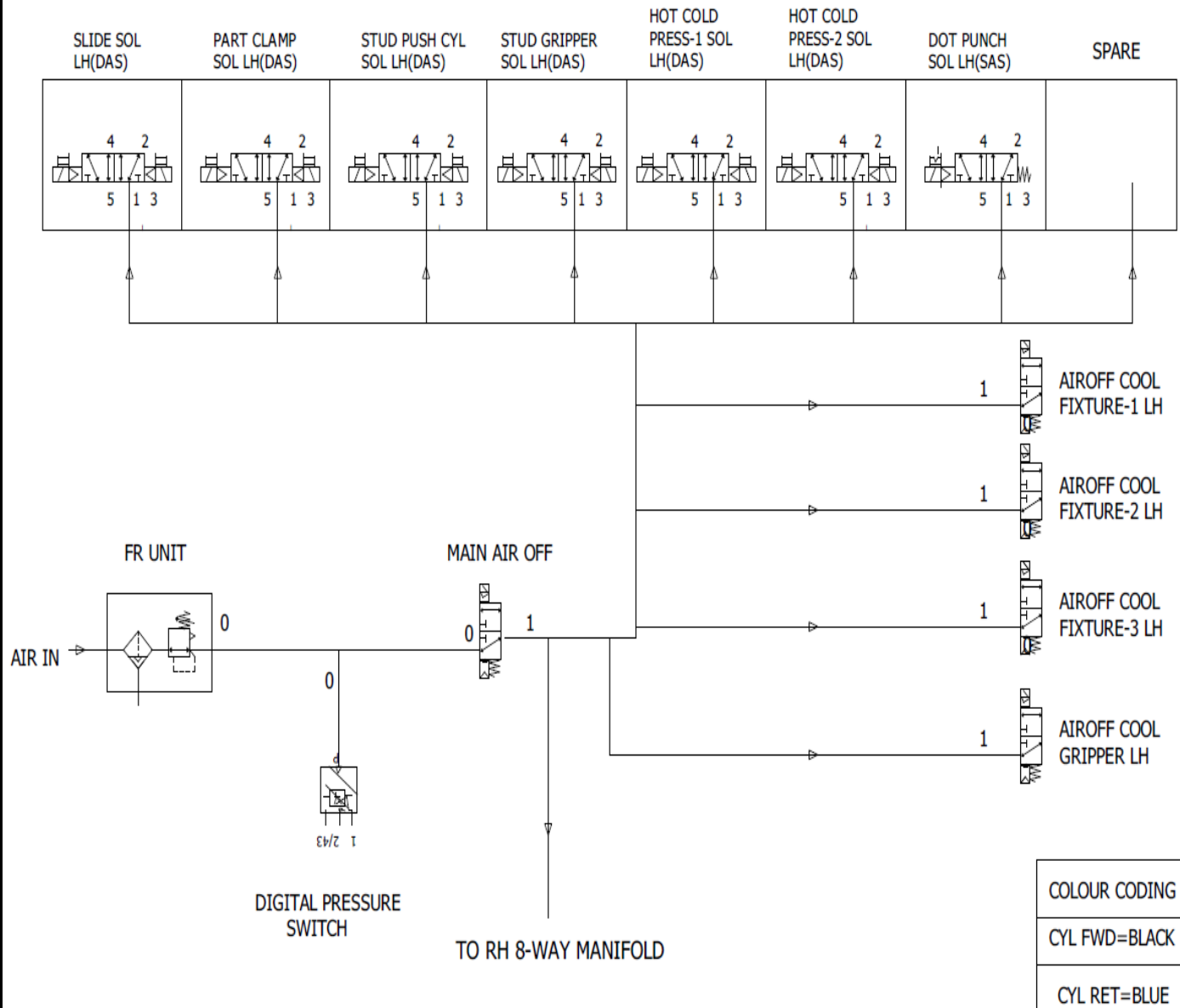
FIXTURE SIDE 24 PIN INDO CONNECTOR - RH

FIXTURE SIDE 24 PIN INDO CONNECTOR(MALE)-RH



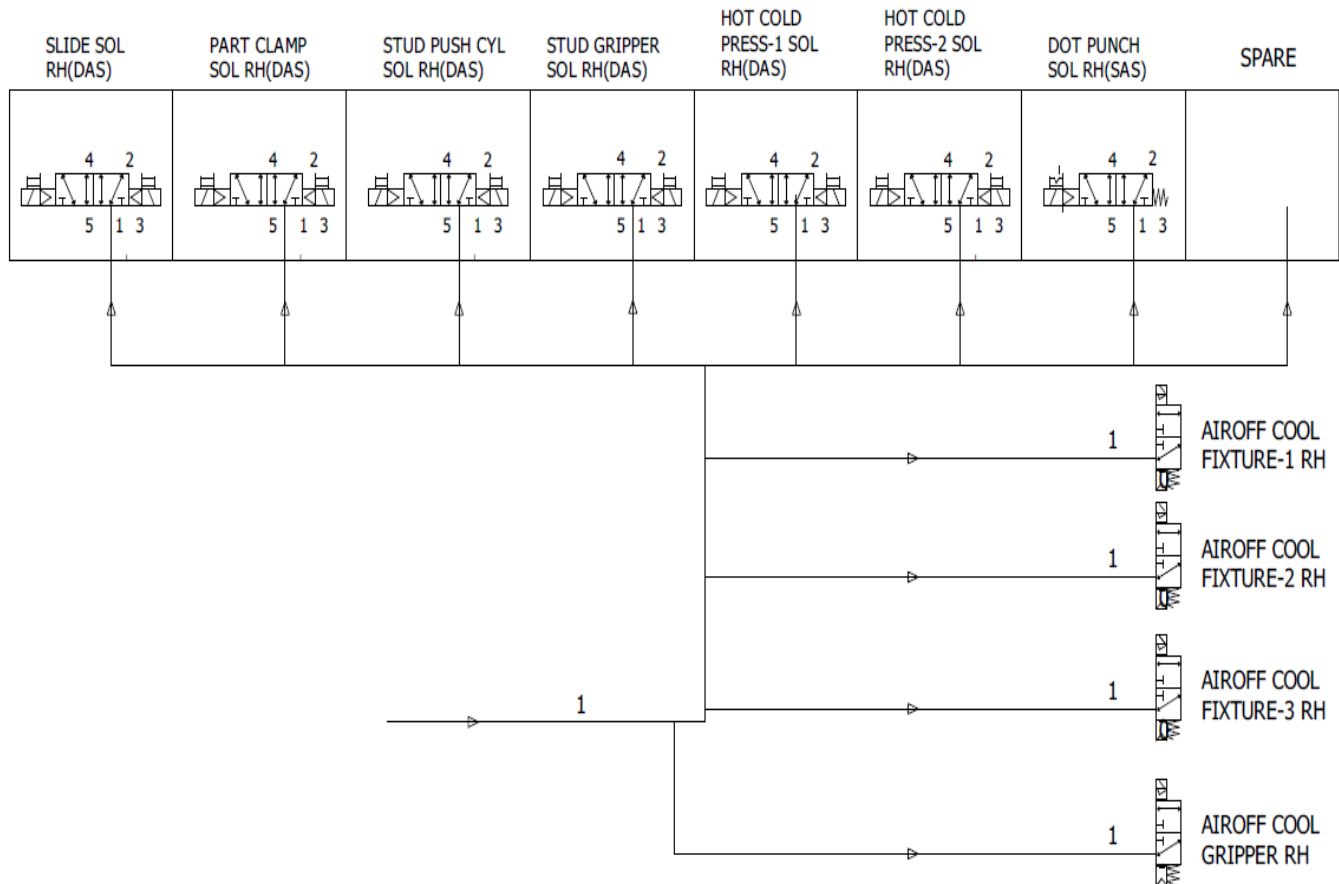
PNEUMATIC CONNECTIONS-LH

PNEUMATIC CONNECTIONS-LH



PNEUMATIC CONNECTOR - RH

PNEUMATIC CONNECTIONS-RH



COLOUR CODING

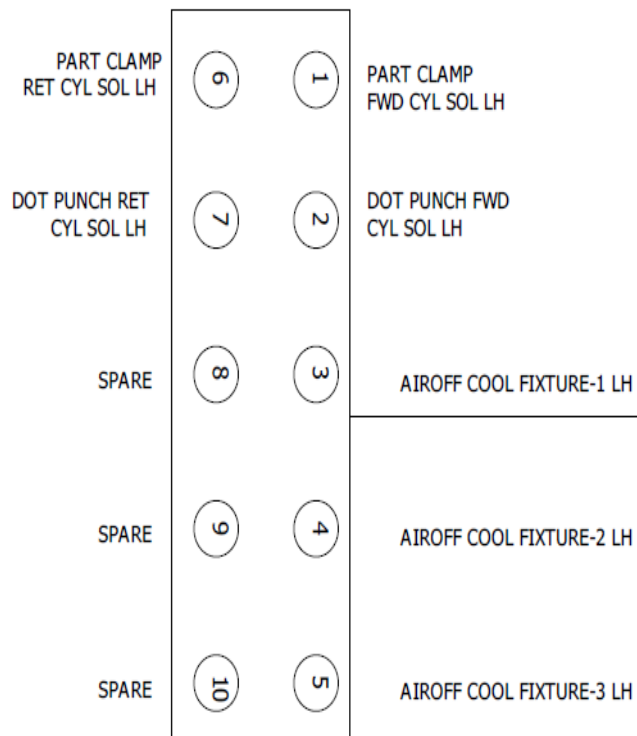
CYL FWD=BLACK

CYL RET=BLUE

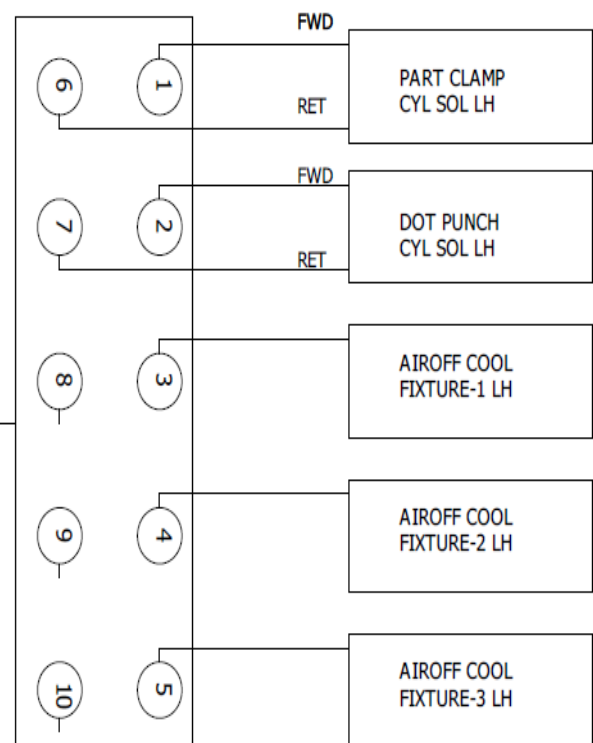
PNEUMATIC CONNECTOR -LH

PNEUMATIC CONNECTOR-LH

PNEUMATIC CONNECTOR (MALE)



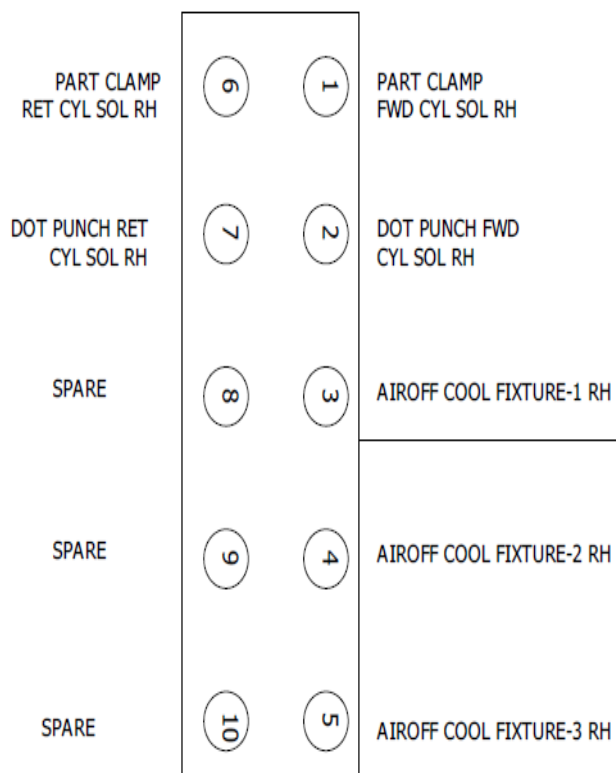
PNEUMATIC CONNECTOR (FEMALE)



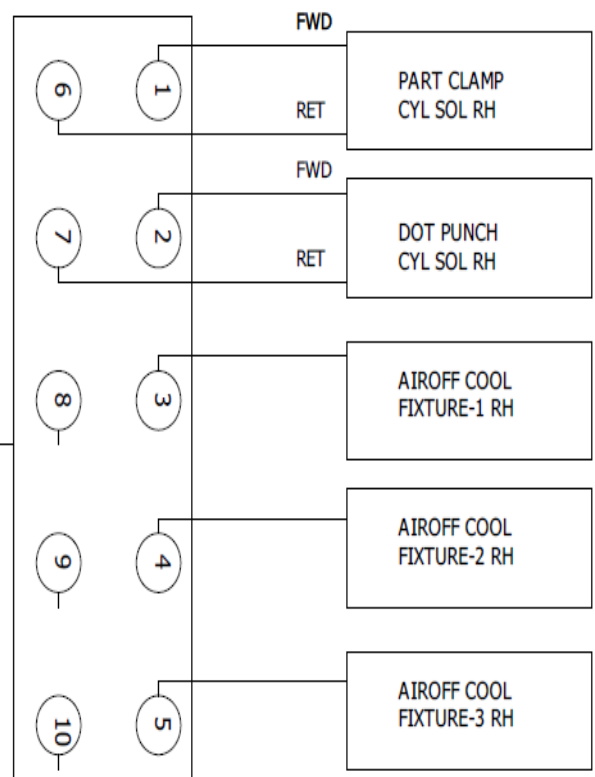
PNEUMATIC CONNECTOR -RH

PNEUMATIC CONNECTOR-RH

PNEUMATIC CONNECTOR (MALE)

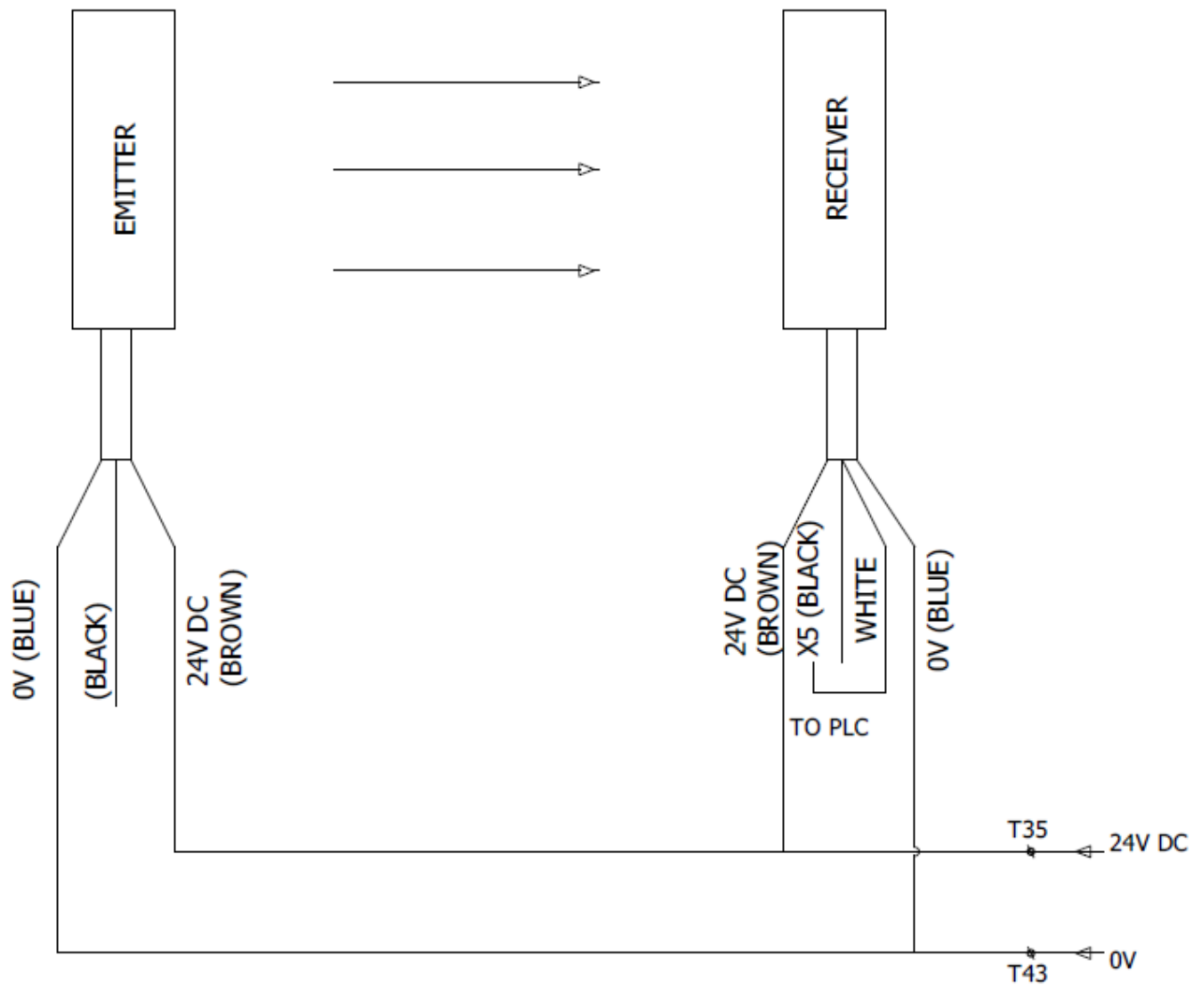


PNEUMATIC CONNECTOR (FEMALE)



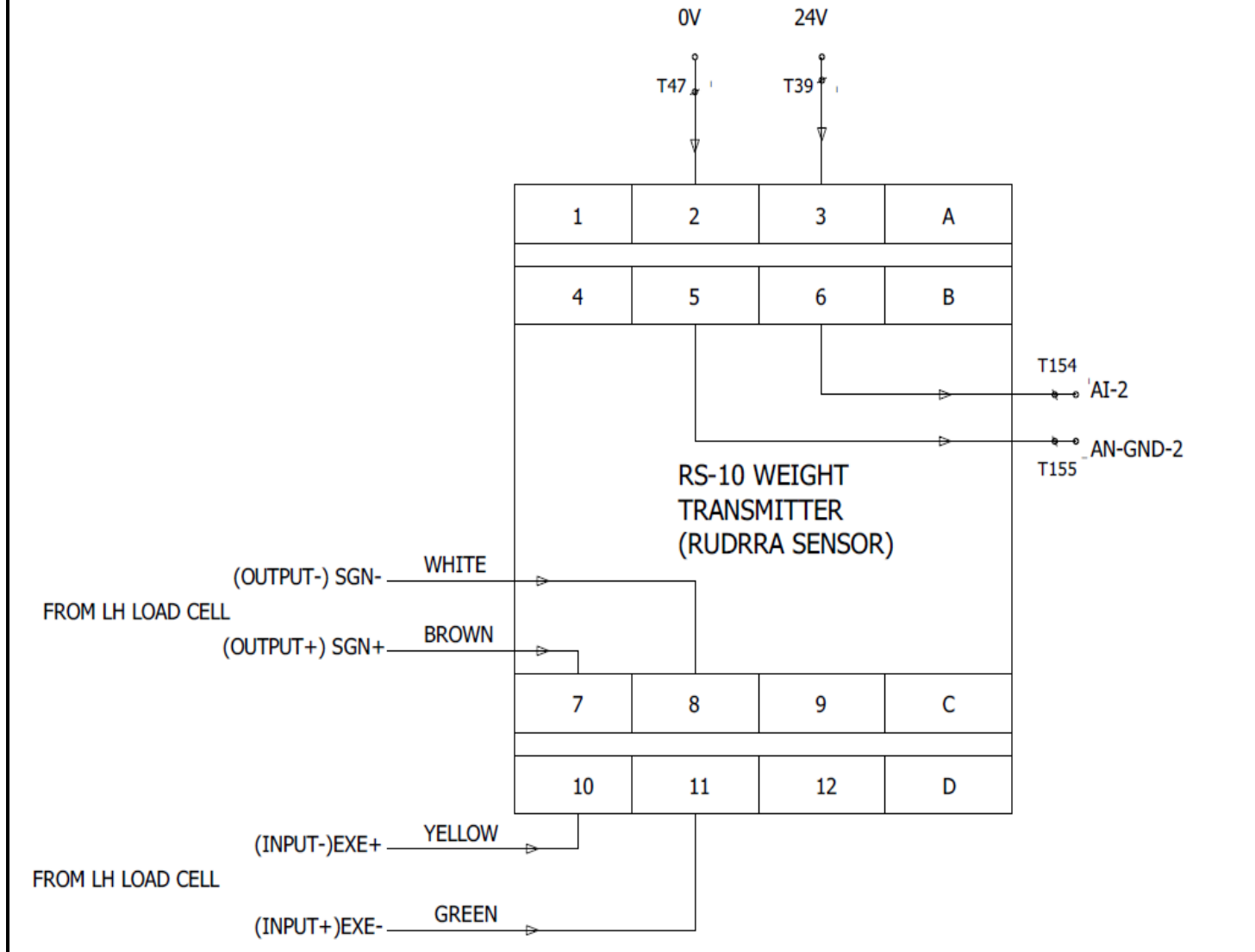
LIGHT CURTAIN CONNECTIONS

LIGHT CURTAIN CONNECTIONS



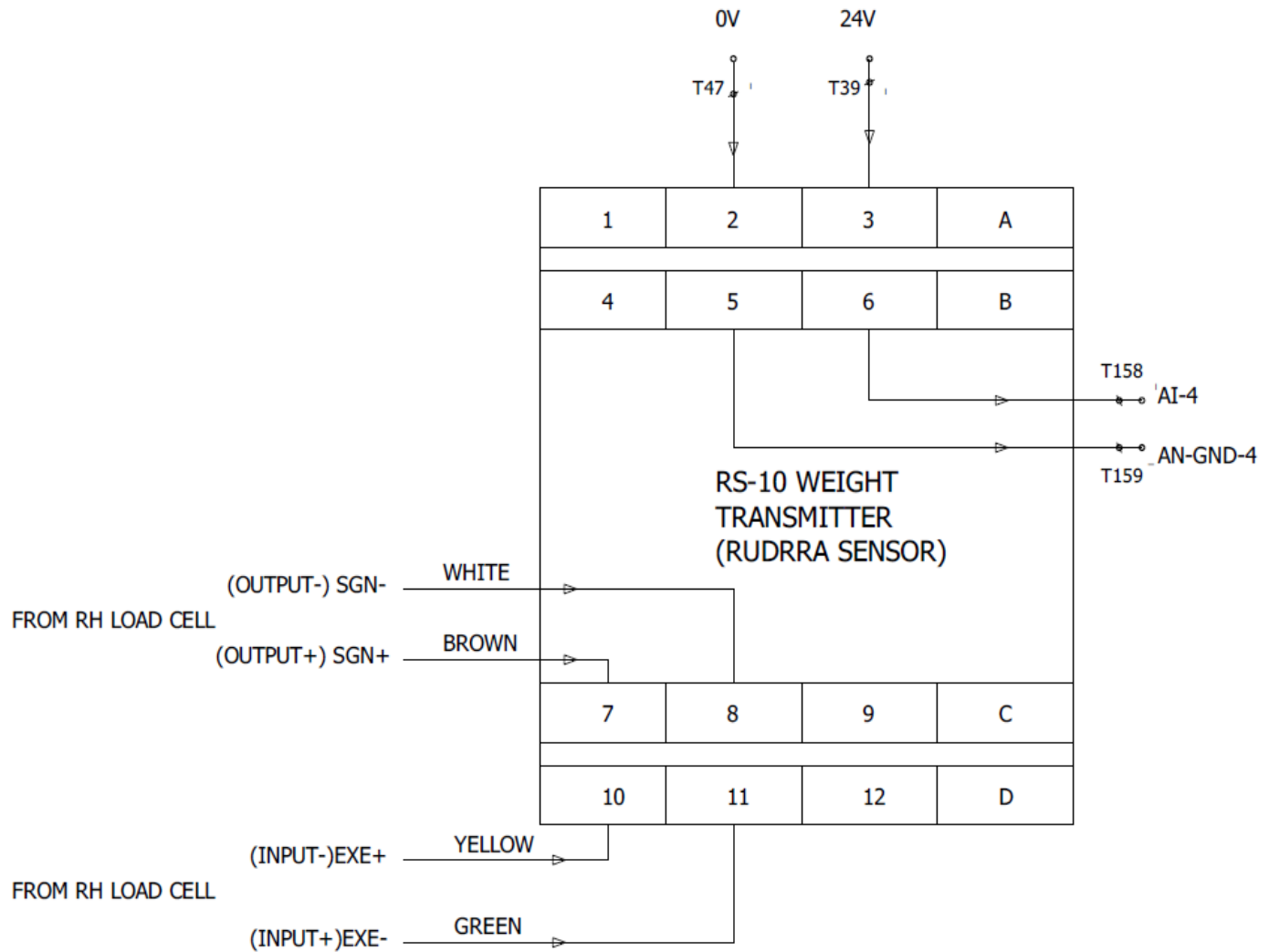
LOAD CELL AMPLIFIER CONNECTIONS -LH

LOAD CELL AMPLIFIER CONNECTIONS-LH



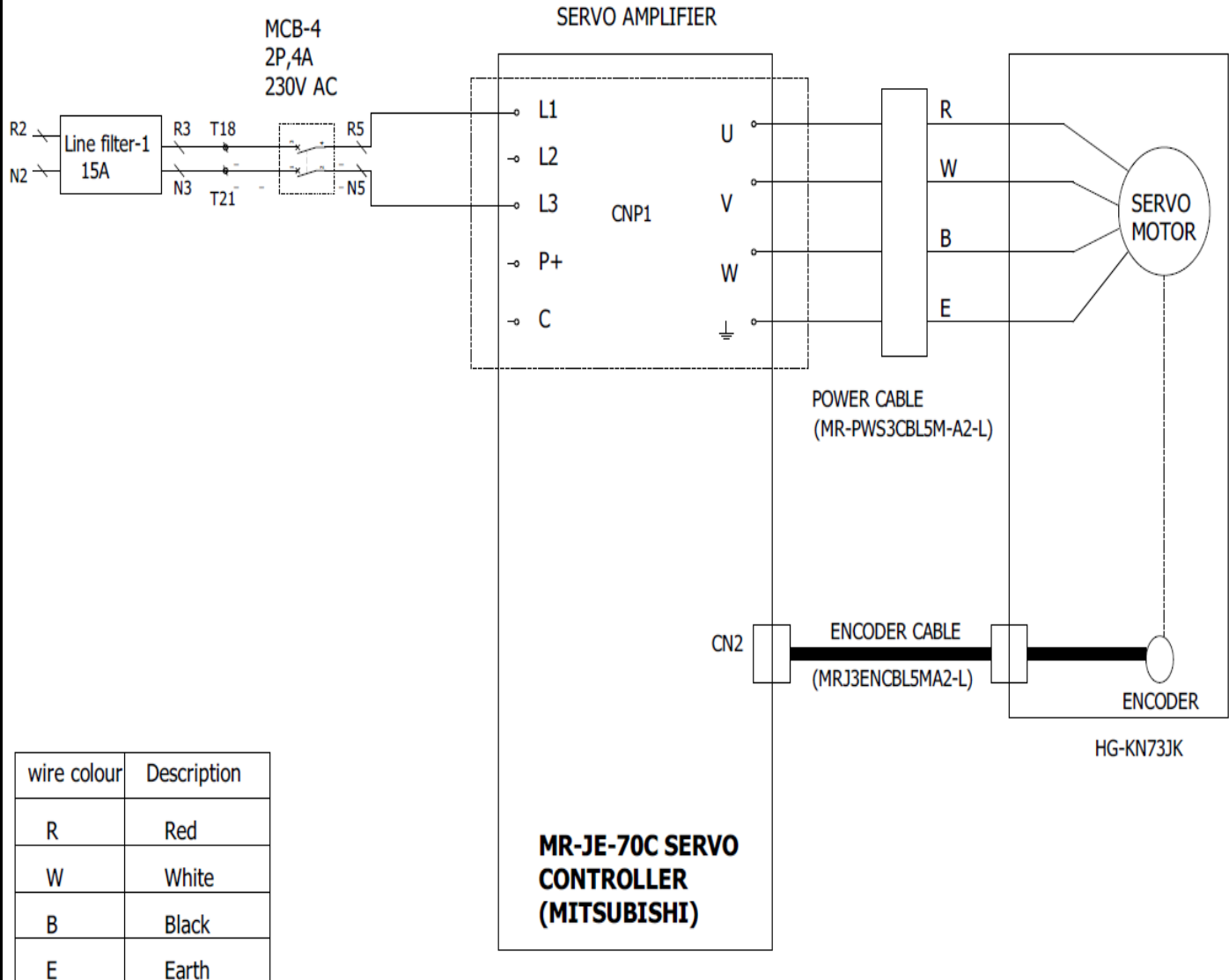
LOAD CELL AMPLIFIER CONNECTIONS-RH

LOAD CELL AMPLIFIER CONNECTIONS-RH



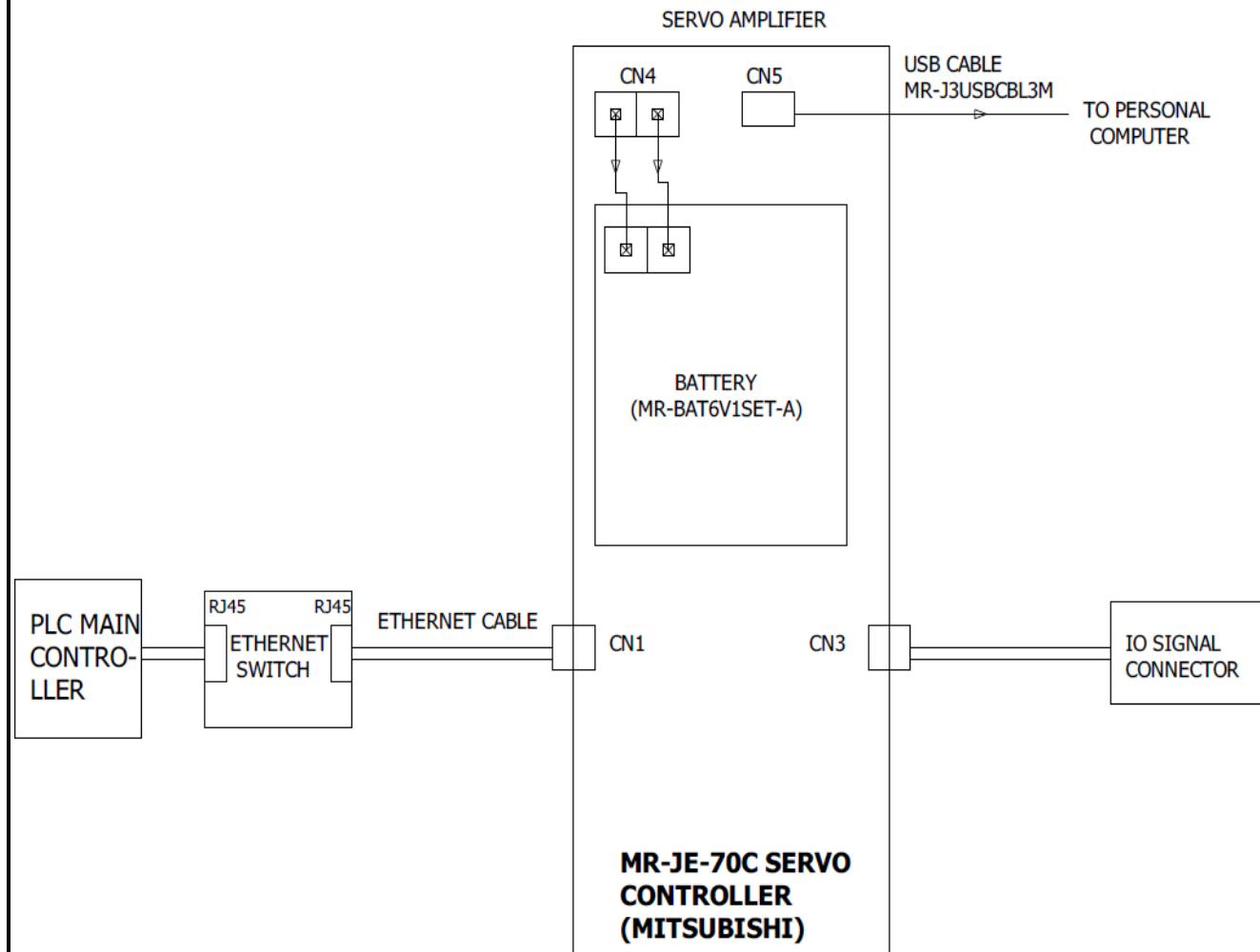
SERVO DRIVE-X AXIS LH POWER CONNECTIONS

SERVO DRIVE-X AXIS LH POWER CONNECTIONS



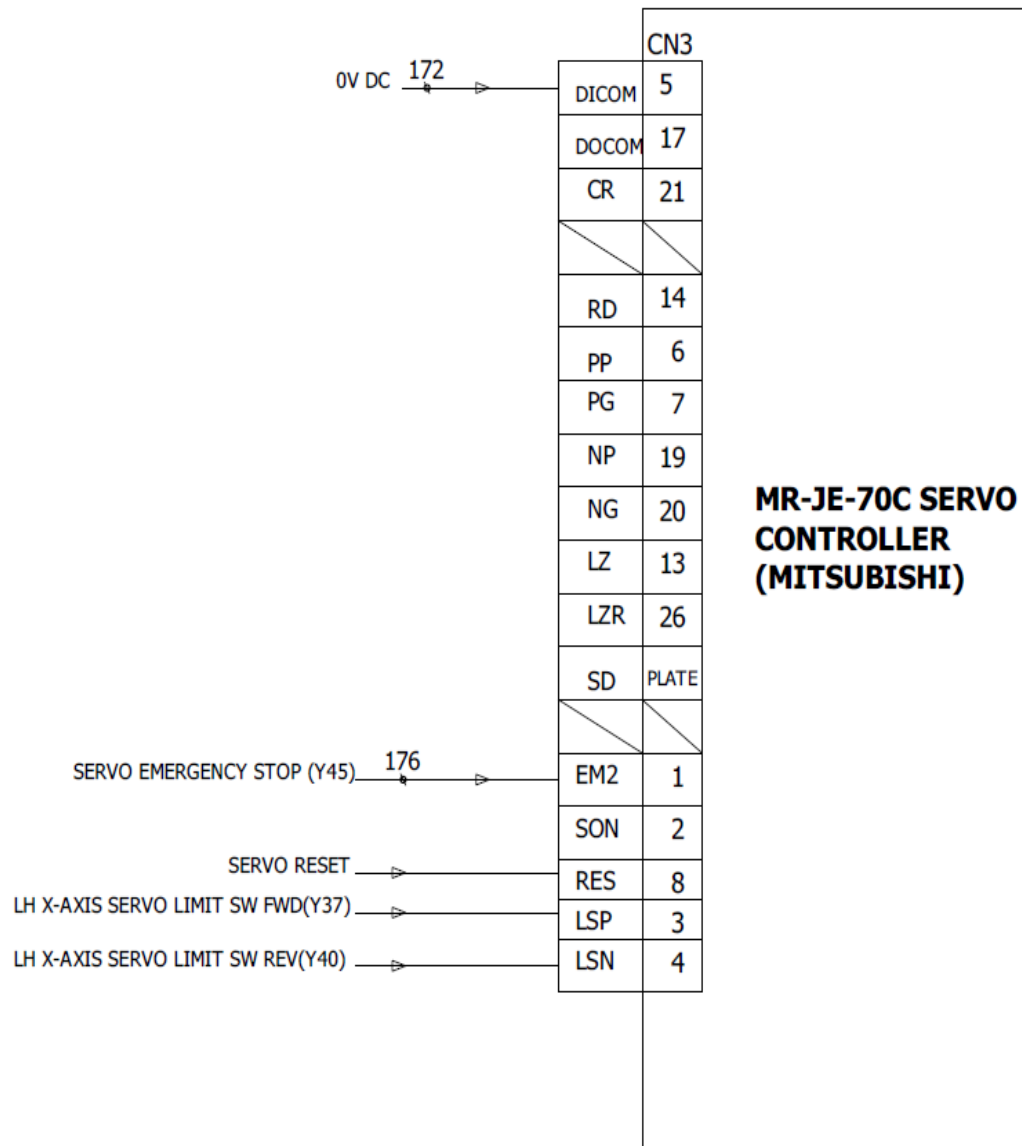
SERVO DRIVE-X AXIS LH I/O CONNECTIONS

SERVO DRIVE-X AXIS LH I/O CONNECTIONS



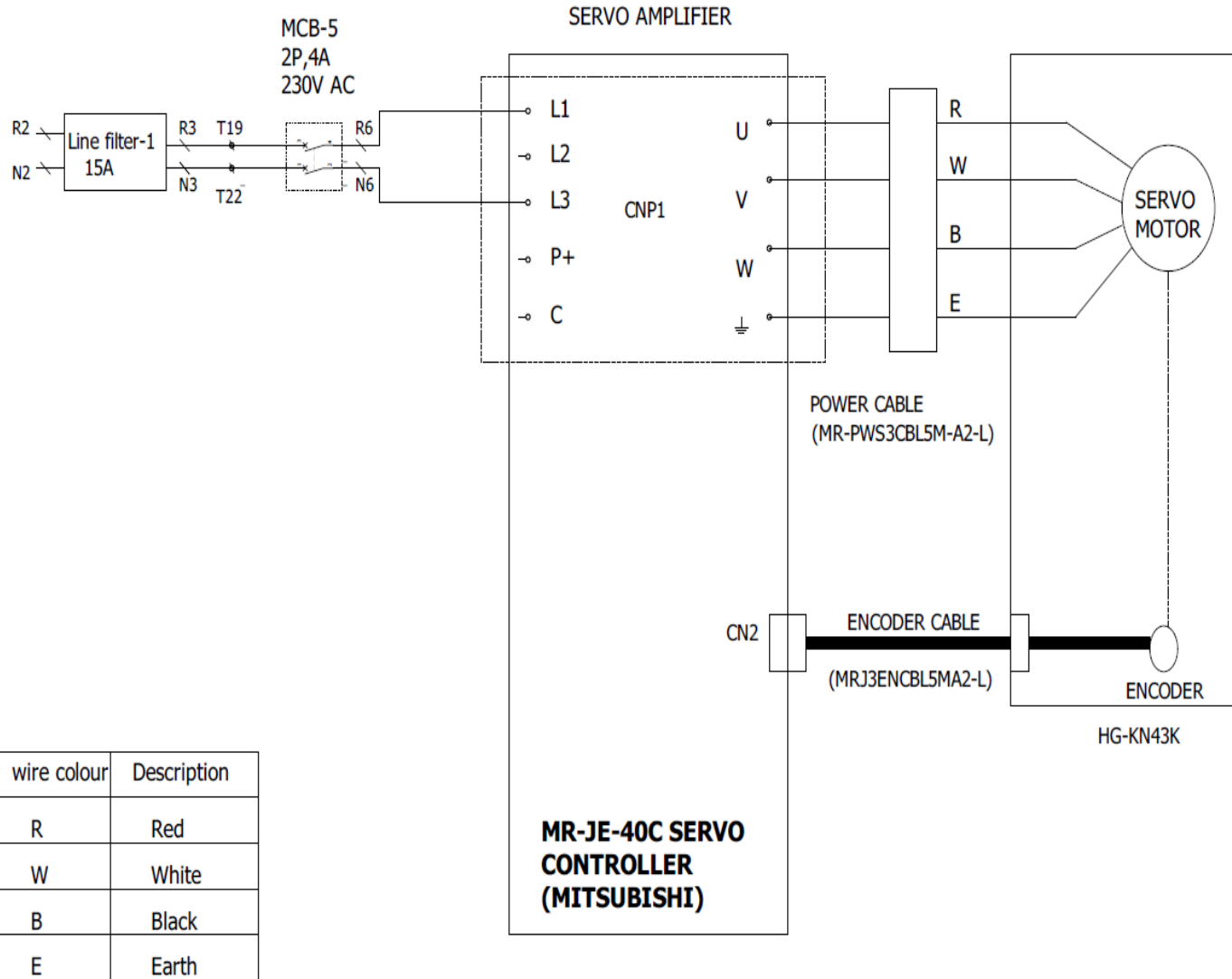
LH X-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIOS

LH X-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



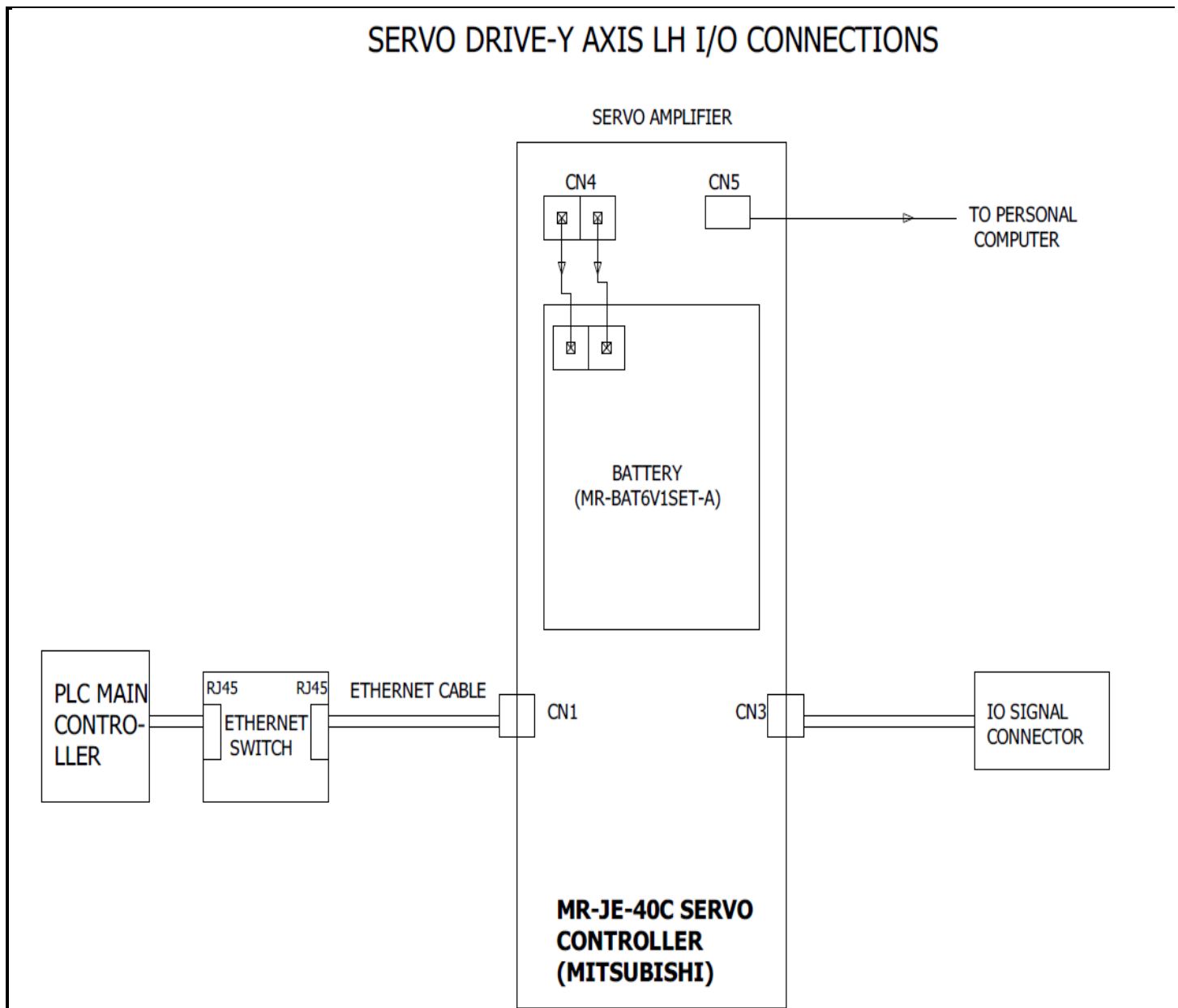
SERVO DRIVE Y AXIS LH POWER CONNECTIONS

SERVO DRIVE-Y AXIS LH POWER CONNECTIONS



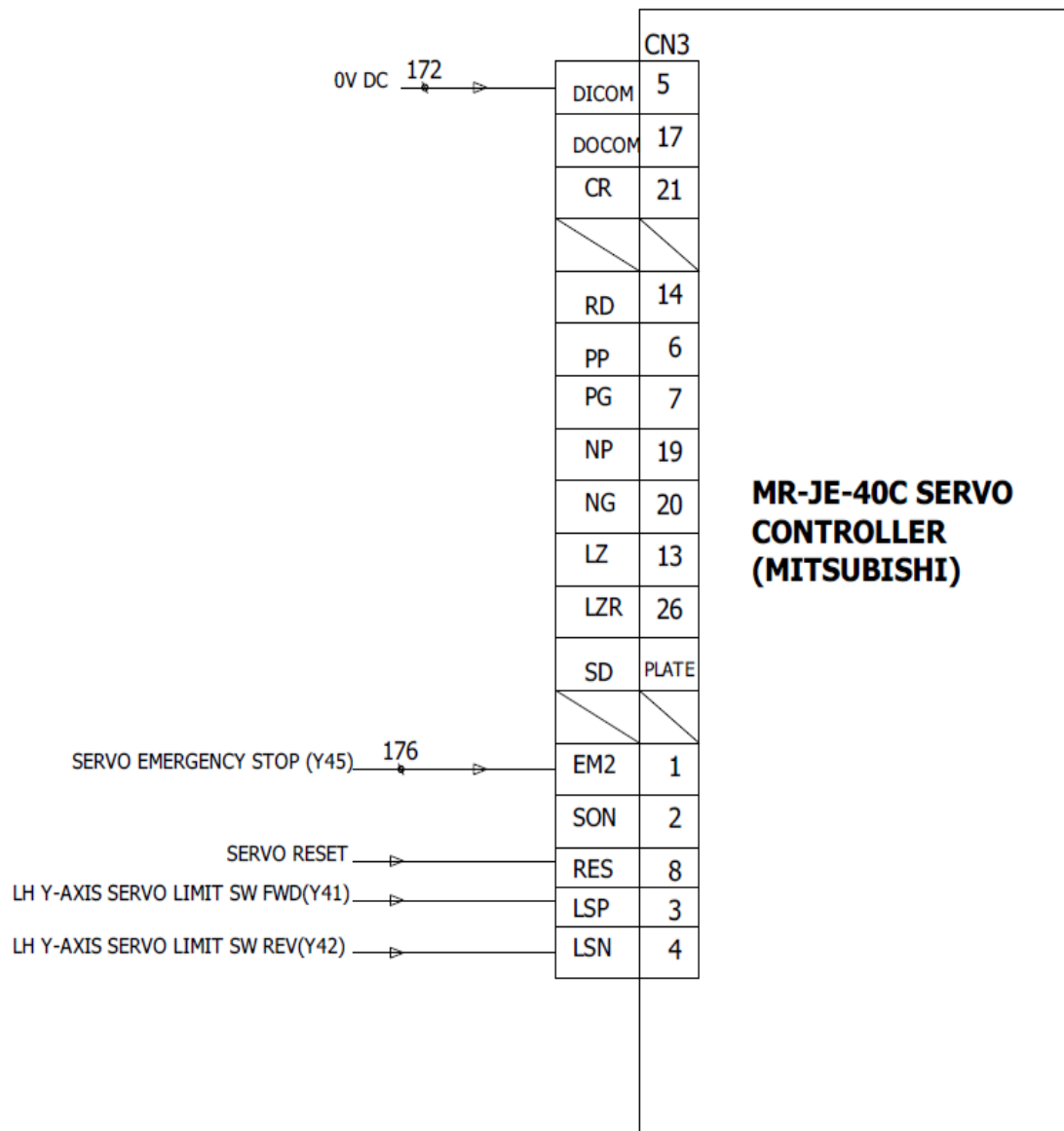
SERVO DRIVE Y AXIS LH I/O CONNECTIONS

SERVO DRIVE-Y AXIS LH I/O CONNECTIONS



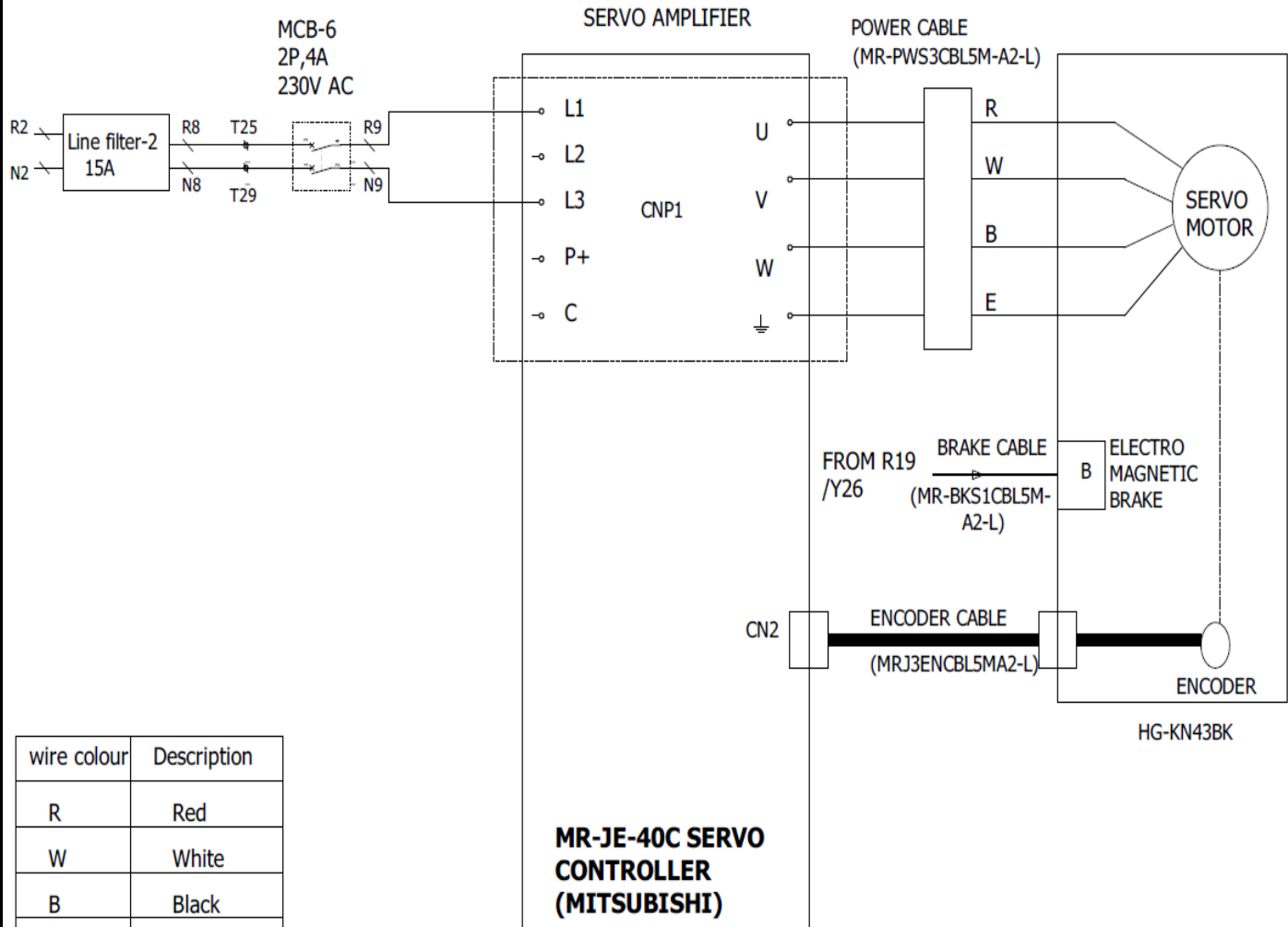
LH Y-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS

LH Y-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



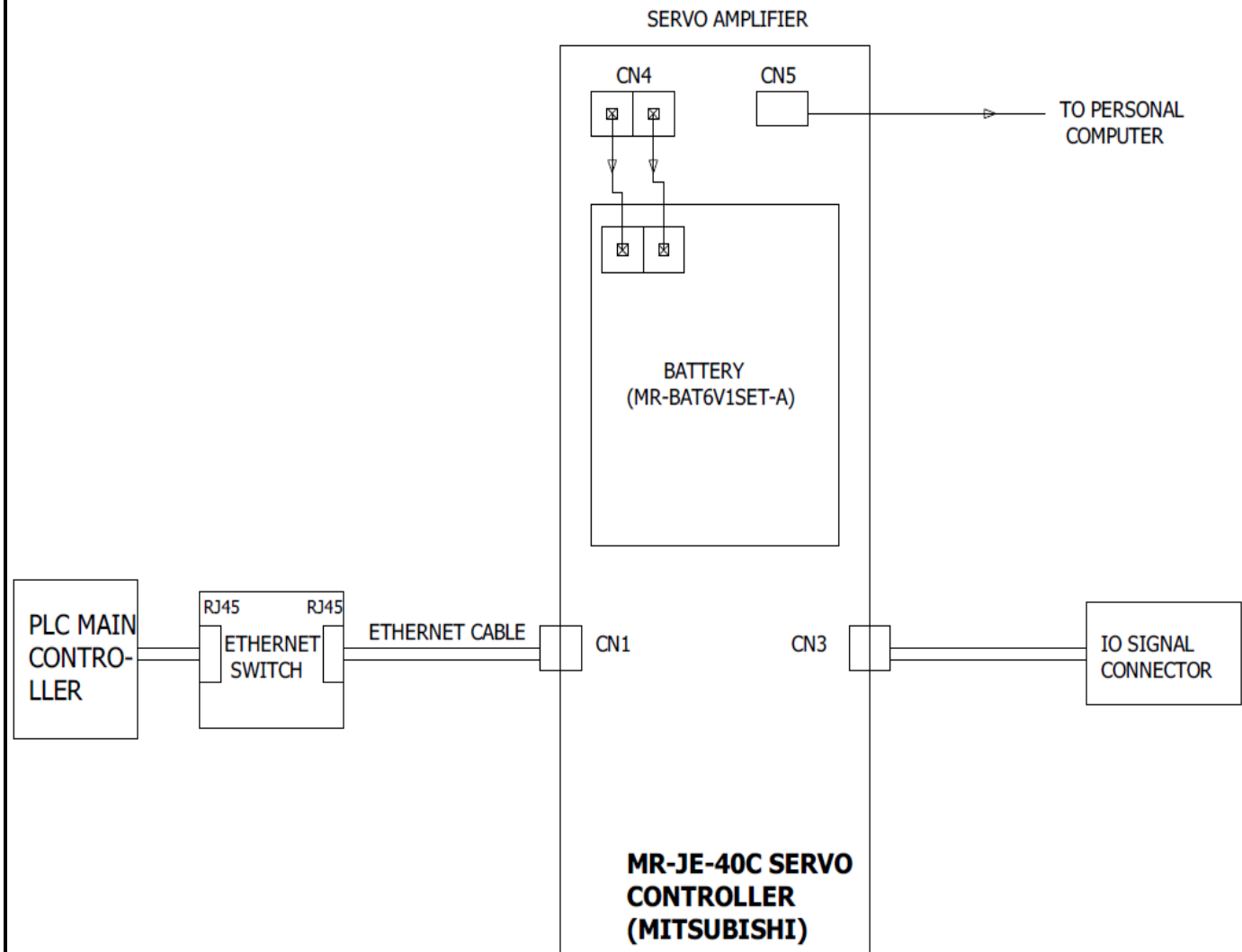
SERVO DRIVE Z AXIS LH POWER CONNECTIONS

SERVO DRIVE-Z AXIS LH POWER CONNECTIONS



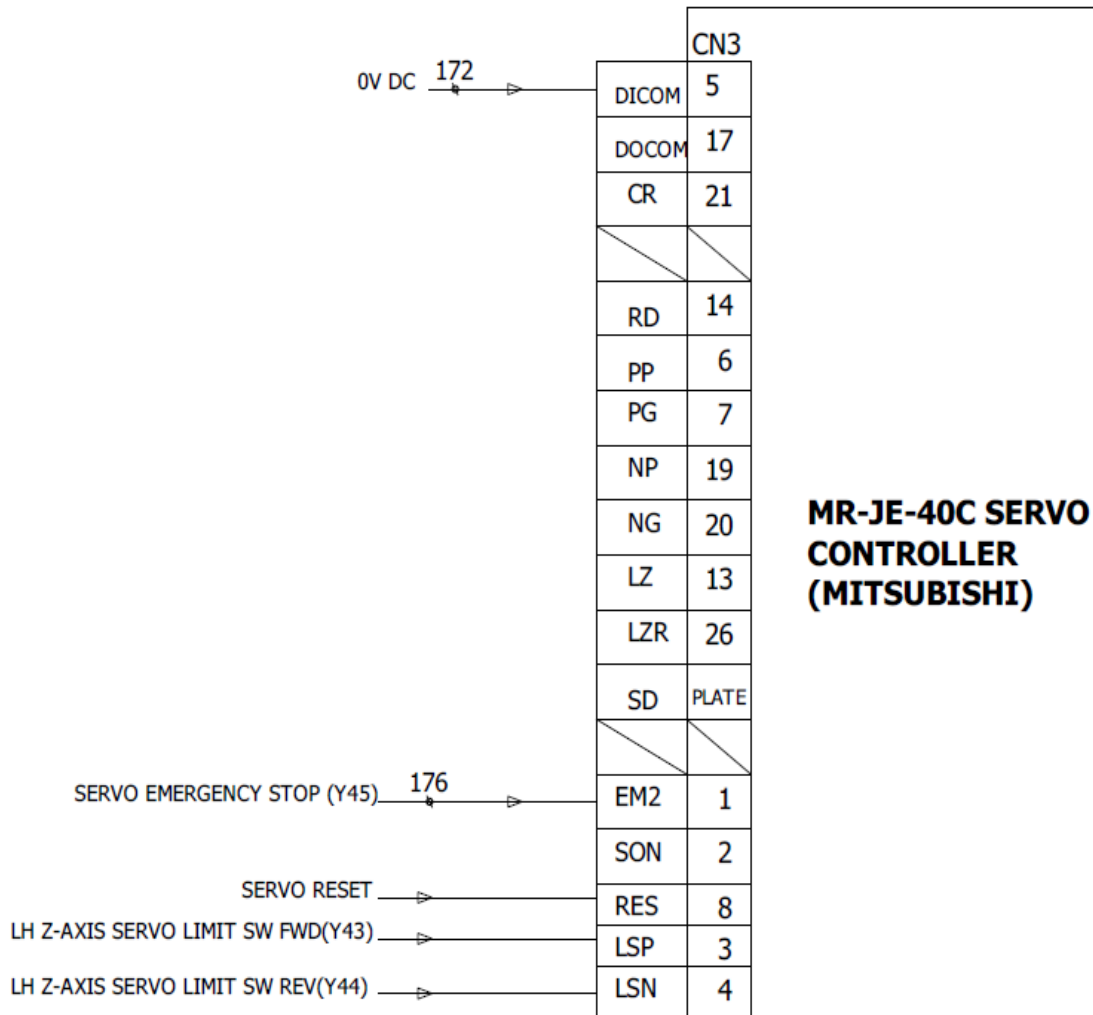
SERVO DRIVE Z AXIS LH I/O CONNECTIONS

SERVO DRIVE-Z AXIS LH I/O CONNECTIONS



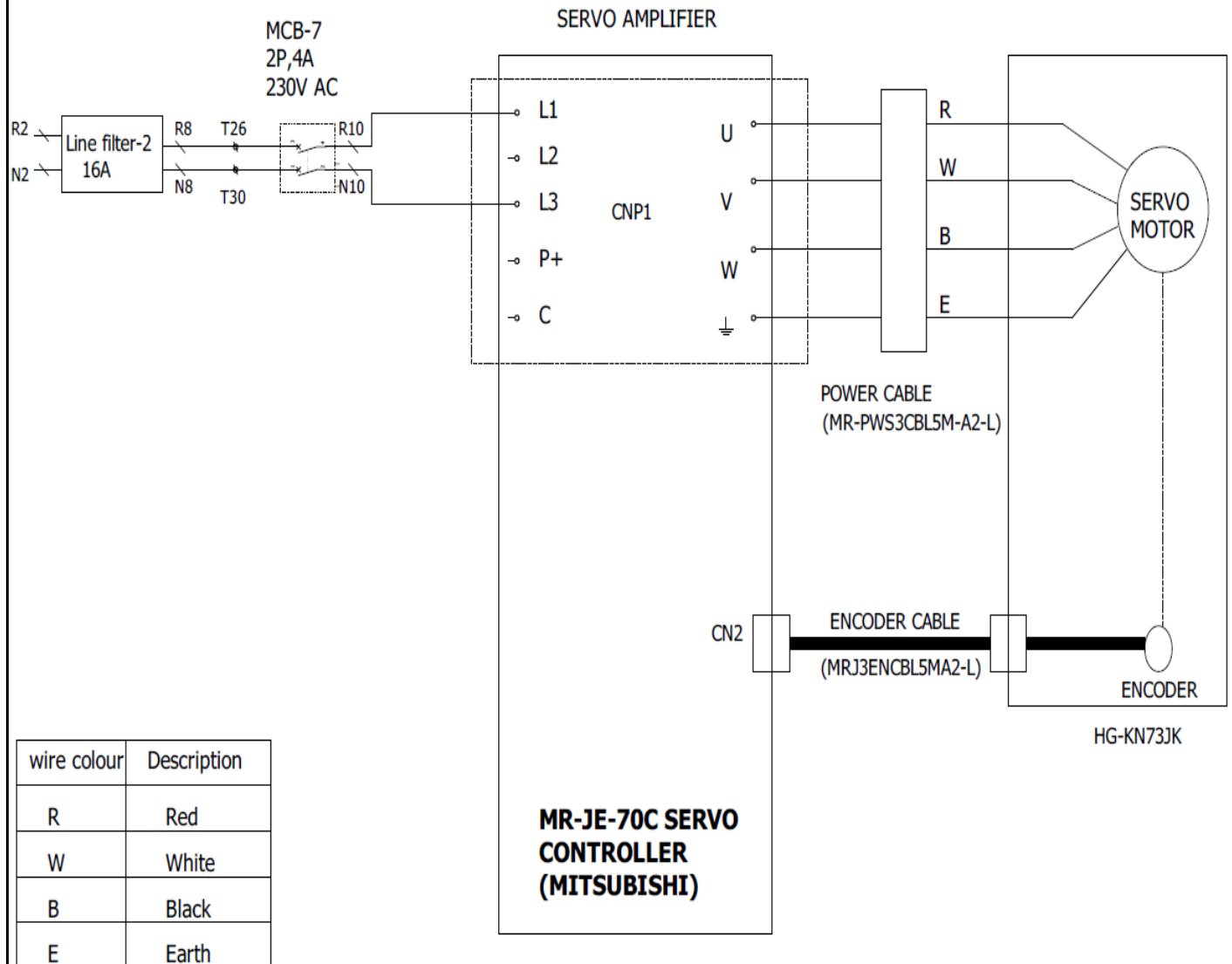
LH Z-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS

LH Z-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



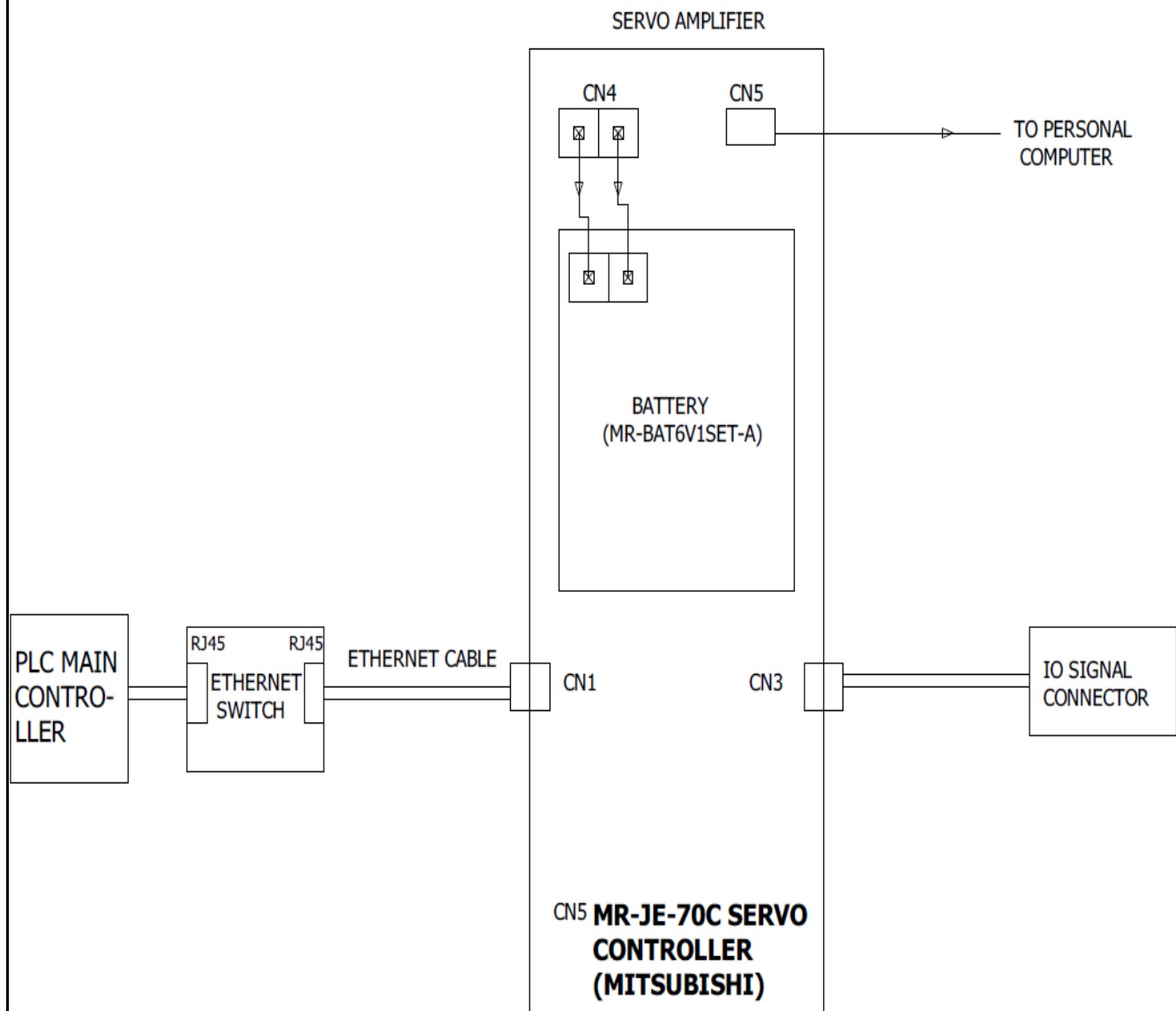
SERVO DRIVE -X AXIS RH POWER CONNECTIONS

SERVO DRIVE-X AXIS RH POWER CONNECTIONS



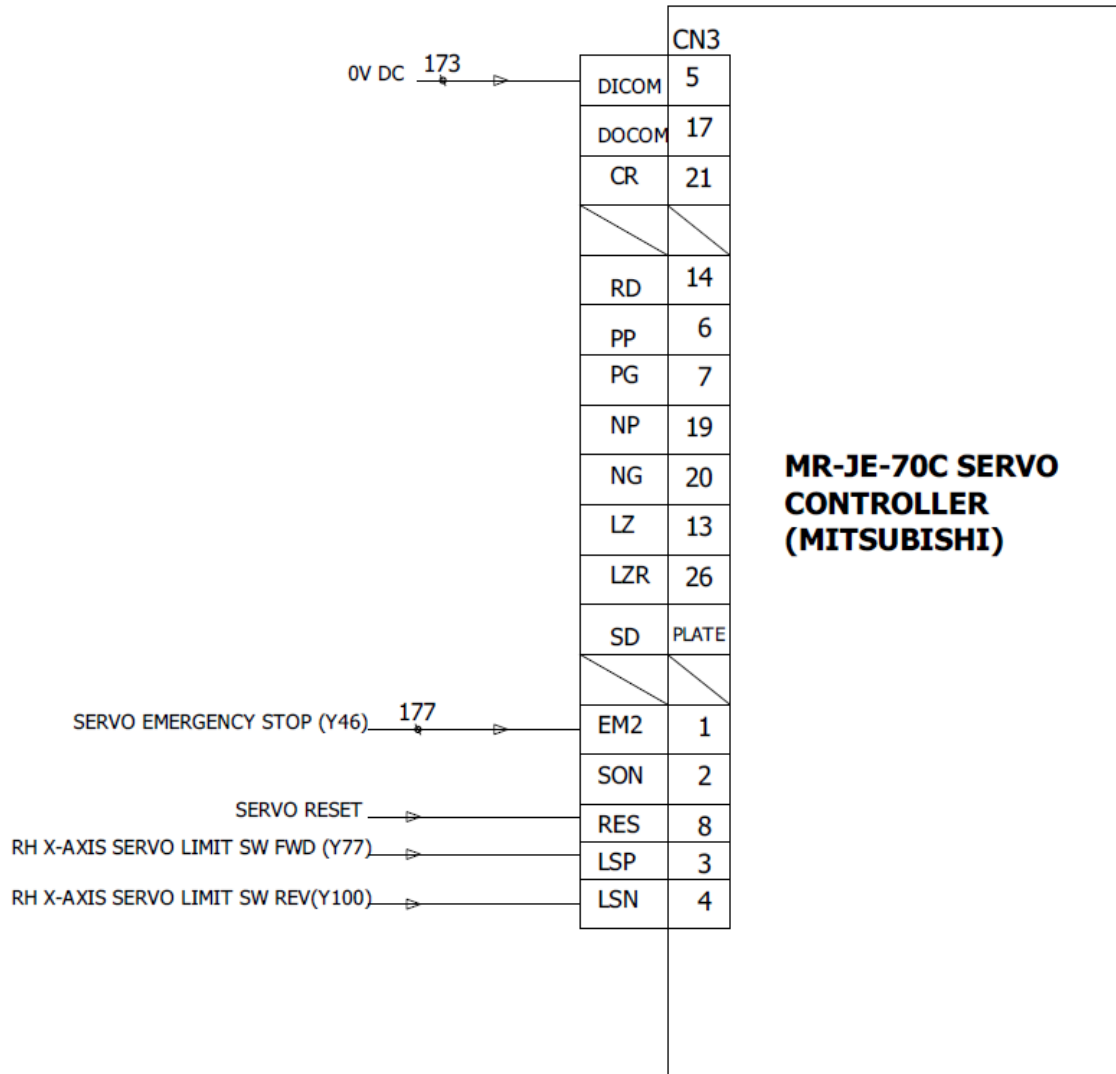
SERVO DRIVE-X AXIS RH I/O CONNECTIONS

SERVO DRIVE-X AXIS RH I/O CONNECTIONS



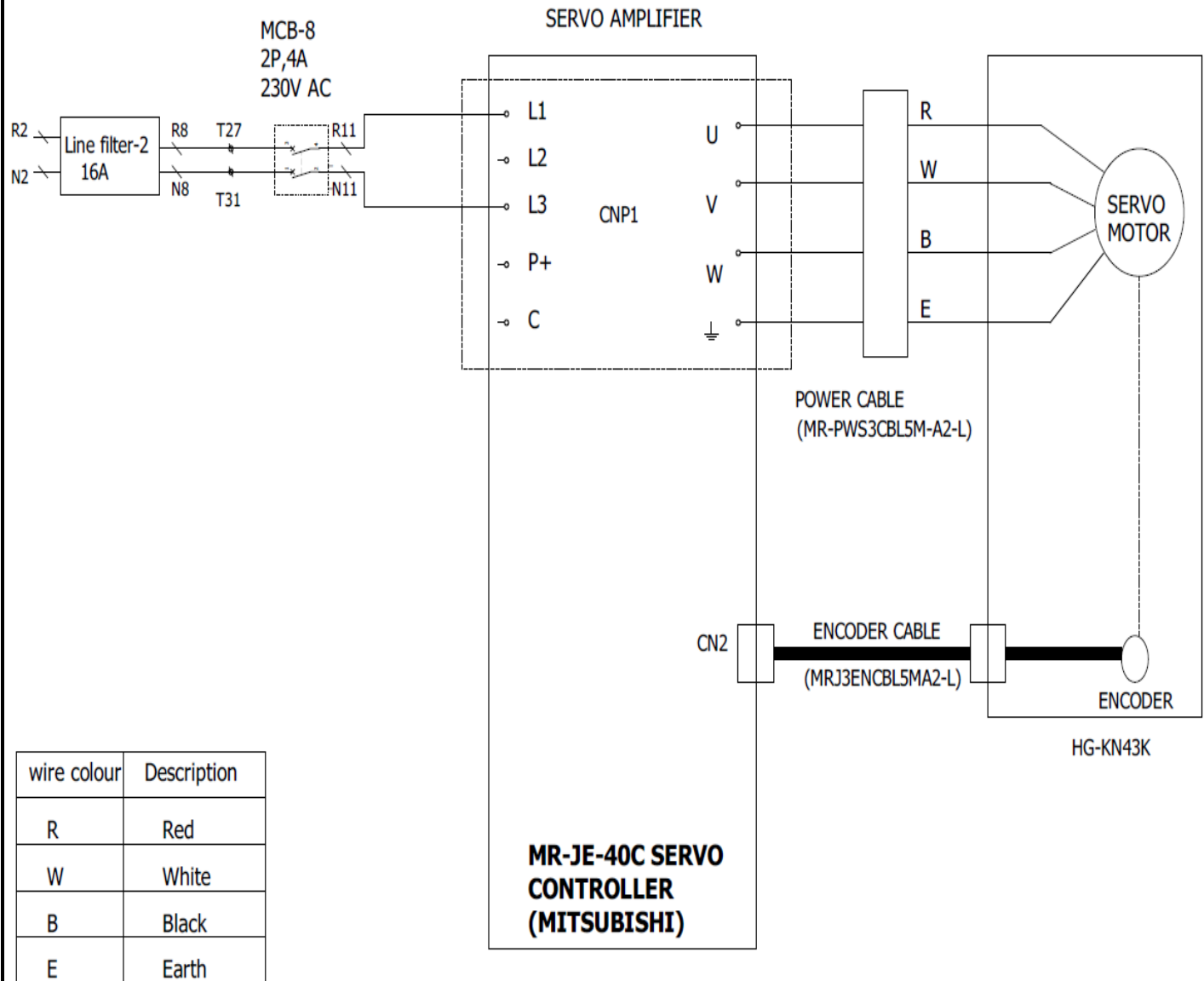
RH X-AXIS SERVO DRIVE I/O CONNECTIONS

RH X-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



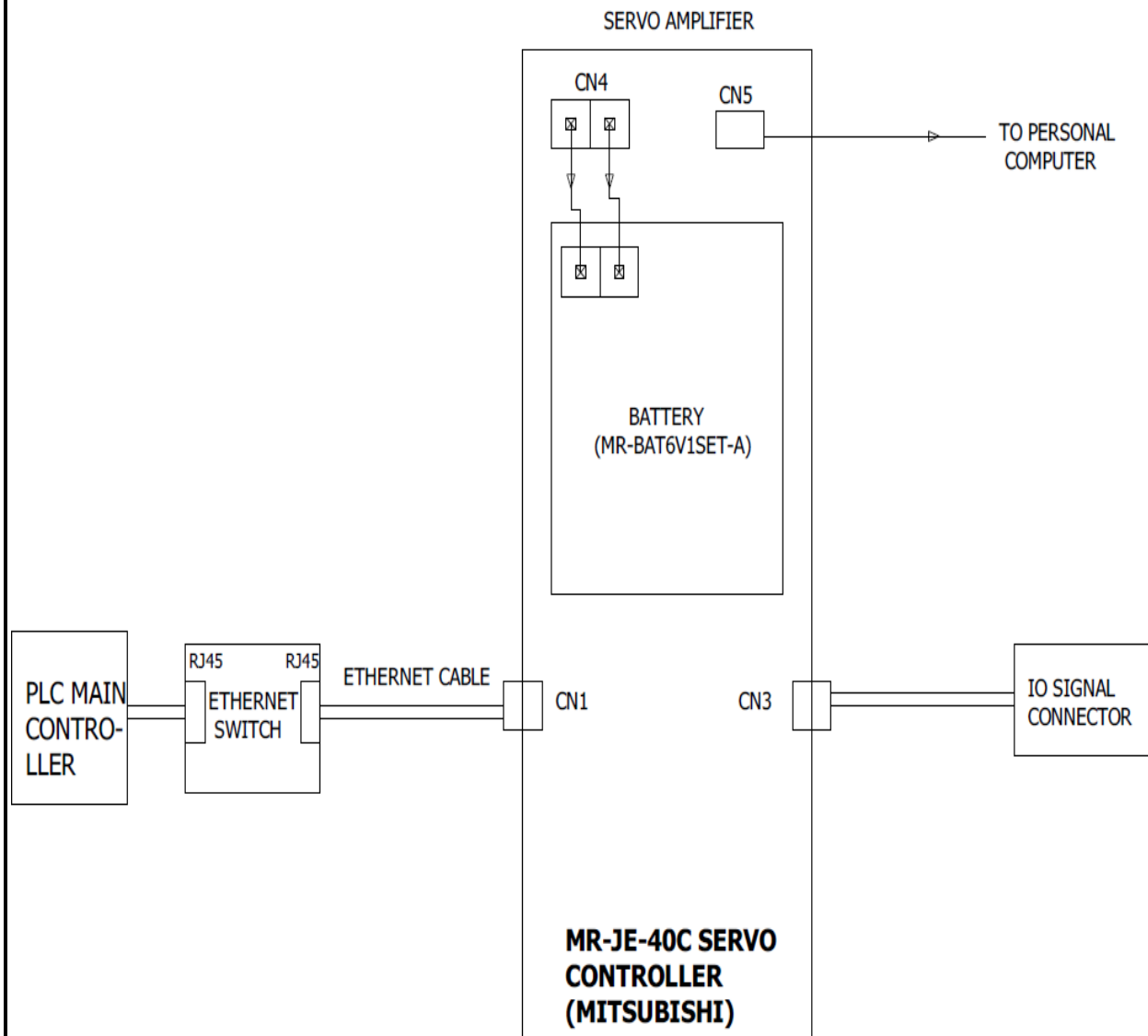
SERVO DRIVE-Y AXIS RH POWER CONNECTIONS

SERVO DRIVE-Y AXIS RH POWER CONNECTIONS



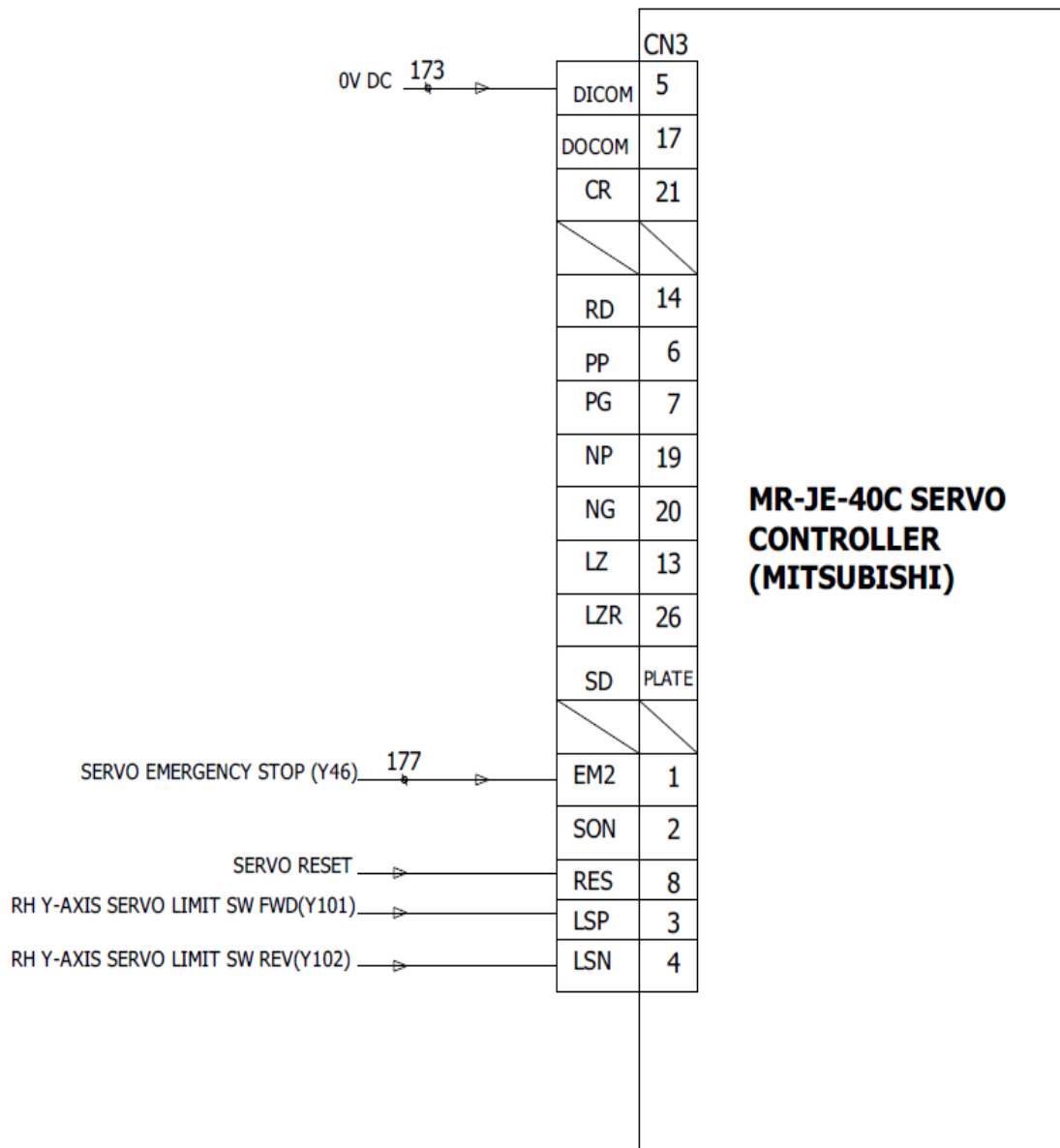
SERVO DRIVE-Y AXIS RH I/O CONNECTIONS

SERVO DRIVE-Y AXIS RH I/O CONNECTIONS



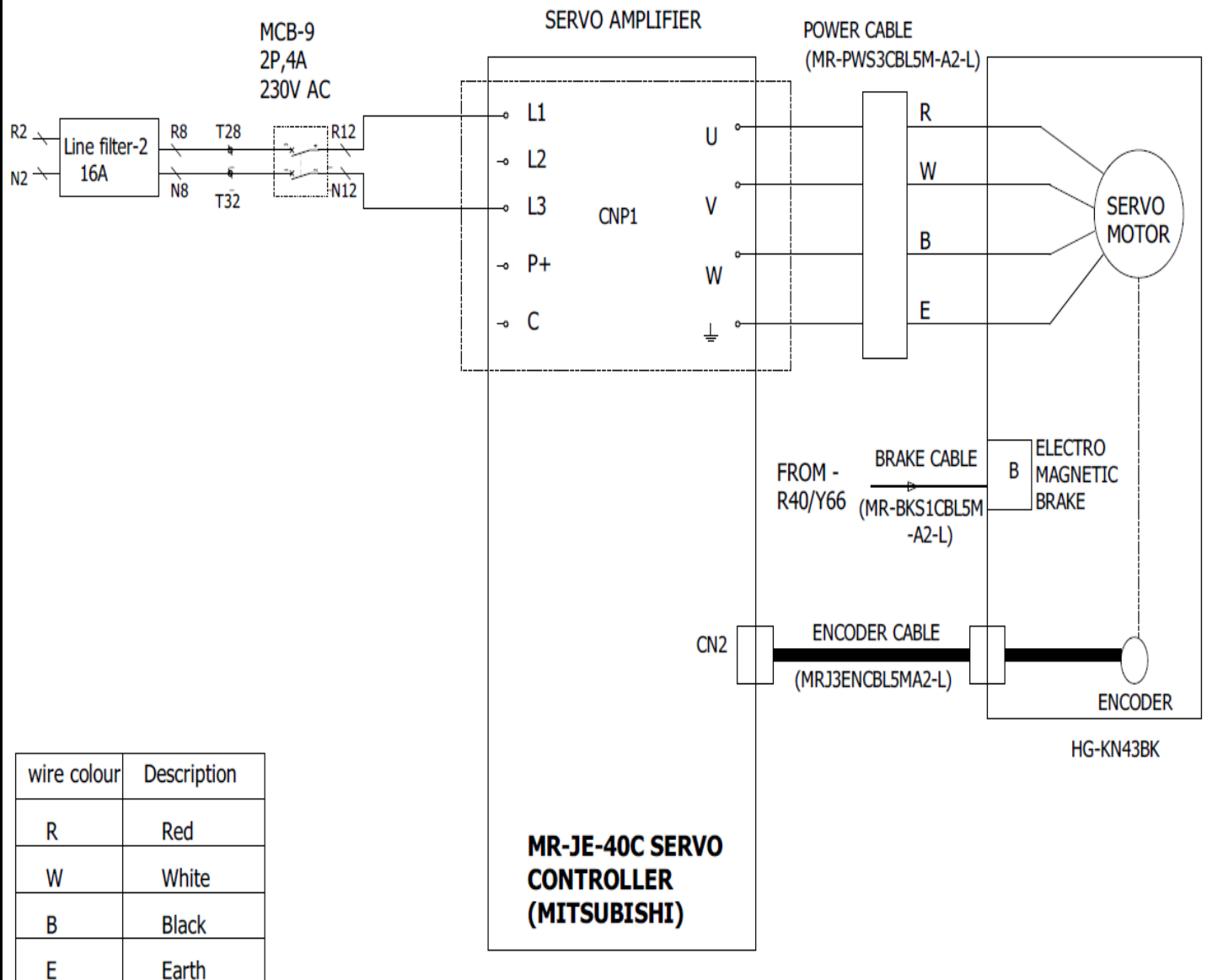
RH Y-AXIS SERVO DRIVE I/O CONNECTIONS

RH Y-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



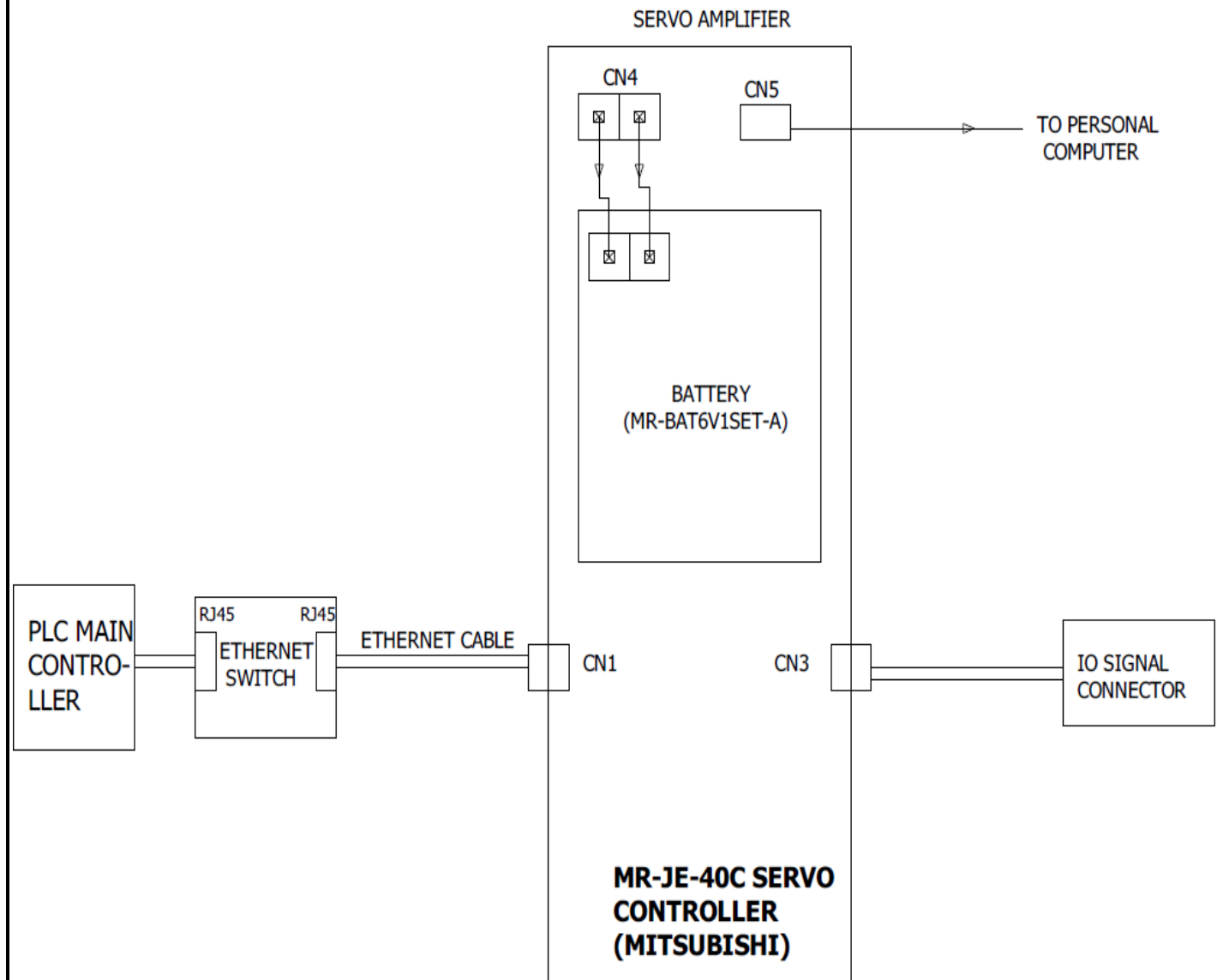
SERVO DRIVE-Z AXIS RH POWER CONNECTIONS

SERVO DRIVE-Z AXIS RH POWER CONNECTIONS



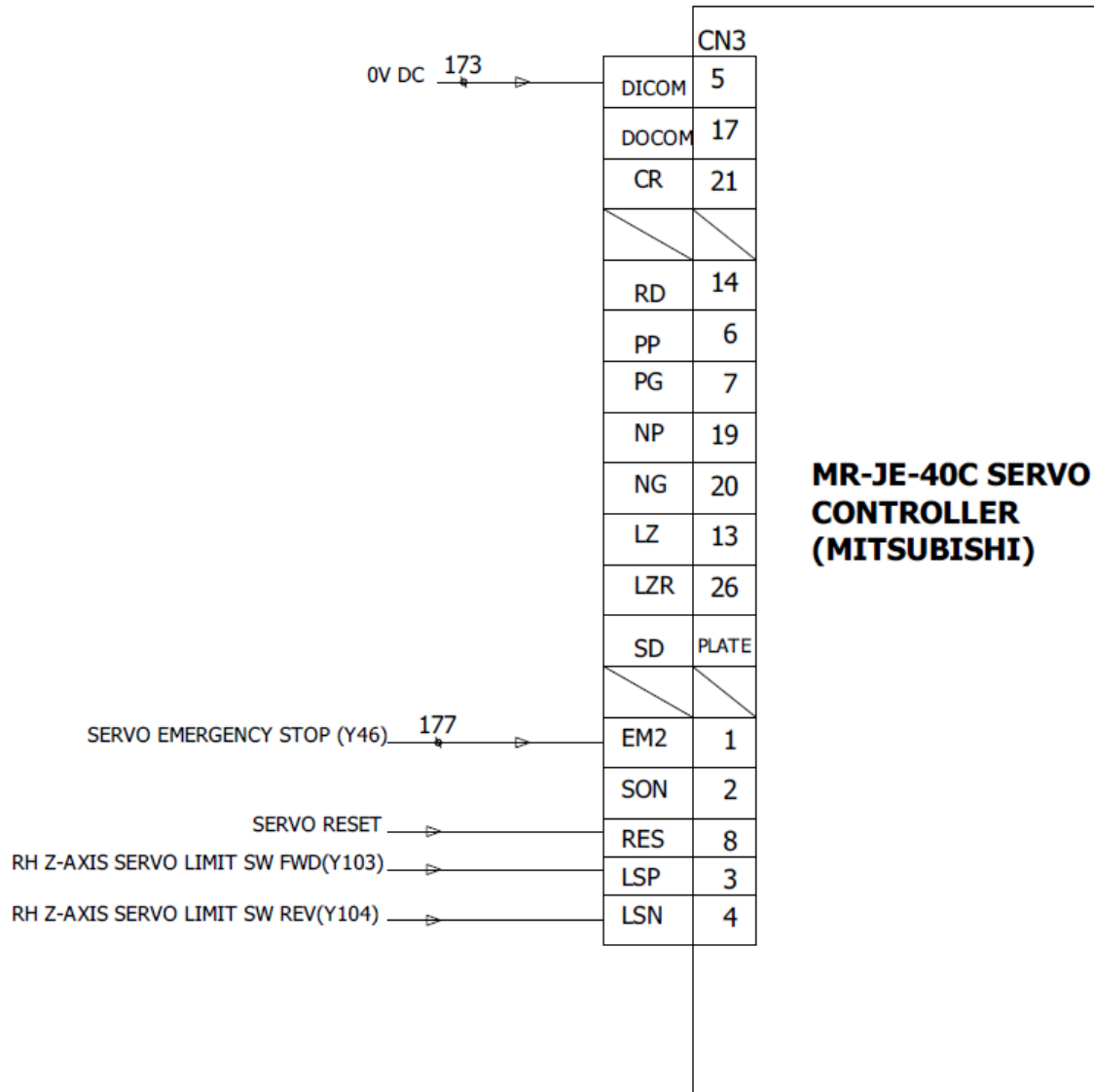
SERVO DRIVR-Z AXIS RH I/O CONNECTIONS

SERVO DRIVE-Z AXIS RH I/O CONNECTIONS



RH Z-AXIS SERVO DRIVE I/O CONNECTIONS

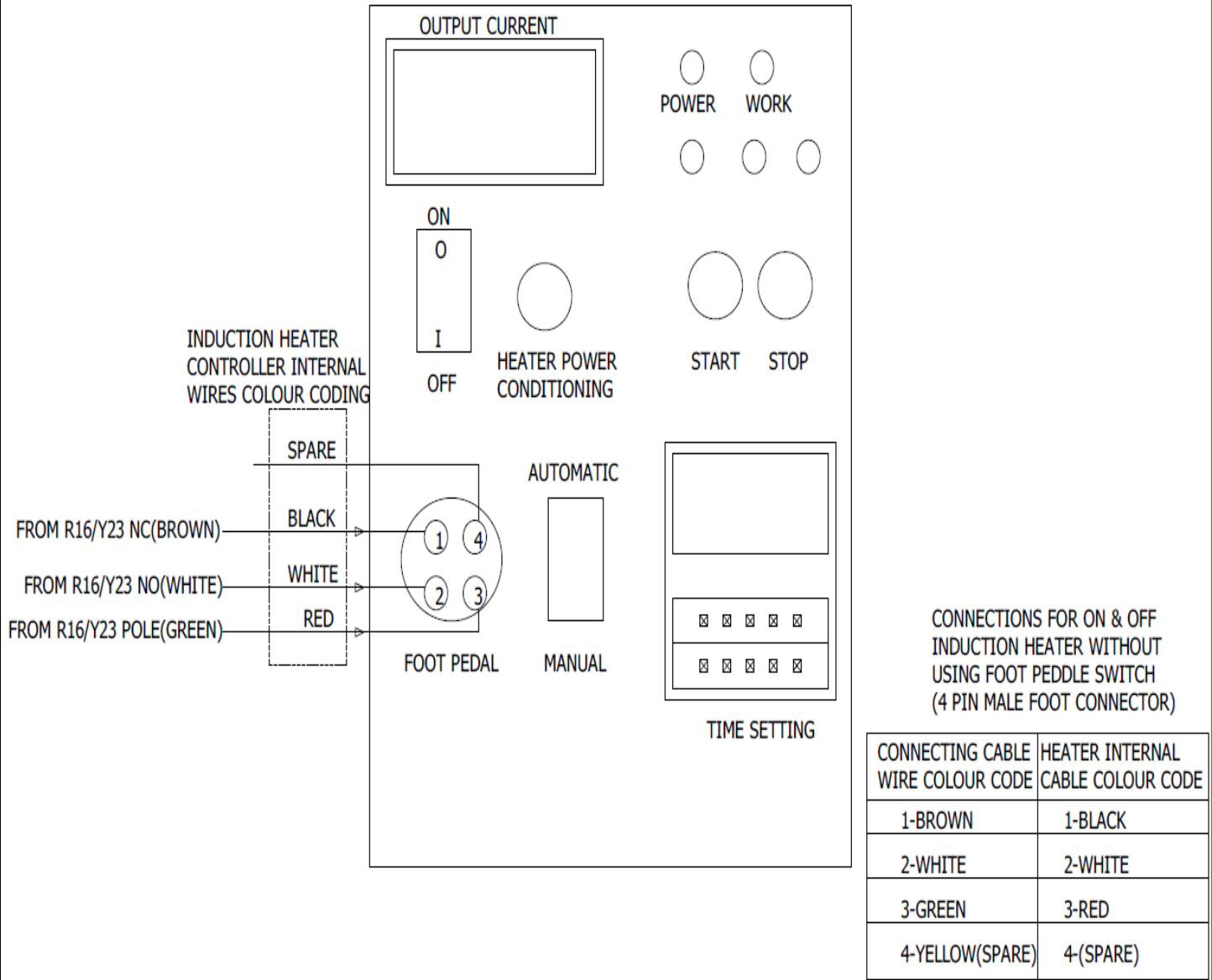
RH Z-AXIS SERVO DRIVE INPUT-OUTPUT CONNECTIONS



LH HEATER CONTROLLER CONNECTIONS DETAILS

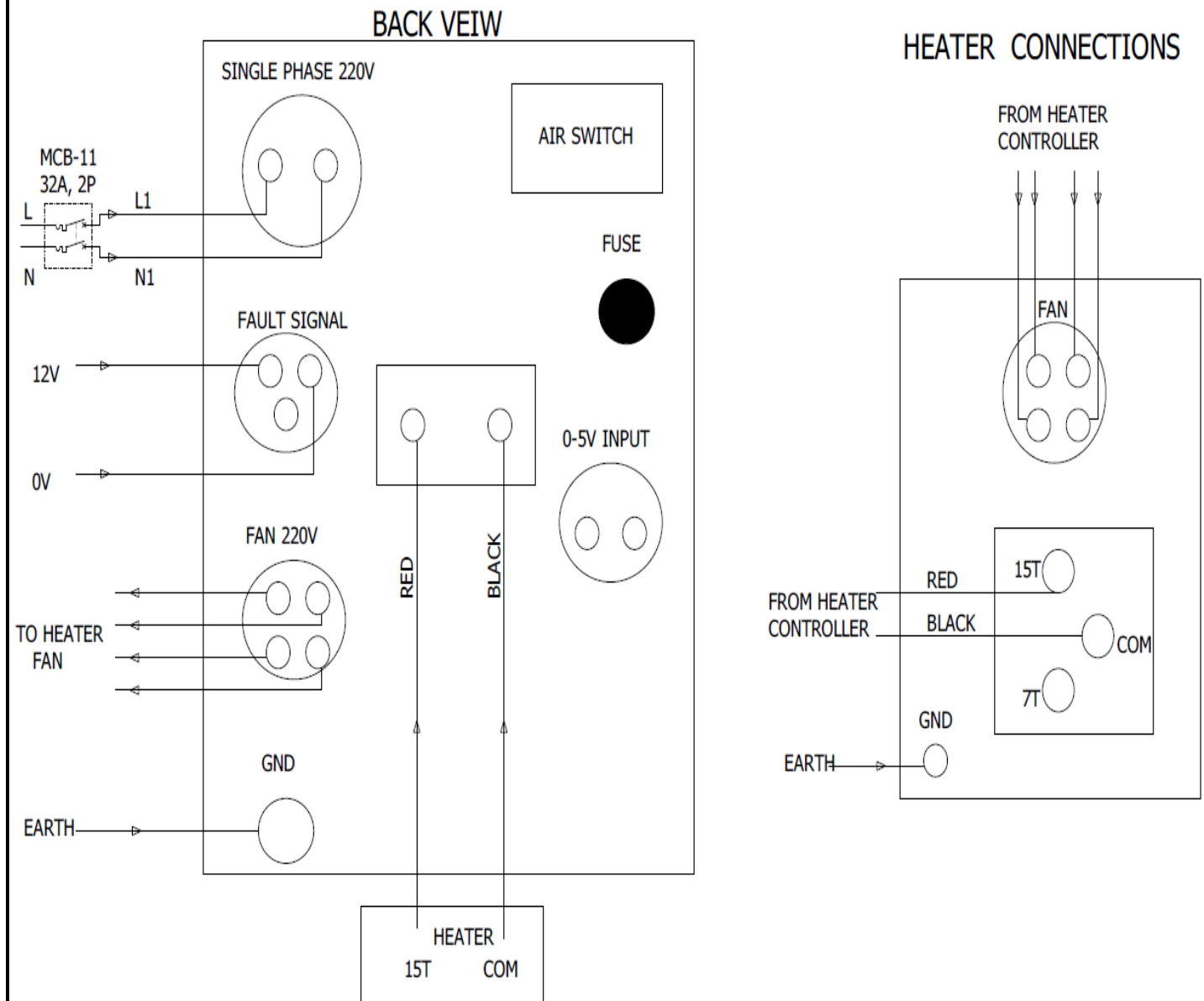
LH HEATER CONTROLLER(FL15KW202007-001) CONNECTIONS DETAILS

FRONT VIEW



LH HEATER CONTROLLER CONNECTIONS DETAILS

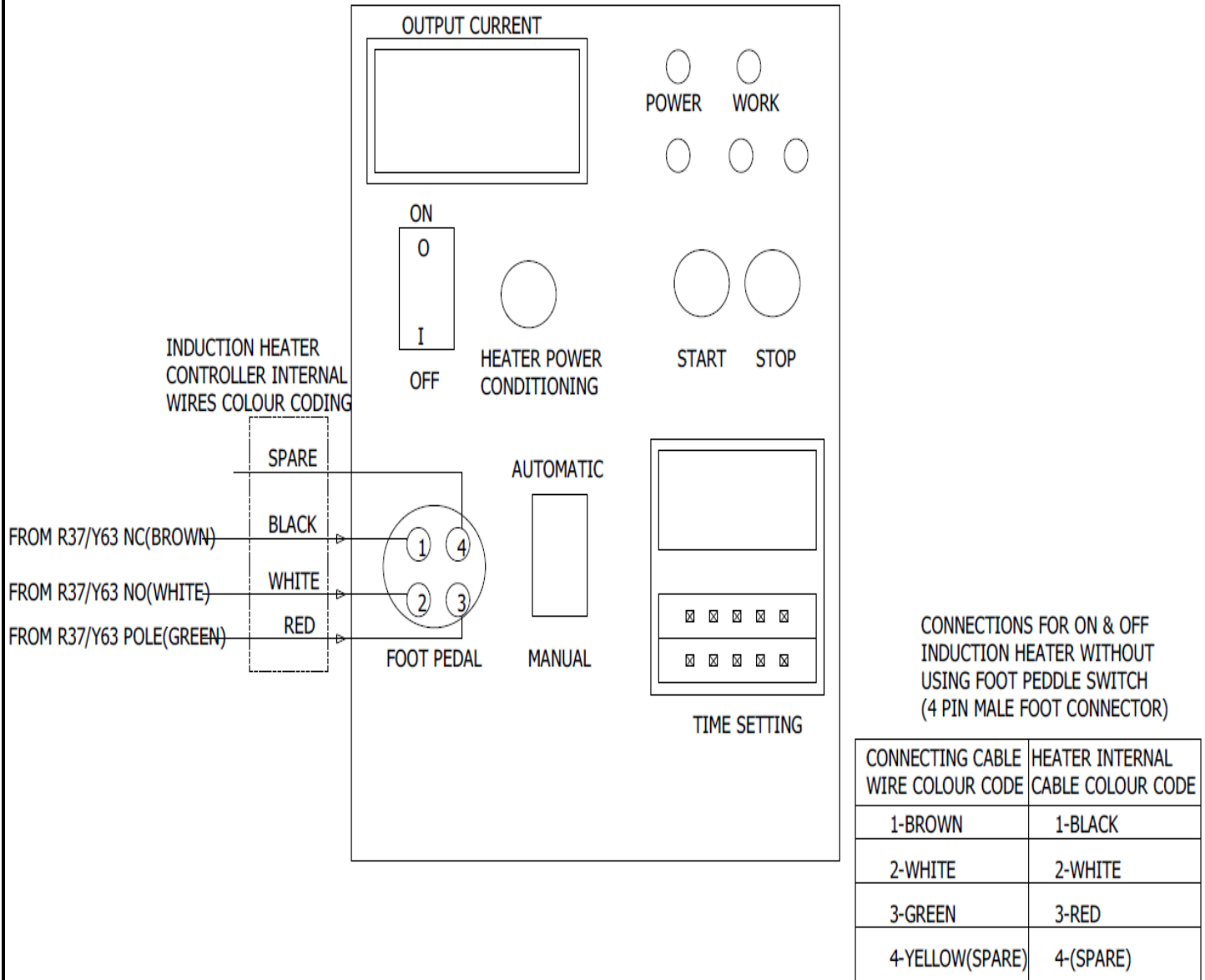
LH HEATER CONTROLLER & HEATER CONNECTIONS DETAILS



RH HEATER CONTROLLER CONNECTIONS DETAILS

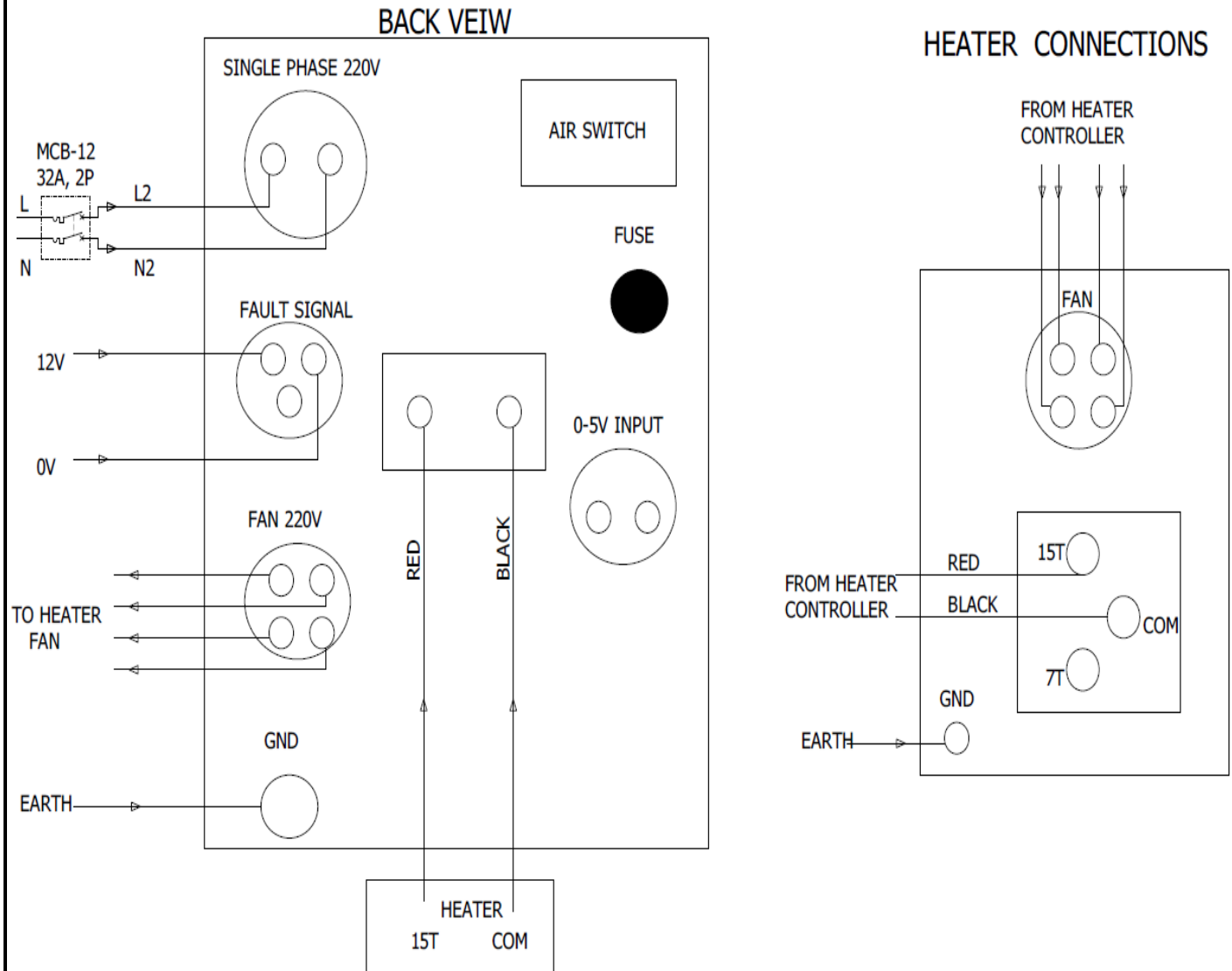
RH HEATER CONTROLLER (FL15KW202007-001) CONNECTIONS DETAILS

FRONT VIEW



RH HEATER CONTROLLER CONNECTIONS DETAILS

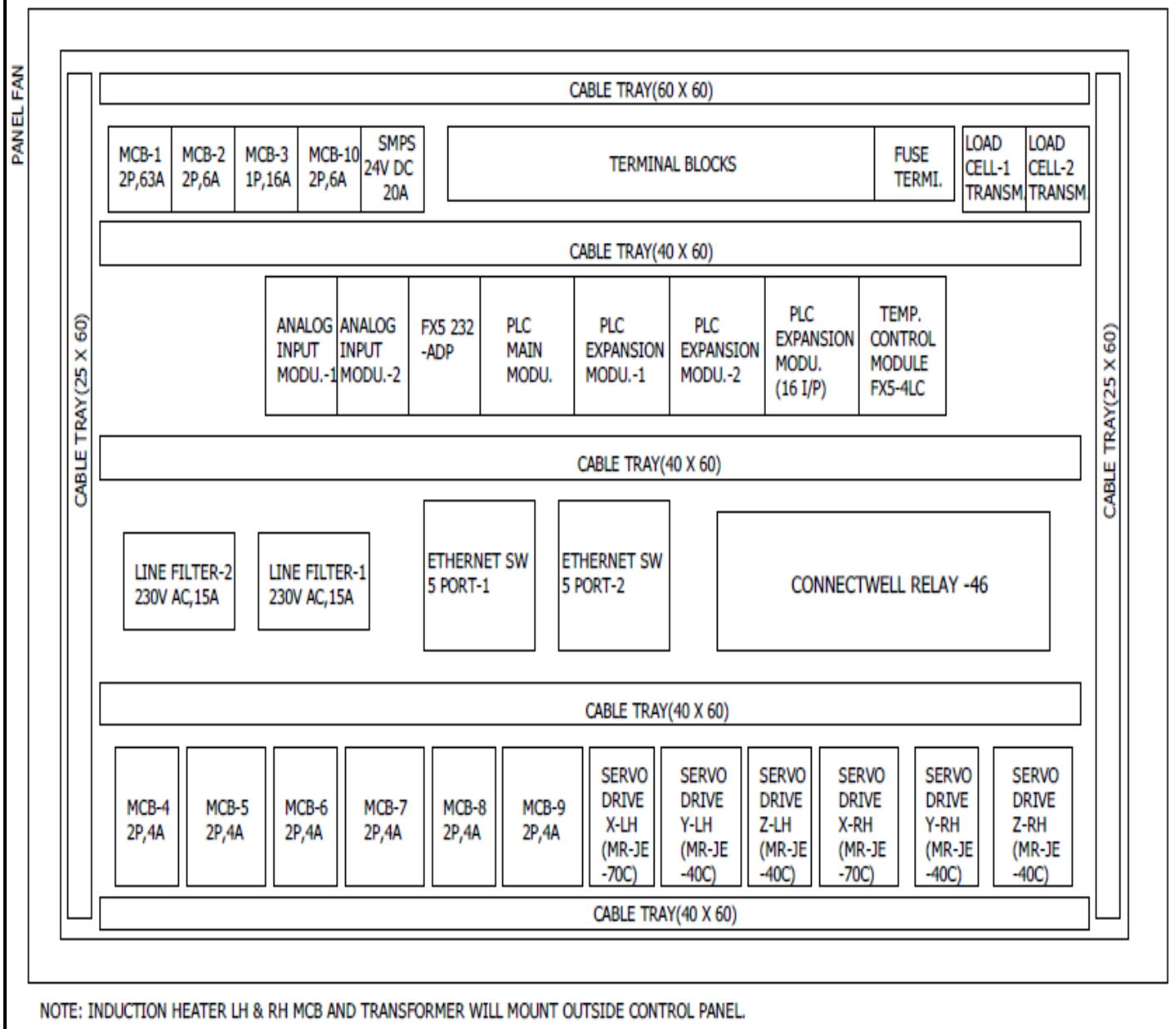
RH HEATER CONTROLLER & HEATER CONNECTIONS DETAILS



COMMUNICATION ARCHITECTURE

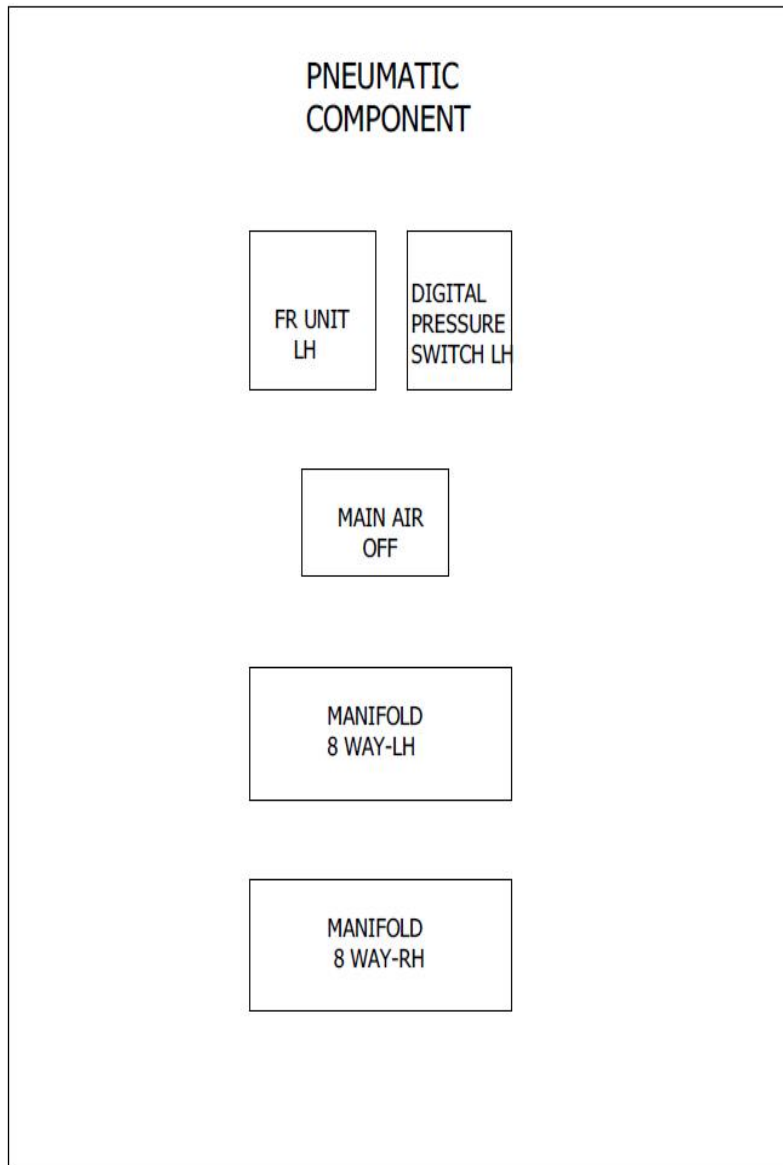
COMPONENT LAYOUT

Control panel size: 1000 X 1000 X 300

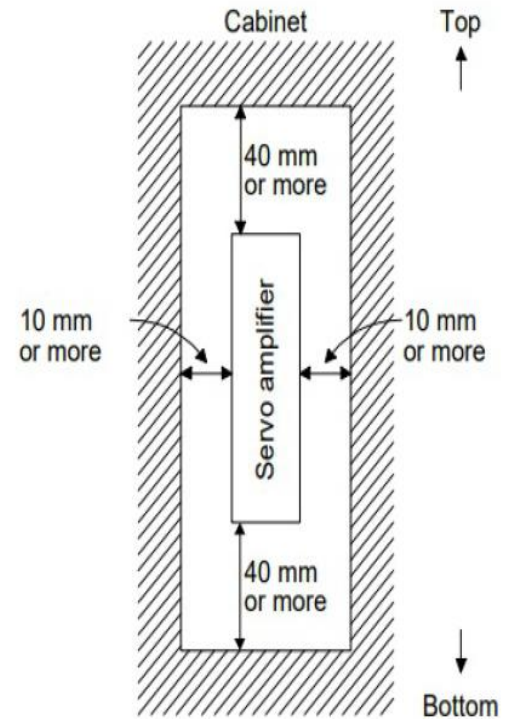


PNEUMATIC COMPONENT LAYOUT

PNEUMATIC COMPONENT LAYOUT



SERVO DRIVE MOUNTING



CABLE DETAILS

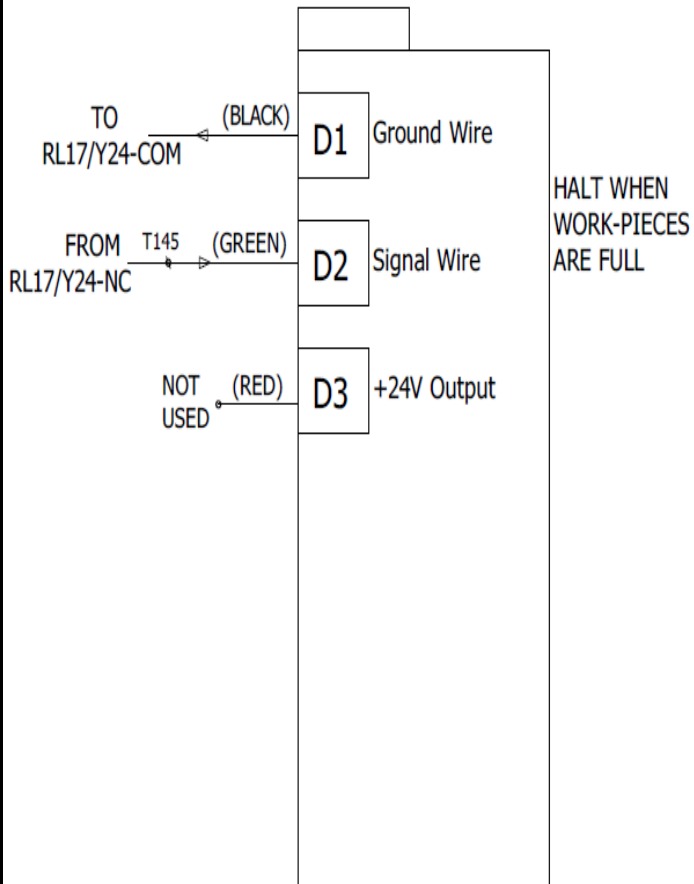
CABLE DETAILS

CABLE	CORE DIAMETER	COLOUR CODE		MAKE
POWER CABLE	2.5 SQ MM 3 CORE FLEXIBLE	RED	LINE	POLYCAB
		BLACK	NEUTRAL	
		YELLOW+GREEN	EARTH	
PANEL WIRING	CONTROL WIRING 0.5 SQ. MM,1 CORE	BLUE	+24V DC	RR CABLE
		WHITE	OV DC	
		GREY	I/P OF PLC	
		ORANEG	OUTPUT OF PLC	
	POWER WIRING 1 SQ. MM,1 CORE	RED	LINE	
		BLACK	NEUTRAL	
		YELLOW+GREEN	EARTH	
TOWER LAMP	0.25 SQ.MM 3 CORE,SHIELED			STANDARD
HMI-PLC CABLE	ETHERNET CABLE	BLACK	COMMUNICATION CABEL	STANDARD
SERVO AMPLIFIER TO MOTOR CABLE	POWER CABLE			STANDARD
SERVO AMPLIFIER TO MOTOR CABLE	ENCODER CABLE			STANDARD
CAMERA CONTROLLER TO CAMERA CABLE				STANDARD

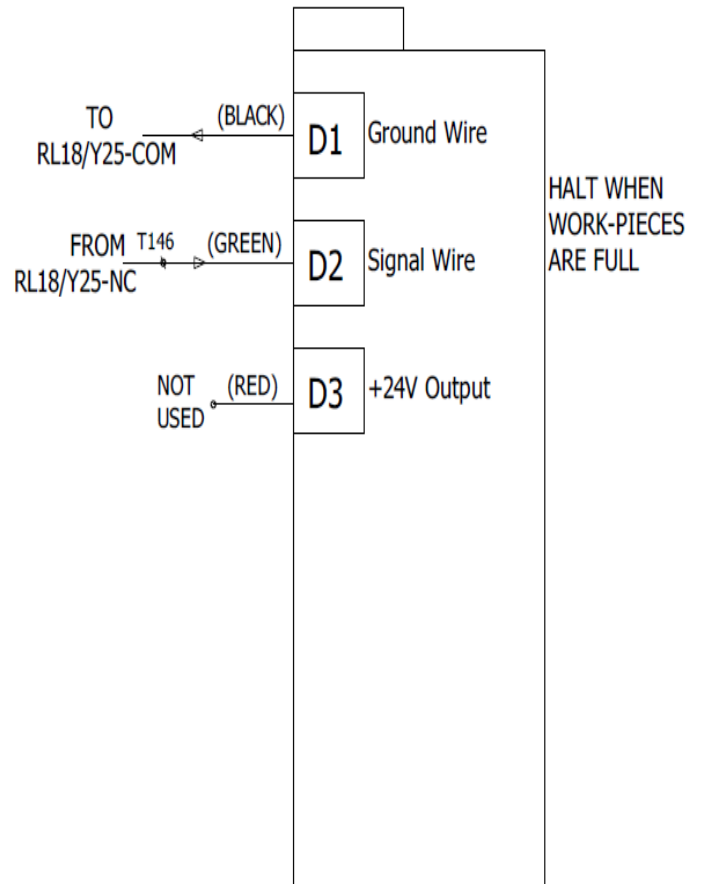
LH SIDE LINEAR AND BOWLFEEDEER CONTROLLER CONNECTIONS

LH SIDE LINEAR AND BOWLFEEDER CONTROLLER CONNECTIONS (SDVC21)

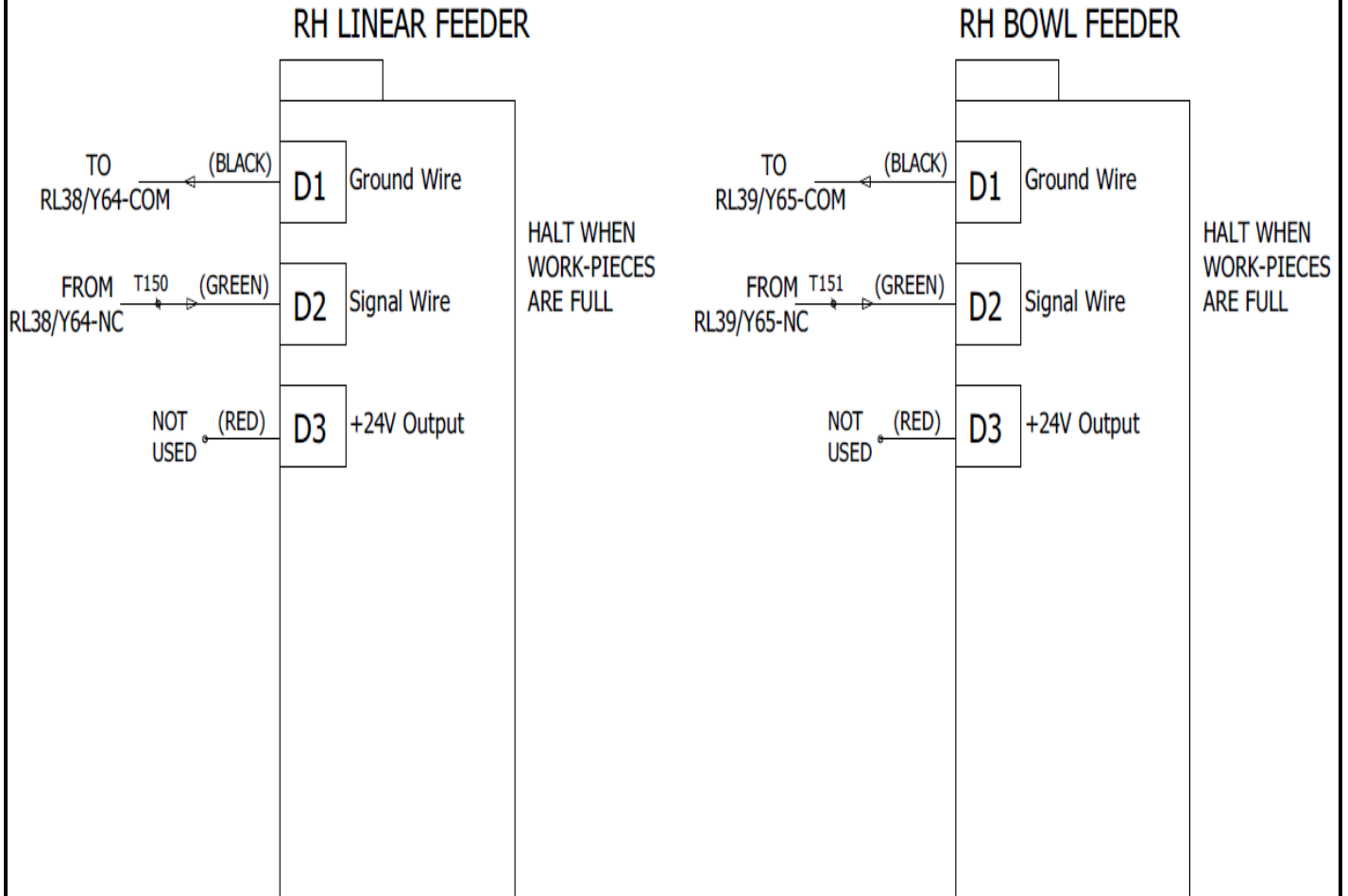
LH LINEAR FEEDER



LH BOWL FEEDER



RH SIDE LINEAR AND BOWLFEEDER CONTROLLER CONNECTIONS (SDVC21)



BILL OF MATERIAL:

YFG TRUNK STUD INSERTION MACHINE					
SR NO.	COMPONENT NAME	PART DESCRIPTION	PART NO.	MAKE	QTY
1	Control panel	1000 X 1000 X 300	MAD1001030R5	ELDON	1

2	PLC	40DI/40DO	FX5U-80MT/ESS	mitsubishi	1
		16DI/DO	FX5-32ER/ES	mitsubishi	2
		16DI	FX5-16EX/ES	mitsubishi	1
3	Temperature control module		FX5-4LC	mitsubishi	1
4	Analog Module	4AI	FX5U - 4AD-ADP	mitsubishi	2
	HMI	10"	GS2110-WTBD	mitsubishi	1
5	COMM. MODULE	232 COMM	FX5 232-ADP	mitsubishi	1
6	SMPS	24VDC,20A	S8VK-C48024	OMRON	1
7	MCB	2P, 63A	5SL62637RC	SIEMENS	1
		2P,6A	5SL62067RC	SIEMENS	2
		2P,4A	5SL62047RC	SIEMENS	6
		1P,16A	5SL61167RC	SIEMENS	1
8	Relay Board	1 CHANNEL		CONNECTWE LL	42
9	ON/Off Switch	230 AC MAIN SWITCH 63A,2P	63A-61060	SALZER	1
10	Line Filter	15A,230V AC	EF-1B015B01A-C02	ELCOM	2
11	PB Box	2 WAY	50MM	TEKNIC	2
12	Push Button (Green)	Mushroom Type	2AM3	TEKNIC	2
13	Reset Button(red)	Mushroom Type	2AM4	TEKNIC	1
14	Emergency Stop	Latched type	2AML4	TEKNIC	1
15	NO Element	24 VDC	S1	TEKNIC	3
16	NC Element	24 VDC	S2	TEKNIC	1
17	Load cell controller	Weight transmitter	RS-10	Rudra sensor	2
18	Servo Drive	230 VAC, 0.75KW	MR-JE-70C	mitsubishi	2
		230 VAC, 0.4KW	MR-JE-40C	mitsubishi	4
19	Servo motor	230 VAC, 750W	HG-KN73JK	mitsubishi	2
		230 VAC, 400W	HG-KN43K	mitsubishi	2
		230 VAC, 400W(BRAKE)	HG-KN43BK	mitsubishi	2
20	Power cable	5 meter	MR-PWS3CBL5M- A2-L	Mitsubishi	6
21	Encoder cable	5 meter	MR-J3ENCBL5M-A2- L	Mitsubishi	6
22	Brake cable	5 meter	MR-BKS1CBL5M- A2-L	Mitsubishi	2
23	Encoder connector set		MR-ECNM	Mitsubishi	6
24	Battery set		MR-BAT6V1SET-A	Mitsubishi	6
25	Tower lamp	24 V DC	LSL0301RAGPB	LEUZE	1
26	panel fan	230VAC	REC 22038 A2T	REXFORD	1
27	Panel Tube light	5w	MTL-2835-150-3W	DSS	1
28	Safety Switch	230V,4A	AZ16-12ZVRK-M16	SCHMERSAL	6



YFG TRUNK STUD INSERTION MACHINE.

Marelli Motherson Auto System Pvt Ltd.
Gat No 148/150, Chakan, Pune-410501(Near
Dwarka School Village Ambethan Taluka Khed)



29	Camera	Vision System	CA-H048MX	Keyence	2
30	Camera Controller	Vision System Controller	CV-X300R	Keyence	1
31	Indo connector	24 pin (M&F)Regular	0110024M011 & 0110024F011	INDO	2
32	Indo connector hood	24 pin		INDO	2
33	Temp sensor(4-20mA)			Keyence	2
34	MONITOR for vision system	100-240V ,60 HZ,0.5 A,13W	D1918H	DELL	1
35	Analog sensor	24v dc	SMAT-8ES50-IU-E-0,3-M8D	Festo	4
36	Bowl feeder controller	5A,1100VA,85-260V AC	SDVC21-S	CUH	2
37	Linear feeder				2
38	Sensor	Photoelectric sensors	O300Gp	Baumer	19
39	Light curtain	24V DC PNP	F3SJ-E0625P25	Omron	1
40	Door L.S	1NO/2NC	AZ16-12ZVRK-M16	Schmersal	3
41	Extension board	Surface Box	674 099	Legrand	2
		Frame 9 Module	6755 69	Legrand	2
		6A 2/3 Pin Socket	6755 50	Legrand	6
		Switch 1M 1 Way SP	6755 11	Legrand	6
42	Fuse with terminal	4A			2
43	Step down Transformer	primary voltage:0-415v ac,2-phase, secondary voltage:-0-230v ac,1 phase, 4 KVA		Truetech	1
44	SSR	200V AC/DC,10A	G3NA-210B	OMRON	2
45	Ethernet sw	24vdc,5 port	FL SW SFNT 5TX-PN	Phoenix	1
46	Ethernet sw	24vdc, 8 port	FL SW SFNT 8TX	Phoenix	1
47	Heater with built in thermocouple(K type)	Single phase,220V,450w	MTF-45KW	Malhotra Tech.	2
48	station fan	60W	-	Hi-speed	1
49	station tube light	22W	-	LORDS	1
50	Indication lamp	230v ac	green		1

PNEUMATIC MATERIAL:

YFG TRUNK STUD INSERTION MACHINE

Sr No	Component Name	Description	Part No.	Make	Qty.
1	Flow Control Valve	L-Shape M5 X 6OD	AS1201F-M5-06A	SMC	1
2	Solenoid Valve	Double Acting	SY7220-5LZ-02	SMC	12



YFG TRUNK STUD INSERTION MACHINE.

Marelli Motherson Auto System Pvt Ltd.
Gat No 148/150, Chakan, Pune-410501(Near
Dwarka School Village Ambethan Taluka Khed)



		Single Acting	SY7120-5LZ-02	SMC	2
3	Manifold	8-Way	SS5Y720-04	SMC	2
4	Manifold Plug	1/4"	PL-02	SMC	16
5	Air Off Solenoid Valve	3-Port	VT307-5G1-02F	SMC	9
6	FR Unit	1/4"	AW20-02BG1	SMC	1
7	Shut Off Valve	1/4"	VHS20-02	SMC	1
8	Shut Off Valve Bkt	Bkt	Y200-A	SMC	1
9	Straight Connector	1/4" X 6OD	KQ2H06-U02	SMC	1
10	Silencer	1/4"	AN15-02	SMC	1
11	Digital Pressure Switch	Pressure Display	ISE30AF-01-P-LA2	SMC	1
12	Pneumatic Tube	Blue 6 OD	TU0604BU	SMC	-
		Black 6 OD	TU0604BK	SMC	-
13	Pneumatic connector	10 pin male	KDM10-06	SMC	2
		10 pin female	KDM10-06	SMC	2
14	Reed Switch	24VDC,3-Wire	D-M9PL-LUG	SMC	32

I/O LIST:

YFG TRUNK STUD INSERTION MACHINE						
Inputs				Outputs		
Mod	Add	Description	Ferrule	Add	Description	Ferrule
FX5U-80MT/ESS	X0	START PB 1	X0	Y0	TOWER LAMP GREEN	Y0
	X1	START PB 2	X1	Y1	TOWER LAMP YELLOW	Y1
	X2	RESET PB	X2	Y2	TOWER LAMP RED	Y2
	X3	EMERGENCY STOP	X3	Y3	TOWER LAMP BUZZER	Y3
	X4	DIGITAL PRESSURE SW	X4	Y4	PANEL FAN-1	Y4
	X5	LIGHT CURTAIN LH	X5	Y5	CAM LIGHT LH	Y5
	X6	DOOR L.S.1 (MIDDLE)	X6	Y6	MAIN AIR OFF	Y6
	X7	DOOR L.S.2 (RH)	X7	Y7	CAM LIGHT RH	Y7
	X10	DOOR L.S.3 (LH)	X10	Y10	SLIDE CYL FWD SOL LH	Y10
	X11	PART PRESENT LH	X11	Y11	SLIDE CYL RET SOL LH	Y11
	X12	CENTRE CHECK SENS 1 LH	X12	Y12	PART CLAMP FWD SOL LH	Y12
	X13	CENTRE CHECK SENS 2 LH	X13	Y13	PART CLAMP RET SOL LH	Y13
	X14	STUD PRESENT SENS LH	X14	Y14	STUD PUSHER CYL FWD SOL LH	Y14
	X15	BOWL FEEDER FULL SENS LH	X15	Y15	STUD PUSHER CYL RET SOL LH	Y15
	X16	STUD 1 PRESENCE SENSOR LH	X16	Y16	STUD GRIPPER FWD SOL LH	Y16
	X17	STUD 2 PRESENCE SENSOR LH	X17	Y17	STUD GRIPPER RET SOL LH	Y17
	X20	STUD 3 PRESENCE SENSOR LH	X20	Y20	DOT PUNCH SOL	Y20

FX5-32ER/ES-1(SINK I/P)	X21	STUD 4 PRESENCE SENSOR LH(PROVISION)	X21	Y21	HOT COLD PRESS CYL-1 FWD SOL-LH	Y21
	X22	REJECTION BIN SENSOR LH	X22	Y22	HOT COLD PRESS CYL-1 RET SOL-LH	Y22
	X23	PART CLAMP FWD RD LH	X23	Y23	INDUCTION HEATER ON/OFF LH	Y23
	X24	PART CLAMP RET RD LH	X24	Y24	LINEAR FEEDER START LH	Y24
	X25	PUSH CYL FWD RD LH	X25	Y25	BOWL FEEDER START LH	Y25
	X26	PUSH CYL RET RD LH	X26	Y26	AXIS-Z BREAK LH	Y26
	X27	STUD GRIPPER FWD RD LH	X27	Y27	HOT COLD PRESS CYL-2 FWD SOL-LH	Y27
	X30	STUD GRIPPER RET RD LH	X30	Y30	HOT COLD PRESS CYL-2 RET SOL-LH	Y30
	X31	DOT PUNCH FWD RD LH	X31	Y31	AIR OFF COOL FIX-1 LH	Y31
	X32	DOT PUNCH RET RD LH	X32	Y32	AIR OFF COOL FIX-2 LH	Y32
	X33	HOT COLD PRESS CYL-1 FWD RD SW-LH	X33	Y33	AIR OFF COOL FIX-3 LH	Y33
	X34	HOT COLD PRESS CYL-1 RET RD SW-LH	X34	Y34	AIR OFF COOL GRIPPER LH	Y34
	X35	HOT COLD PRESS CYL-2 FWD RD SW-LH	X35	Y35	LH LOCK FWD SOL	Y35
	X36	HOT COLD PRESS CYL-2RET RD SW-LH	X36	Y36	LH LOCK RET SOL	Y36
	X37	SLIDE CYL FWD RD SW	X37	Y37	LH X-AXIS SERVO LIMIT SW FWD	Y37
	X40	SLIDE CYL RET RD SW	X40	Y40	LH X-AXIS SERVO LIMIT SW REV	Y40
	X41	LH FIX SELN IN-1	X41	Y41	LH Y-AXIS SERVO LIMIT SW FWD	Y41
	X42	LH FIX SELN IN-2	X42	Y42	LH Y-AXIS SERVO LIMIT SW REV	Y42
	X43	LH FIX SELN IN-3	X43	Y43	LH Z-AXIS SERVO LIMIT SW FWD	Y43
	X44	LH FIX SELN IN-4	X44	Y44	LH Z-AXIS SERVO LIMIT SW REV	Y44
	X45	STUD PRESENT SENSOR RH	X45	Y45	SERVO EMERGENCY STOP LH	Y45
	X46	BOWL FEEDER FULL SENSOR RH	X46	Y46	SERVO EMERGENCY STOP RH	Y46
	X47	LH HEATER FAULT SIGNAL (BROWN)	X47	Y47	SPARE	Y47
	X50	AXIS-X END LS-1 LH	X50	Y50	SLIDE CYL FWD SOL RH	Y50
	X51	AXIS-X END LS-2 LH	X51	Y51	SLIDE CYL RET SOL RH	Y51
	X52	AXIS-X HOME LS LH	X52	Y52	PART CLAMP FWD SOL RH	Y52
	X53	AXIS-Y END LS-1 LH	X53	Y53	PART CLAMP RET SOL RH	Y53
	X54	AXIS-Y END LS-2 LH	X54	Y54	STUD PUSHER CYL FWD SOL RH	Y54
	X55	AXIS-Y HOME LS LH	X55	Y55	STUD PUSHER CYL RET SOL RH	Y55
	X56	AXIS-Z END LS-1 LH	X56	Y56	STUD GRIPPER FWD SOL RH	Y56

	X57	AXIS-Z END LS-2 LH	X57	Y57	STUD GRIPPER RET SOL RH	Y57
	X60	AXIS-Z HOME LS LH	X60	Y60	DOT PUNCH SOL	Y60
	X61	LH LOCK FWD RD	X61	Y61	HOT COLD PRESS CYL-1 FWD SOL-RH	Y61
	X62	LH LOCK RET RD	X62	Y62	HOT COLD PRESS CYL-1 RET SOL-RH	Y62
	X63	RH HEATER FAULT SIGNAL	X63	Y63	INDUCTION HEATER ON/OFF RH	Y63
	X64	RH FIX SEL IN-1	X64	Y64	LINEAR FEEDER START RH	Y64
	X65	RH FIX SEL IN-2	X65	Y65	BOWL FEEDER START RH	Y65
	X66	RH FIX SEL IN-3	X66	Y66	AXIS-Z BREAK RH	Y66
	X67	RH FIX SEL IN-4	X67	Y67	HOT COLD PRESS CYL-2 FWD SOL-RH	Y67
FX5-32ER/ES-2	X70	PART PRESENT RH	X70	Y70	HOT COLD PRESS CYL-2 RET SOL RH	Y70
	X71	CENTRE CHECK SENS 1 RH	X71	Y71	AIR OF COOL FIXTURE-1 RH	Y71
	X72	CENTRE CHECK SENS 2 RH	X72	Y72	AIROFF COOL FIXTURE-2 RH	Y72
	X73	STUD 1 PRESENT SENSOR RH	X73	Y73	AIROFF COOL FIXTURE-3 RH	Y73
	X74	STUD 2 PRESENT SENSOR RH	X74	Y74	AIR OFF COOL GRIPPER RH	Y74
	X75	STUD 3 PRESENT SEONOR RH	X75	Y75	RH LOCK CYL FWD SOL	Y75
	X76	STUD 4 PRESENCE SENSOR RH(PROVISION)	X76	Y76	RH LOCK CYL RET SOL	Y76
	X77	PART CLAMP FWD RD RH	X77	Y77	RH-X AXIS SERVO LIMIT SW FWD	Y77
	X100	PART CLAMP RET RD RH	X100	Y100	RH- X AXIS SERVO LIMIT SW REV	Y100
	X101	PUSH CYL FWD RD RH	X101	Y101	RH -Y AXIS SERVO LIMIT SW FWD	Y101
	X102	PUSH CYL RET RD RH	X102	Y102	RH -Y AXIS SERVO LIMIT SW REV	Y102
	X103	STUD GRIPPER FWD RD RH	X103	Y103	RH- Z AXIS SERVO LIMIT SW FWD	Y103
	X104	STUD GRIPPER RET RD RH	X104	Y104	RH- Z AXIS SERVO LIMIT SW REV	Y104
	X105	DOT PUNCH FWD RD RH	X105	Y105	OZO BANG DISPENCER	Y105
	X106	STUD GRIPPER RET RD RH	X106	Y106	SPARE	Y106
	X107	HOT COLD PRESS CYL-1 FWD RD SW-RH	X107	Y107	SPARE	Y107
FX5-16EX/ES(SINK I/P)	X110	HOT COLD PRESS CYL-1 RET RD SW-RH	X110			
	X111	HOT COLD PRESS CYL-2 FWD RD SW-RH	X111			
	X112	HOT COLD PRESS CYL-2 RET RD SW-RH	X112			
	X113	SLIDE CYL FWD RD SW-RH	X113			
	X114	SLIDE CYL RET RD SW-RH	X114			
	X115	AXIS-X END LS-1 RH	X115			

	X116	AXIS-X END LS-2 RH	X116
	X117	AXIS-X HOME LS RH	X117
	X120	AXIS-Y END LS-1 RH	X120
	X121	AXIS-Y END LS-2 RH	X121
	X122	AXIS-Y HOME LS RH	X122
	X123	AXIS-Z END LS-1 RH	X123
	X124	AXIS-Z END LS-2 RH	X124
	X125	AXIS-Z HOME LS RH	X125
	X126	RH LOCK CYL FWD RD	X126
	X127	RH LOCK CYL RET RD	X127
FX5-4AD-ADP-1	V1+	TEMEPRATURE SENSOR (4-20mA)_LH	AI-1
	I1+		
	COM	ANALOG GND	AN-GND1
	V2+	LOAD CELL (4-20mA)_LH	AI-2
	I2+		
	COM	ANALOG GND	AN-GND2
	V3+	TEMEPRATURE SENSOR (4-20mA)_RH	AI-3
	I3+		
	COM	ANALOG GND	AN-GND3
	V4+	LOAD CELL (4-20mA)_RH	AI-4
	I4+		
FX5-4AD-ADP-2	COM	ANALOG GND	AN-GND4
	V1+	SMAT SENSOR-LH	AI-4
	I1+		
	COM	ANALOG GND	AN-GND1
	V2+	SMAT SENSOR-RH	AI-5
	I2+		
	COM	ANALOG GND	AN-GND2
	V3+	SPARE	AI-6
	I3+		
	COM	ANALOG GND	AN-GND3



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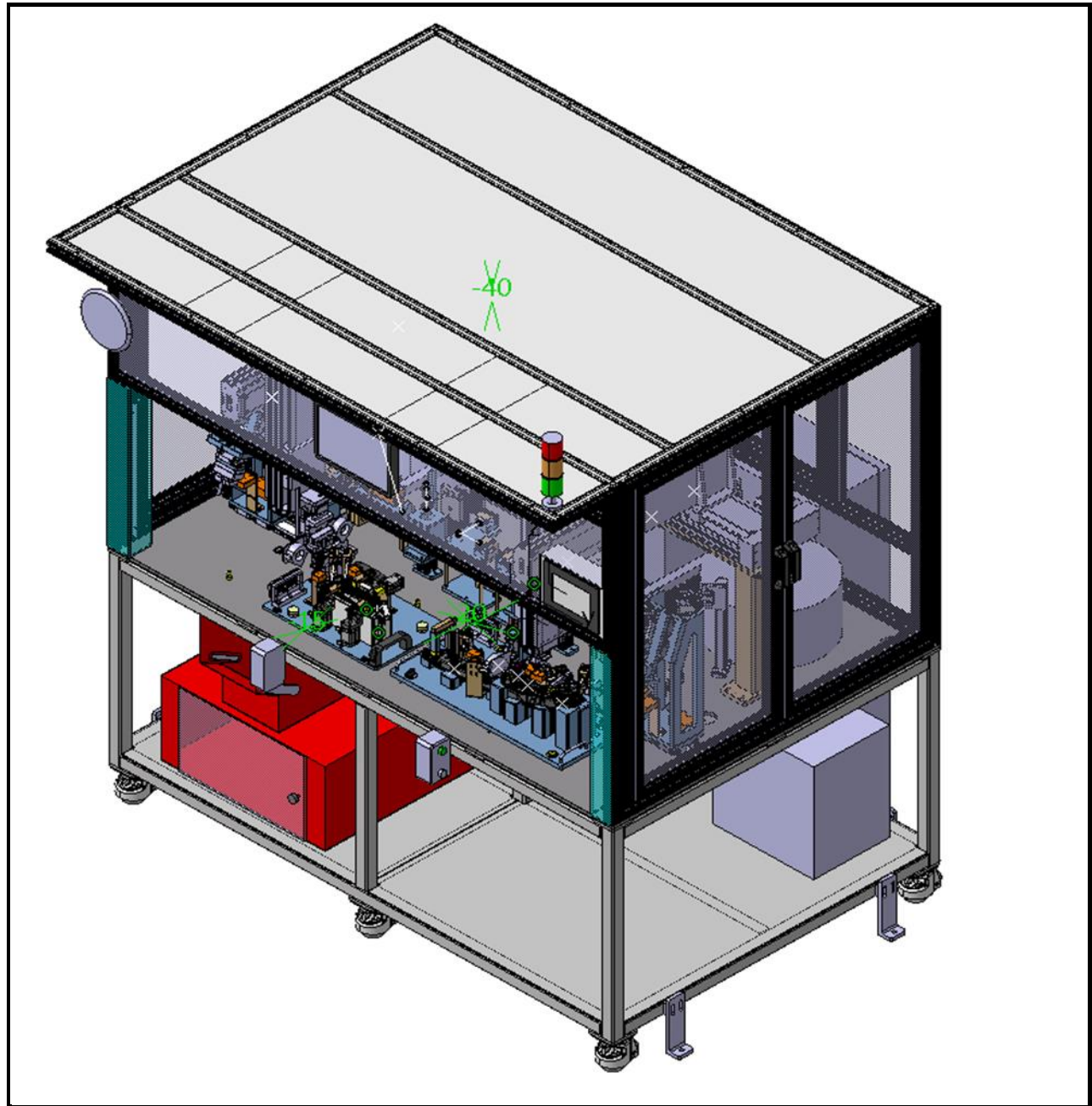
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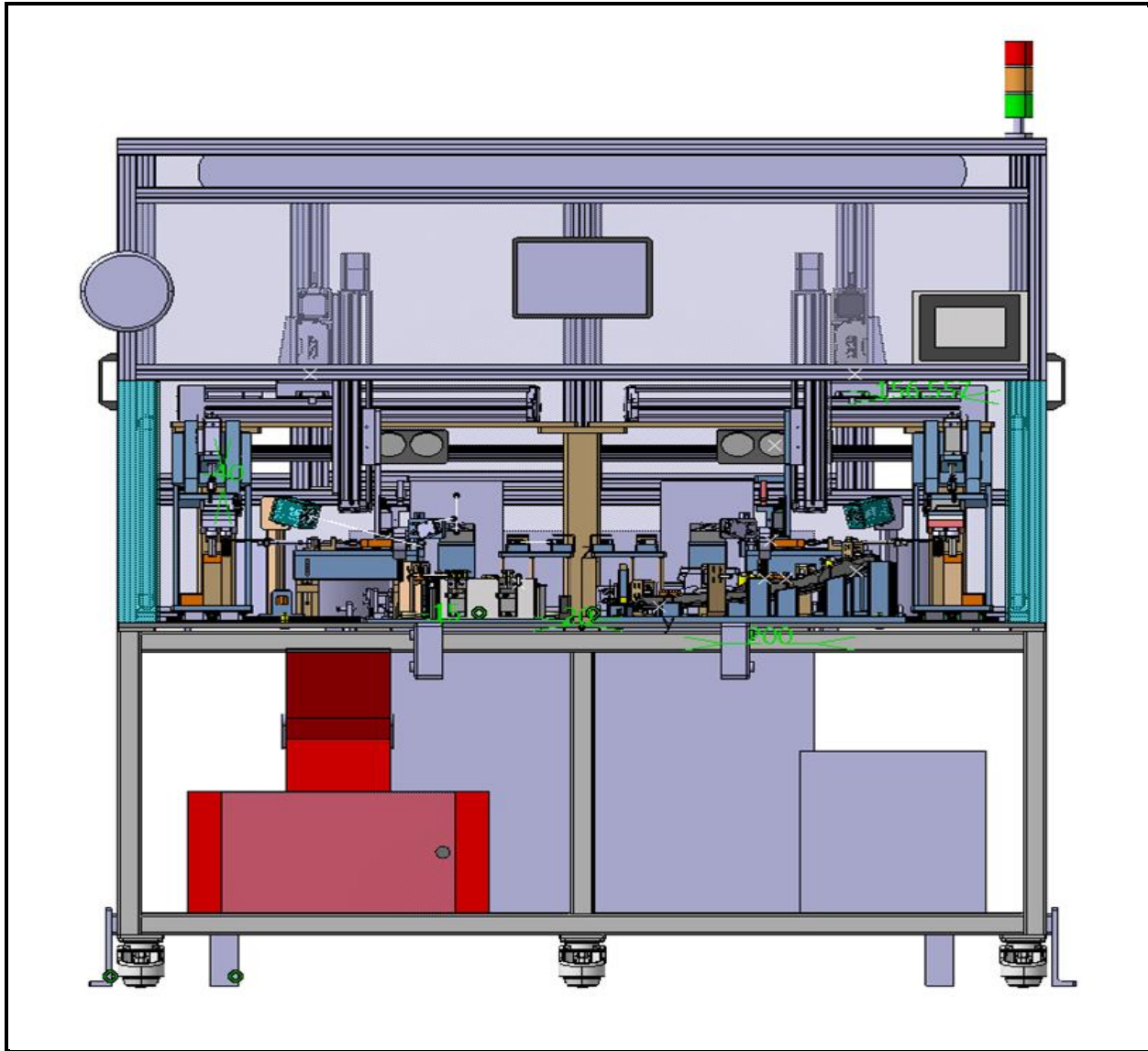
	V4+	SPARE	AI-7
	I4+		
	COM	ANALOG GND	AN-GND4

WORKSTATION DETAILS:

STUD PRESSING SYSTEM

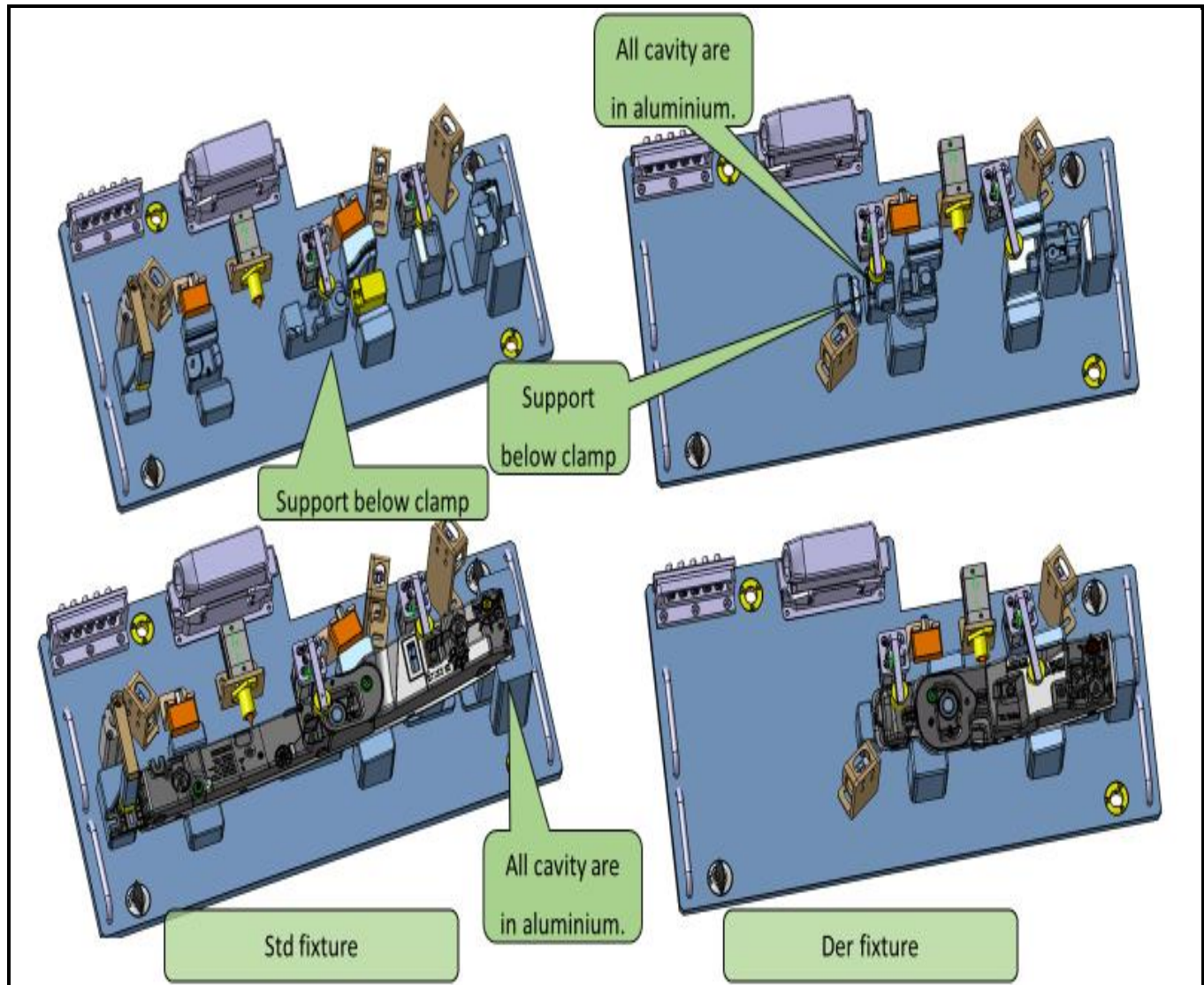


FRONT VIEW



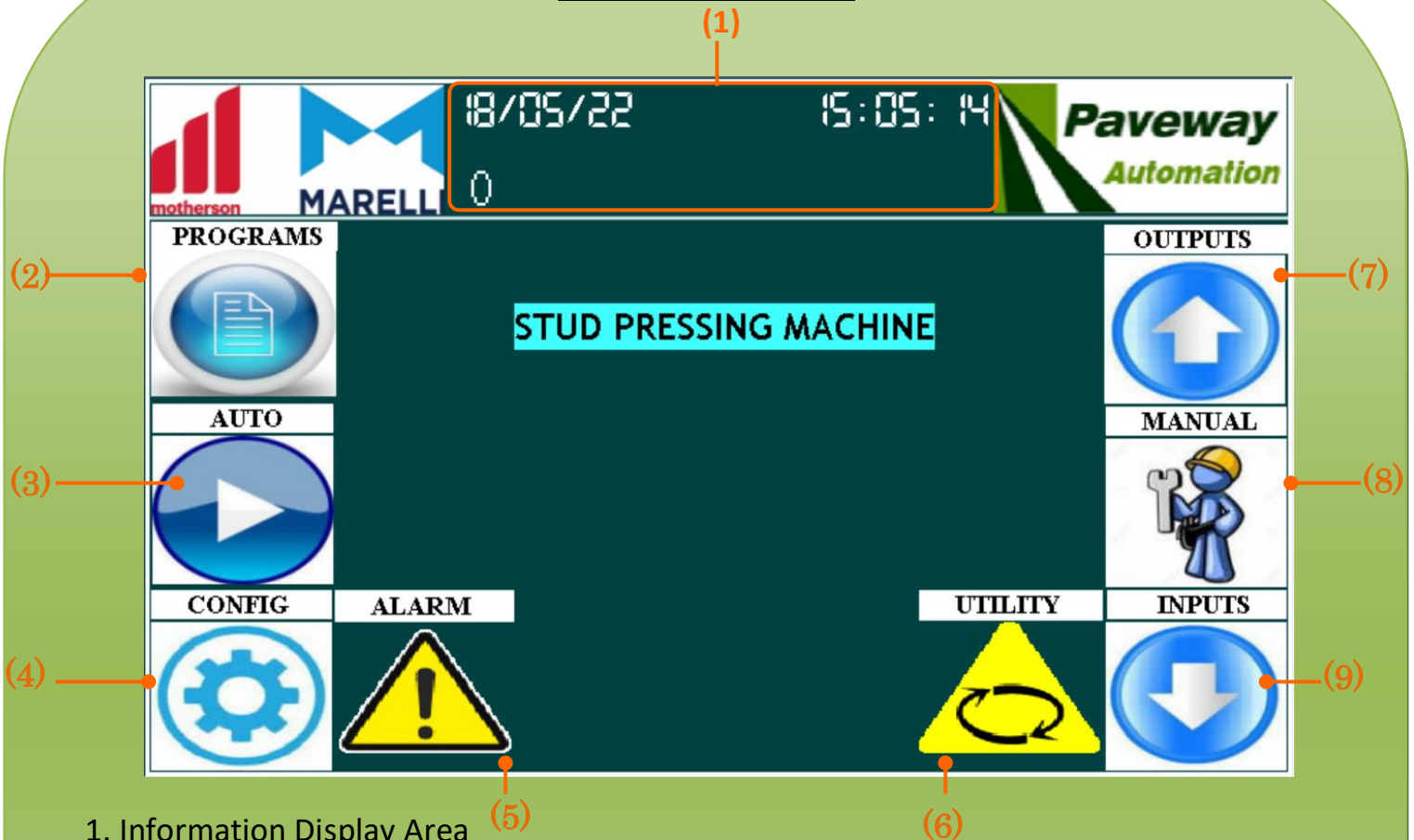
NOTE: DIMENSIONS ARE IN MM.

FIXTURE DETAILS



HMI SCREEN DETAILS:

HOME SCREEN



1. Information Display Area
Displays the Program name, Date & Time.

2. Program screen Navigation button .

3. Auto Screen Navigation button.

4. Configuration screen Navigation button.

5. Alarm Screen Navigation button.

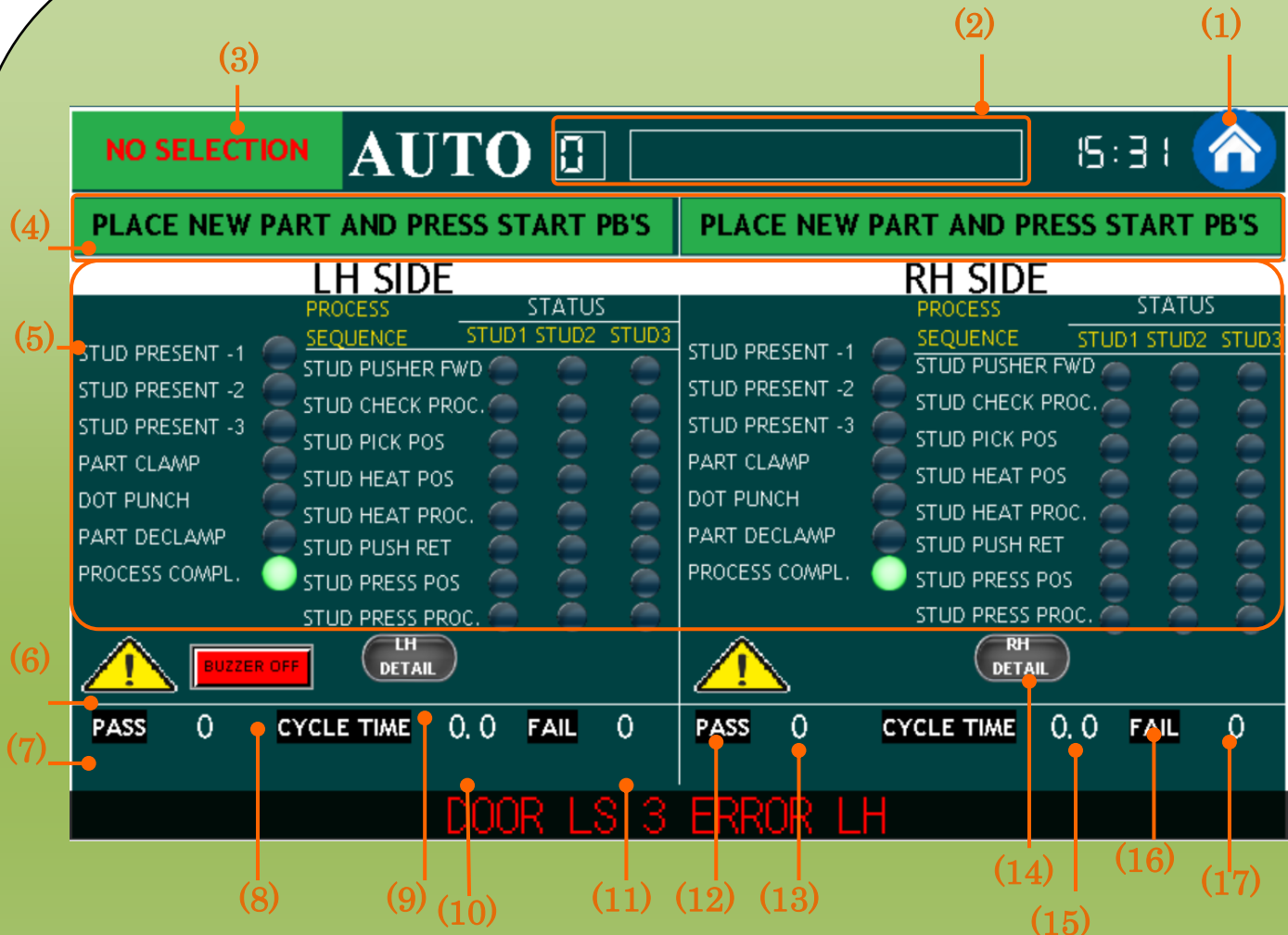
6. Utility Screen Navigaton Button.

7. Digital Output screen Navigation button.

8. Manual operation screen button.

9. Digital Input Monitoring screen button.

AUTO MODE SCREEN



- | | | |
|----------------------------------|-----------------------|-----------------------------|
| 1.Home screen navigation button. | 8.Buzzer OFF | 9.LH Detail Screen. |
| 2. Program name& Number. | 10.LH Cycle time | 11.LH Part NOK Count. |
| 3.LH & RH Selection | 12.RH Alarm Screen. | 13.RH Part OK Count |
| 4. LH & RH Running Cycle Status | 13.RH Part OK Count. | 14. RH Detail Screen |
| 5.Operation status display- | 15.RH Cycle Time | 16.Present Alarm Of Machine |
| 6.LH Alarm Screen | 17. RH Part NOK Count | |
| 7. LH Part OK Count. | | |

PROGRAM SCREEN

PROGRAMS		RST COUNT	
1	0	9	
2		10	
3		11	
4		12	
5		13	
6		14	
7		15	
8			

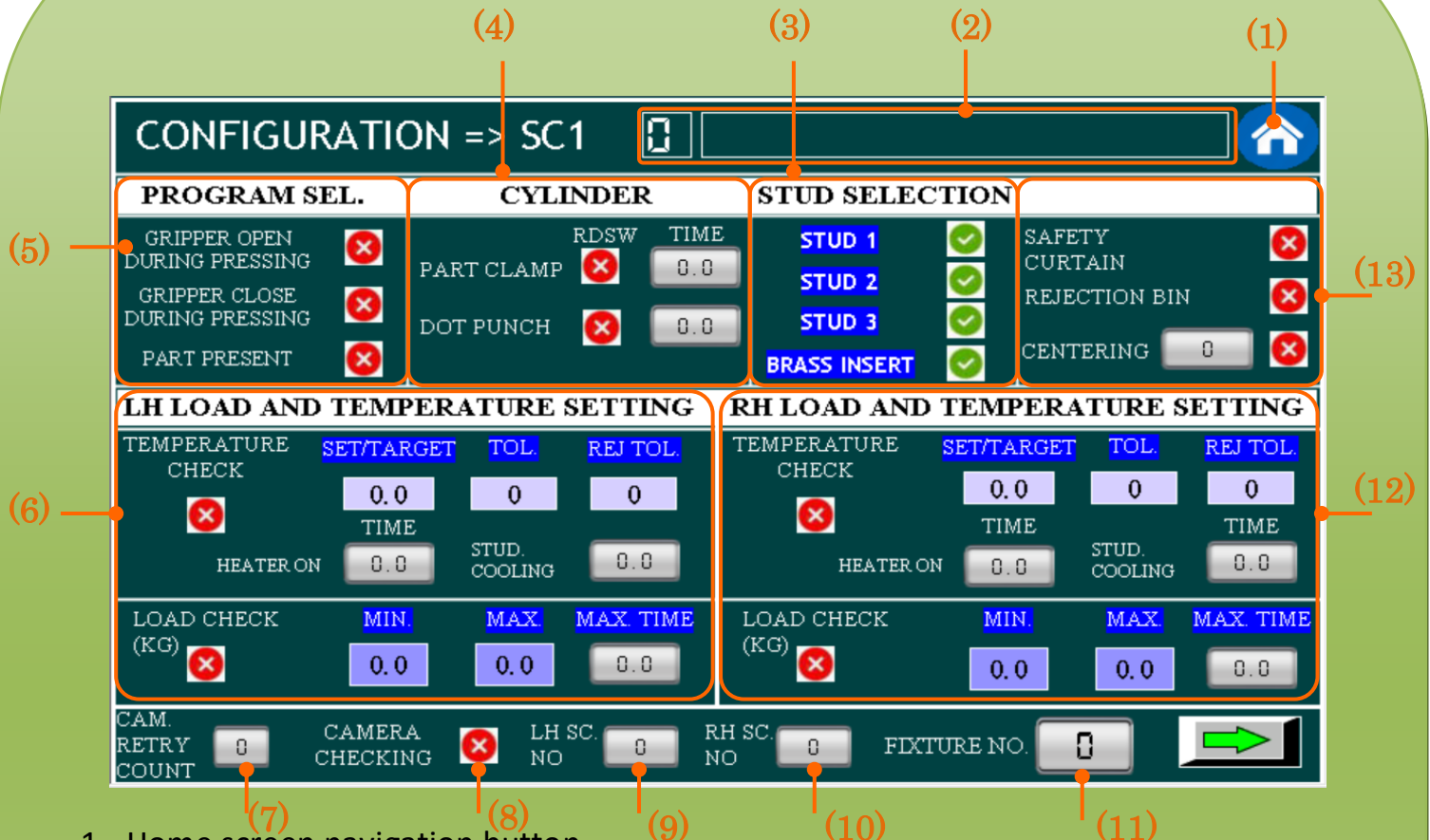
1. Navigate to Home Screen.

2. **RST COUNT** Reset Pass & Fail Count

3. Auto Running Program No Form Fixture Code.

4. **PROGRAMS 0** Program Number from Fixture code(1=Standard,2=Derative).

MAIN CONFIGURATION SCREEN



The screenshot shows the 'CONFIGURATION => SC1' screen. It is divided into several sections: 'PROGRAM SEL.' with three red 'X' icons; 'CYLINDER' with 'RDSW' and 'PART CLAMP' (both red 'X' icons) and 'DOT PUNCH' (red 'X' icon) with time inputs; 'STUD SELECTION' with 'STUD 1', 'STUD 2', 'STUD 3' (all green checkmarks) and 'BRASS INSERT' (green checkmark); 'LH LOAD AND TEMPERATURE SETTING' and 'RH LOAD AND TEMPERATURE SETTING' each with 'TEMPERATURE CHECK' (red 'X' icon), 'HEATER ON' (time input), 'LOAD CHECK (KG)' (red 'X' icon), and 'MIN.', 'MAX.', 'MAX. TIME' inputs; and a bottom bar with 'CAM. RETRY COUNT' (0), 'CAMERA CHECKING' (red 'X' icon), 'LH SC. NO.' (0), 'RH SC. NO.' (0), 'FIXTURE NO.' (0), and a green arrow button. Numbered callouts (1) through (13) point to specific elements: (1) Home button, (2) Program No & Name, (3) Stud & Brass Selection, (4) Cylinder Reed Switch Bypass & Check time, (5) Program Selection, (6) LH Load and Temperature Setting, (7) Cam. Retry Count, (8) Camera Checking, (9) LH SC. NO., (10) RH SC. NO., (11) Fixture No., (12) RH Load and Temperature Setting, and (13) Safety Curtain, Rejection Bin, and Centering.

1. Home screen navigation button
2. Program No & Name(Program Name editable),For both LH & RH.
3. Stud & Brass Selection, For both LH & RH.
4. Cylinder Reed Switch Bypass & Check time, For both LH & RH.
5. Program Selection=Gripper operation in auto & Part Present Check bit, for both LH & RH.
6. Auto SET/TARGET =Temperature set point, TOL=Tolerance Temp set point, REJ TOL=Reject Tolerance set point, HEATER ON TIME, STUD COOLING= On Set time.
Load Check =MIN & Max Set Point, Load Check Max time, For LH.



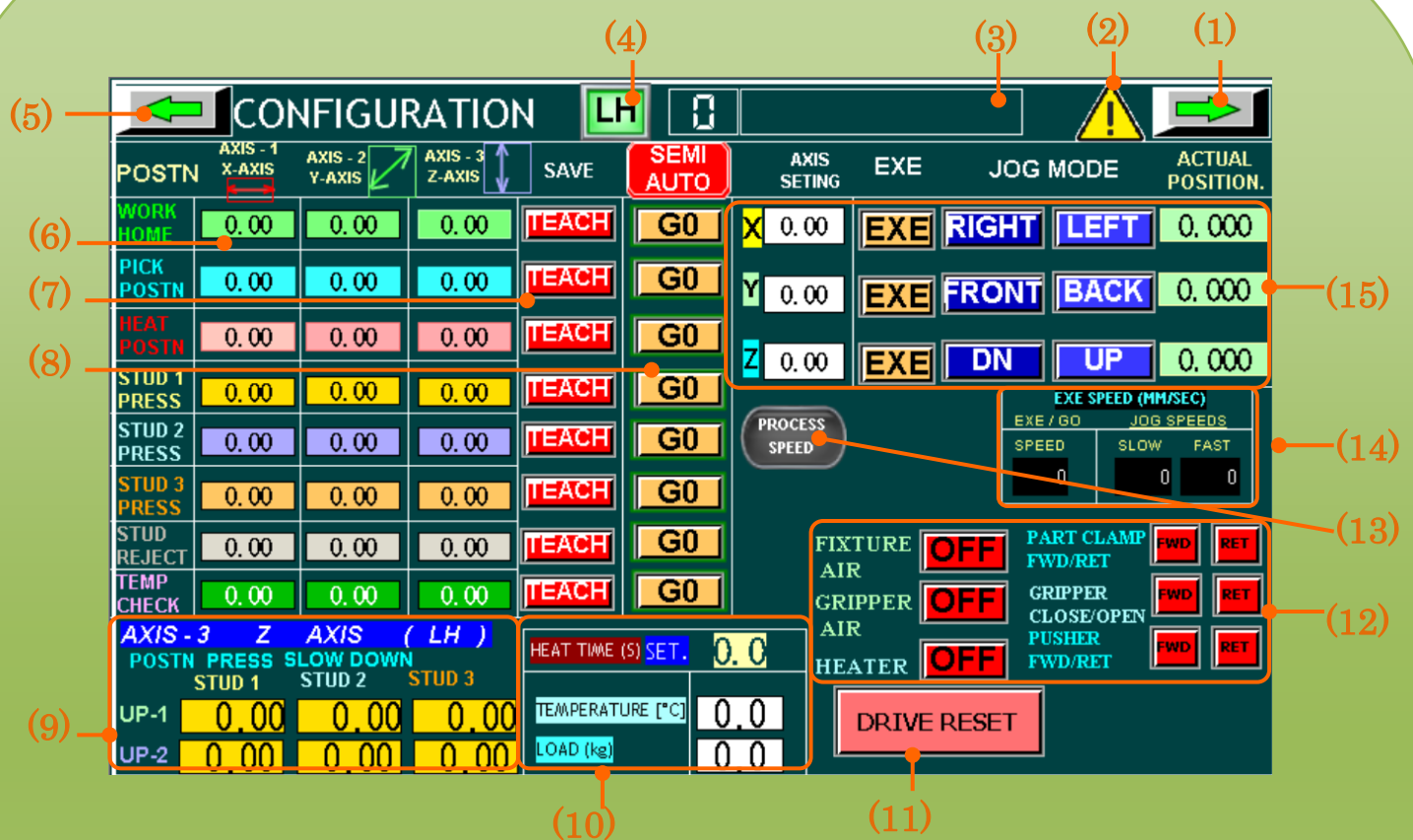
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7. Camera retry count in auto.
8. Camera Check Bypass.
9. LH Side Camera Scene Number.
10. RH Side Camera Scene Number.
11. Fixture Number, For both LH & RH.
12. Auto SET/TARGET =Temperature set point, TOL=Tolerance Temp set point,
REJ TOL=Reject Tolerance set point, HEATER ON TIME, STUD COOLING= On Set time.
Load Check =MIN & Max Set Point, Load Check Max time, For RH.
13. Safety Curtain & Rejection Bin Check bit, Start Centering Operation set point & Check bit.

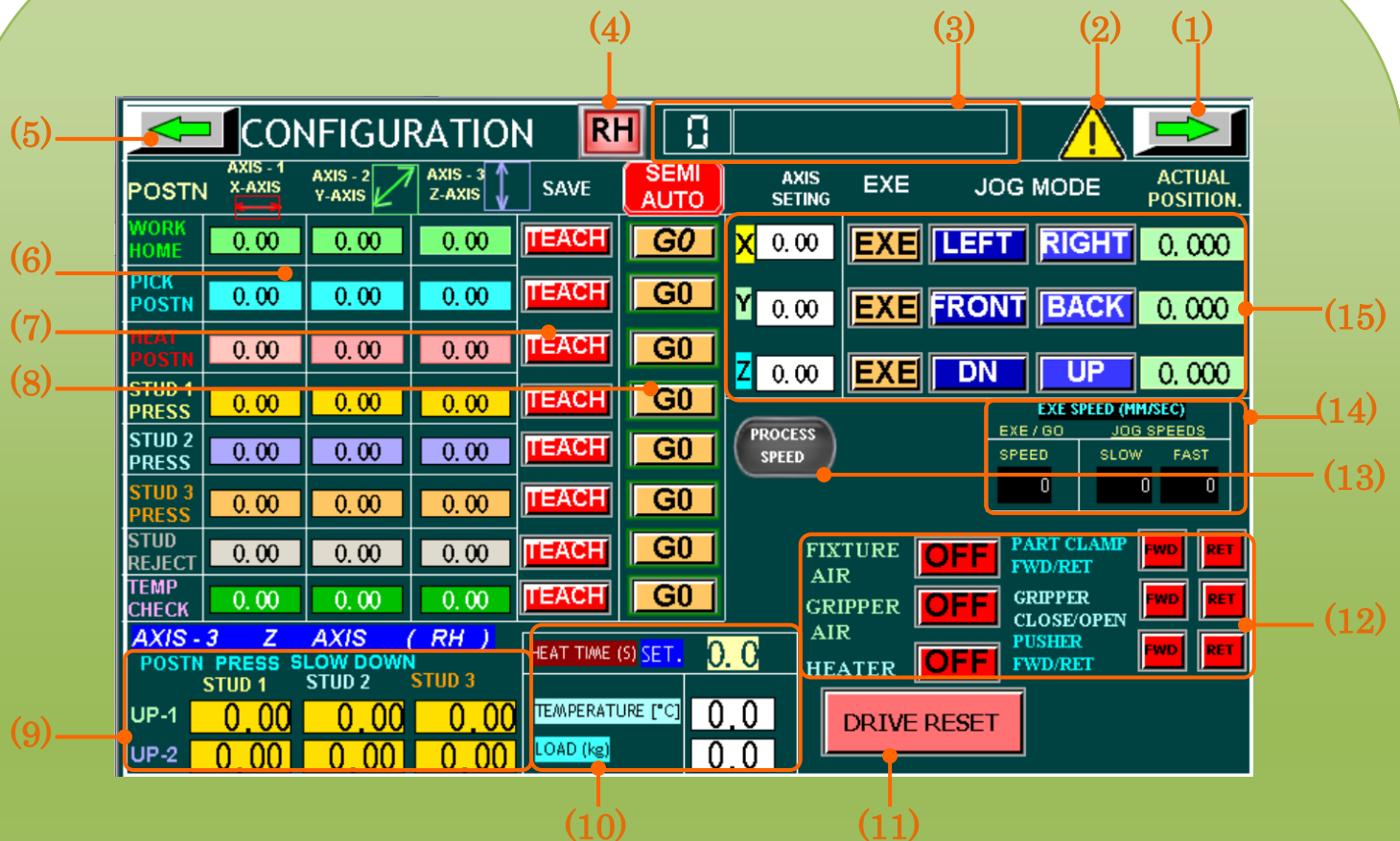
CONFIGURATION SCREEN LH



1. Next screen navigation button.
2. Alarm screen navigation button.
3. Program Number & Name Display.
4. LH Semiauto Selection Button.
5. Previous screen navigation button.
6. X, Y & Z Axis teach set point.
7. Teach Set Point Button.
8. STUD 1 PRESS
9. AXIS - 3 Z AXIS (LH)
10. Heat On Time & Temp & Load Actual Value
11. Reset All LH Drive.
12. Semiauto Operation.
13. Process speed navigation button.
14. Semiauto Speed set point.
15. Semiauto drive operation in jog mode & in Positioning mode.

8. Semi Auto cmd for Position.
9. Stud UP-1 & UP-2 set point.

CONFIGURATION SCREEN RH



The screenshot shows the 'CONFIGURATION SCREEN RH' with various controls and data fields. Numbered callouts identify specific features:

- (1) Next screen navigation button (right arrow)
- (2) Alarm screen navigation button (warning icon)
- (3) Program Number & Name Display (empty box)
- (4) RH Semiauto Selection Button (RH button)
- (5) Previous screen navigation button (left arrow)
- (6) WORK HOME button
- (7) PICK POSTN button
- (8) TEAT POSTN button
- (9) STUD 1 PRESS button
- (10) HEAT TIME (S) SET. 0.0
- (11) DRIVE RESET button
- (12) SEMI AUTO button
- (13) PROCESS SPEED button
- (14) EXE SPEED (MM/SEC) display
- (15) JOG MODE buttons (LEFT, RIGHT, FRONT, BACK, DN, UP)

POSTN	AXIS - 1 X-AXIS	AXIS - 2 Y-AXIS	AXIS - 3 Z-AXIS	SAVE	SEMI AUTO	AXIS SETTING	EXE	JOG MODE	ACTUAL POSITION.
WORK HOME	0.00	0.00	0.00	TEACH	GO	X 0.00	EXE	LEFT RIGHT	0.000
PICK POSTN	0.00	0.00	0.00	TEACH	GO	Y 0.00	EXE	FRONT BACK	0.000
TEAT POSTN	0.00	0.00	0.00	TEACH	GO	Z 0.00	EXE	DN UP	0.000
STUD 1 PRESS	0.00	0.00	0.00	TEACH	GO				
STUD 2 PRESS	0.00	0.00	0.00	TEACH	GO				
STUD 3 PRESS	0.00	0.00	0.00	TEACH	GO				
STUD REJECT	0.00	0.00	0.00	TEACH	GO				
TEMP CHECK	0.00	0.00	0.00	TEACH	GO				

AXIS - 3 Z AXIS (RH)

POSTN	PRESS	SLOW DOWN	STUD 1	STUD 2	STUD 3
UP-1	0.00	0.00	0.00	0.00	0.00
UP-2	0.00	0.00	0.00	0.00	0.00

HEAT TIME (S) SET. 0.0

TEMPERATURE [°C] 0.0

LOAD (kg) 0.0

EXE SPEED (MM/SEC)

EXE / GO	JOG SPEEDS
SPEED	SLOW FAST
0	0 0

FIXTURE AIR OFF PART CLAMP FWD/RET FWD RET

GRIPPER AIR OFF GRIPPER CLOSE/OPEN FWD RET

HEATER OFF PUSHER FWD/RET FWD RET

1. Next screen navigation button.
2. Alarm screen navigation button.
3. Program Number & Name Display.
4. RH Semiauto Selection Button.
10. Heat On Time & Temp & Load Actual Value
11. Reset All LH Drive.
12. Semiauto Operation.
13. Process speed navigation button.



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5. Previous screen navigation button.
6. X, Y & Z Axis teach set point.
7. Teach Set Point Button.
8. Semi Auto cmd for Position.
9. Stud UP-1 & UP-2 set point.
14. Semiauto Speed set point.
15. Semiauto drive operation in jog mode & in Positioning mode.

Configuration Screen (LH & RH Semiauto): -

➤ Check point before Semiauto Operation

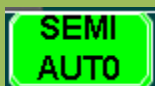
- Before Start semiauto operation check no any alarm On HMI & All drive are healthy.
- If alarm on drive not reset, then use  Button.
- Do Homing of all drive if required.
- Check Speed before start.

➤ Teaching Procedure in Semiauto: -

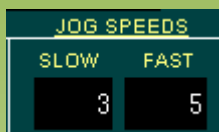
- Select Semiauto



- After Selection Button status changes.



- Enter jog speed MM/Sec as require.





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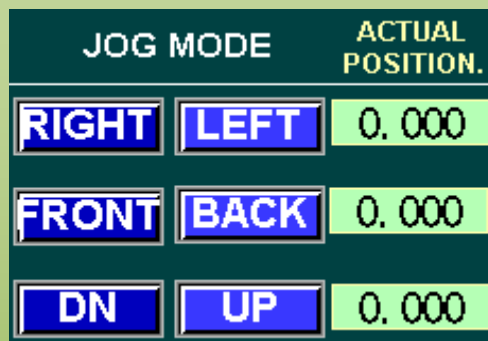
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- Clamp part & Gripper Close.



- Use JOG MODE & Go to Stud position of X, Y & Z axis.



- After reach to stud position X, Y & Z Axis.
- Before Pressing **TEACH** Button Please check physical position of all three axis X, Y & Z is at
 - WORK HOME
 - PICK POSTN
 - HEAT POSTN



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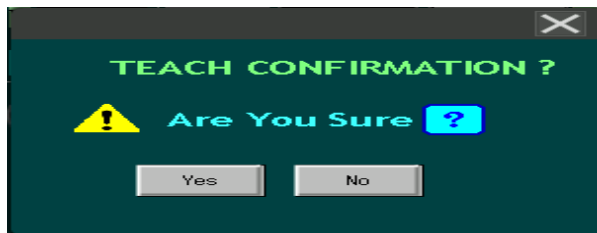


- d) STUD 1 PRESS
- e) STUD 2 PRESS
- f) STUD 3 PRESS
- g) STUD REJECT
- h) TEMP CHECK Position.

➤ Then Press Button



➤ Open pop up



- Press on Yes required position Teach. Then actual position of X, Y & Z AXIS move to **Teach position set point** respectively, Close pop up.

ACTUAL POSITION.
0.000
0.000
0.000

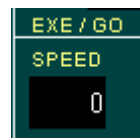
- Cross check Actual Position & Teach Position Set Point are same.
- **Teach Position Set point.**

POSTN	AXIS - 1 X-AXIS	AXIS - 2 Y-AXIS	AXIS - 3 Z-AXIS
WORK HOME	0.00	0.00	0.00
PICK POSTN	0.00	0.00	0.00
HEAT POSTN	0.00	0.00	0.00
STUD 1 PRESS	0.00	0.00	0.00
STUD 2 PRESS	0.00	0.00	0.00
STUD 3 PRESS	0.00	0.00	0.00
STUD REJECT	0.00	0.00	0.00
TEMP CHECK	0.00	0.00	0.00



Use Button to go to the teach position.

➤ Before Move Please Check



Speed.

1) Operations in Semiauto: -

1) **Position Command: -** Operate single axis in Position Mode.

- ❖ **Note: -**
- ❖ Check Z-Axis is at safe position to Move X & Y Axis.
- ❖ Before operate in Semiauto position mode physical position respective axis check, to avoid mechanical damage.



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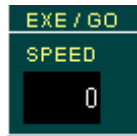
- Select Semiauto



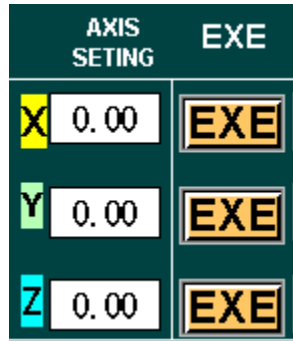
- After Selection Button status changes.



- Set required Speed



- Set Axis Position and press 'EXE' For Command.



2) Cylinder & Heater Operations: -

- Select Semiauto

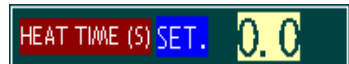


- After Selection Button status changes.



- Semiauto Operations of Fixture Air-Gripper Air (Solenoid valve single acting) & Part Clamp, Gripper Close/Open, Pusher Fwd./Ret. (Double acting solenoid valve), i.e. For single solenoid continuous on-off operation & for double solenoid valve pulse on-off operation.



Heater Operation-Set Heater On Time in sec (Limit 0 to 5  Sec).

Then press Heater On-Off, After Complete settable time heater gets off.

3) Set Point & Display Value: -

- Stud-1 ,2,3 UP-1 & 2 Position Set point.

Note: - These set points are used in auto during pressing.

AXIS - 3	Z	AXIS	(LH)
POSTN	PRESS	SLOW	DOWN
STUD 1	STUD 2	STUD 3	
UP-1	0.00	0.00	0.00
UP-2	0.00	0.00	0.00

- Actual Value of Temperature & Load (Display Only)

TEMPERATURE [°C]	0.0
LOAD (kg)	0.0

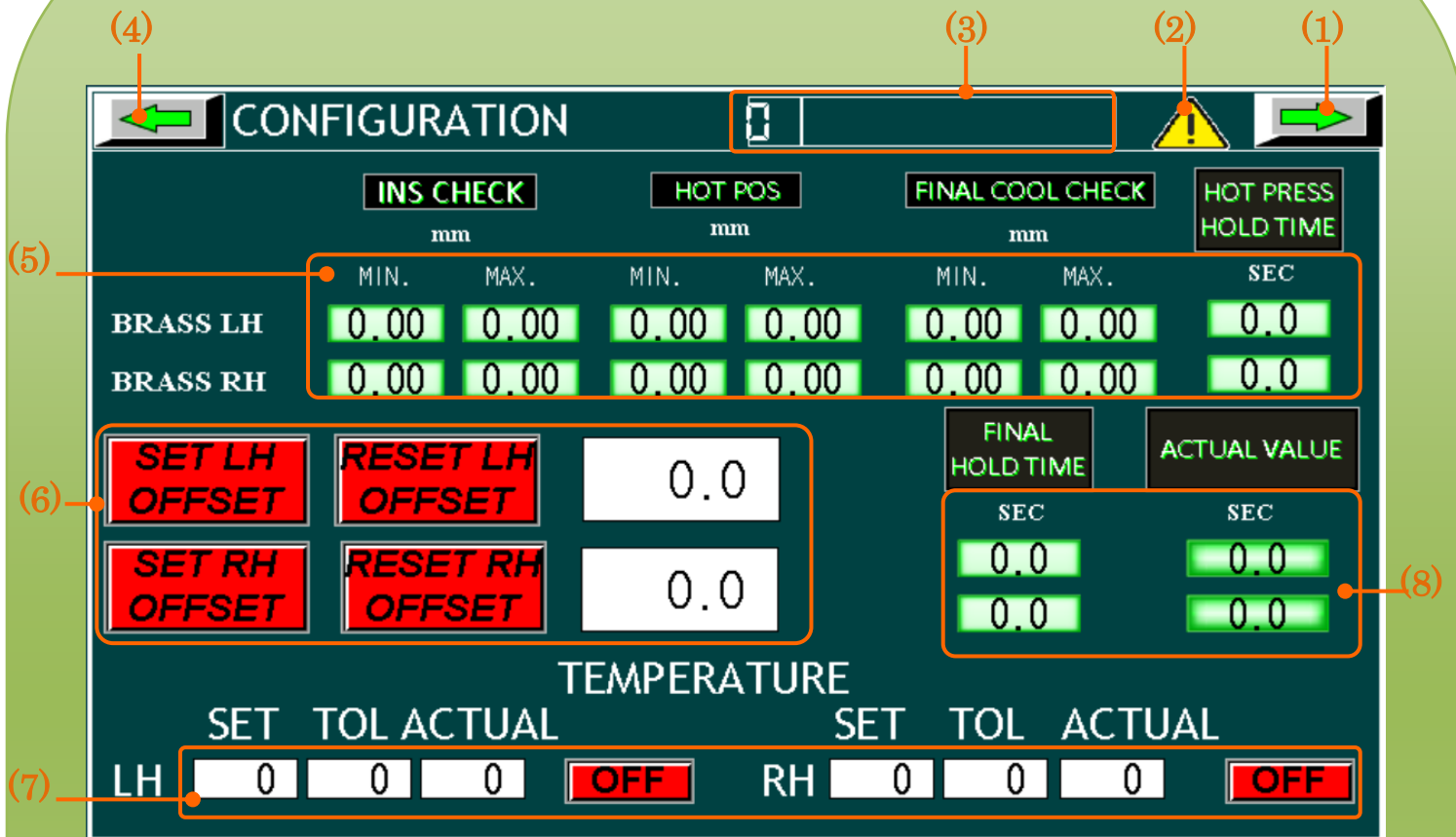
- Used to navigate Auto & Semiauto Speed set point.



PROCESS SPEED

← AUTO PROCESS SPEED X,Y,Z AXIS				SEMI AUTO SPEED X,Y,Z AXIS			
PROCESS			SPEED (MM/SEC)	PROCESS			SPEED (MM/SEC)
WORK HOME	TO	PICK POSTN		WORK HOME	TO	PICK POSTN	
PICK POSTN	TO	HEAT POSTN		PICK POSTN	TO	HEAT POSTN	
HEAT POSTN	TO	STUD PRESS UP1		HEAT POSTN	TO	STUD PRESS UP1	
STUD PRESS UP1	TO	STUD PRESS UP2		STUD PRESS UP1	TO	STUD PRESS UP2	
STUD PRESS UP2	TO	STUD PRESS FNL		STUD PRESS UP2	TO	STUD PRESS FNL	
STUD PRESS FNL	TO	WORK HOME		STUD PRESS FNL	TO	WORK HOME	
Z AXIS SAFE UP POSTN				Z AXIS SAFE UP POSTN			

BRASS CONFIGURATION SCREEN



The screenshot shows the 'BRASS CONFIGURATION SCREEN' with the following elements:

- (1)** Next page navigation button (green arrow pointing right).
- (2)** Alarm screen navigation button (yellow triangle with exclamation mark).
- (3)** Program Name & Number display (empty box).
- (4)** Previous page navigation button (green arrow pointing left).
- (5)** Configuration table with columns for INS CHECK, HOT POS, FINAL COOL CHECK, and HOT PRESS HOLD TIME. Each column has MIN. and MAX. values in mm, except for HOT PRESS HOLD TIME which is in SEC.
- (6)** Offset settings for LH and RH, including SET, RESET, and OFFSET buttons, and a display showing 0.0.
- (7)** Temperature settings for LH and RH, including SET, TOL, and ACTUAL values, and an OFF button.
- (8)** Final Hold Time and Actual Value settings, each with a display showing 0.0 and a button.

	INS CHECK		HOT POS		FINAL COOL CHECK		HOT PRESS HOLD TIME
	mm	mm	mm	mm	mm	mm	SEC
BRASS LH	0.00	0.00	0.00	0.00	0.00	0.00	0.0
BRASS RH	0.00	0.00	0.00	0.00	0.00	0.00	0.0

SET LH OFFSET		RESET LH OFFSET		0.0	FINAL HOLD TIME		ACTUAL VALUE	
SEC	SEC	SEC	SEC	SEC	SEC	SEC	SEC	
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	

TEMPERATURE			
SET	TOL	ACTUAL	
LH	0	0	0
RH	0	0	0

1. Next page navigation button.
2. Alarm screen navigation button.
3. Program Name & Number display.



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4. Previous screen navigation button.
5. INS CHECK=Initial Cold check Min & Max SP, HOT POS = Hot press min-max sp,
FINAL COOL CHECK = Brass final cool sp, Hot Press hold time.
6. Calibration window-Brass press cyl.
7. Temp set point & tolerance for brass heater. (OFF button not used)
8. Final cold press holds time & Brass vertical cylinder sensor value.

LH & RH Brass Setting Screen: -

1) Set Point Details: -

- INS CHECK: -Min & Max position set point for initial cold check or brass check.
- HOT POS: -Min & Max position set point for hot position of brass.
- FINAL COOL CHECK: -Min & Max position set point for final cool press after hot press completed.
- HOT PRESS HOLD TIME: - Hold time of cylinder after 'HOT POS' Completed.

	INS CHECK		HOT POS		FINAL COOL CHECK		HOT PRESS HOLD TIME
	mm		mm		mm		SEC
	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	
BRASS LH	0.00	0.00	0.00	0.00	0.00	0.00	0.0
BRASS RH	0.00	0.00	0.00	0.00	0.00	0.00	0.0

- FINAL HOLD TIME: - Hold time of cylinder after 'FINAL COOL CHECK' Completed.



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➤ ACTUAL VALUE: - Actual value of position sensor in 'MM'.

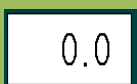


2) Calibration Procedure: -



➤ Vertical cylinder operates forward then actual position value update in i/o field.

➤ If updated value is 12.00 MM and press



The 12.00 MM value considered as Zero or calibration complete.



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If required to reset calibration, then press



Button, Then Actual value 12.00 MM.

- Same procedure follows for both LH & RH side.

3) Brass Heater or PID: -

TEMPERATURE									
	SET	TOL	ACTUAL		SET	TOL	ACTUAL		
LH	0	0	0	OFF	RH	0	0	0	OFF

- SET: - Temperature Set point of Brass heater.
- TOL: - Plus and Minus Tolerance set point of 'SET' Temperature set point.



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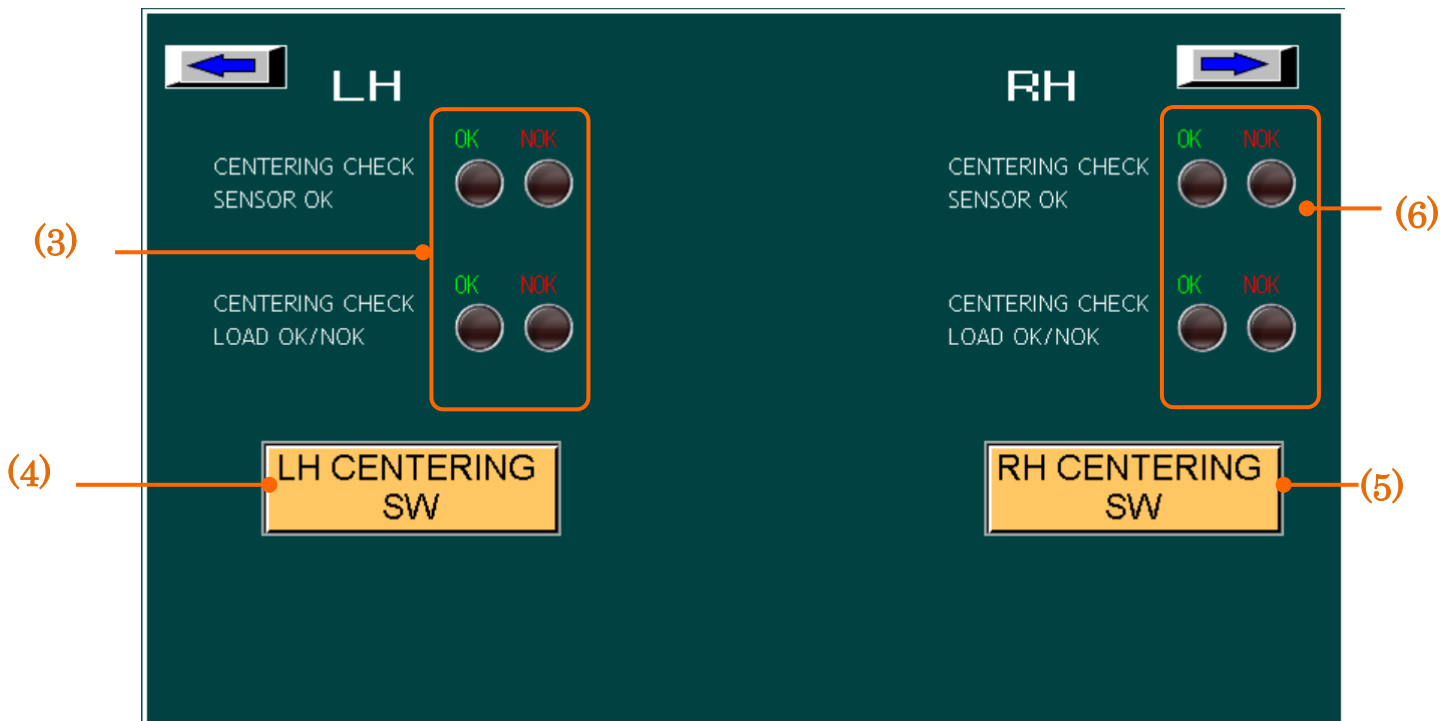


- ACTUAL: - Actual temperature of Brass heater. Same for LH & RH Side. 'OFF'
Not used.

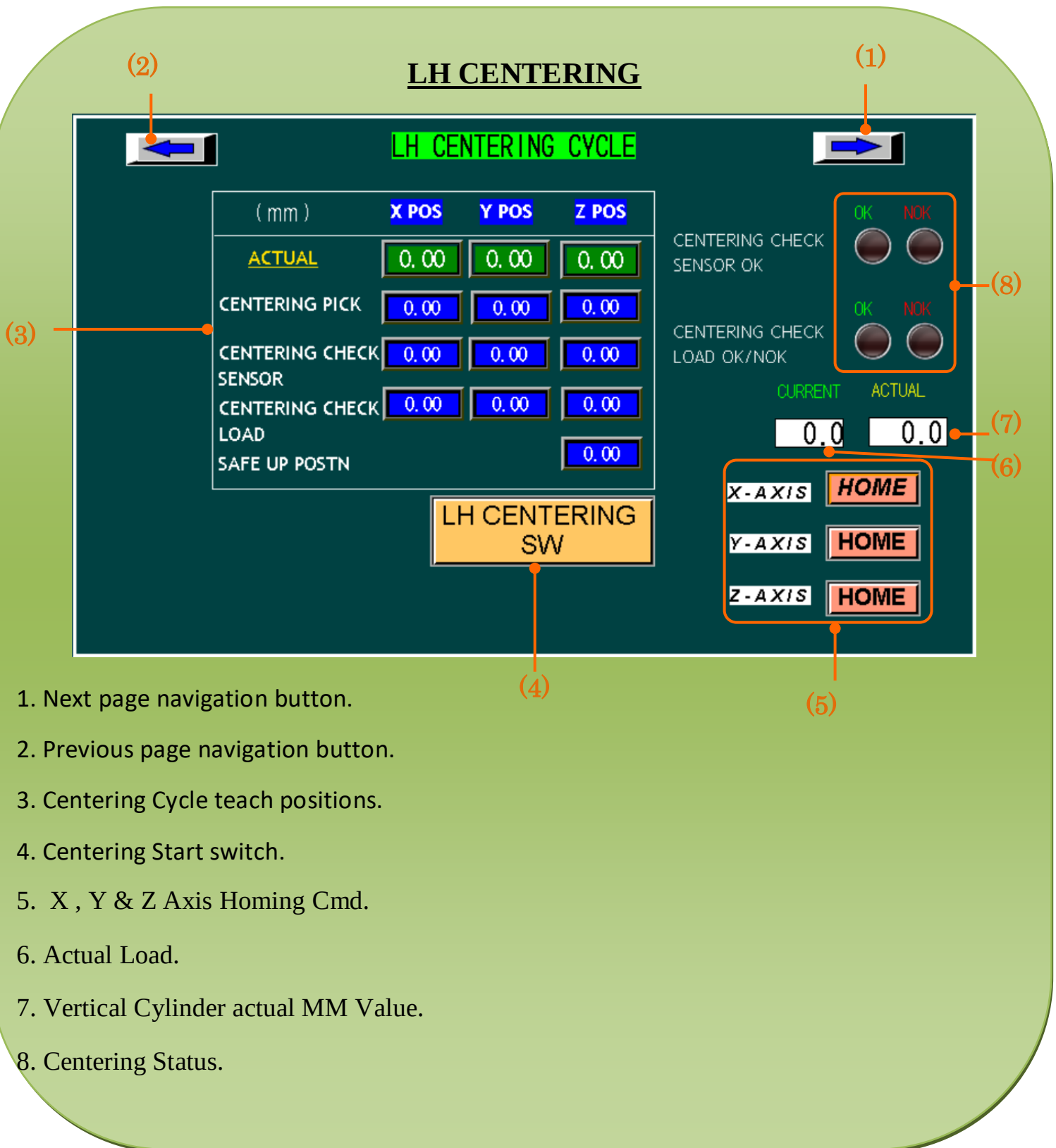
CENTERING START & STATUS SCREEN

(2)

(1)

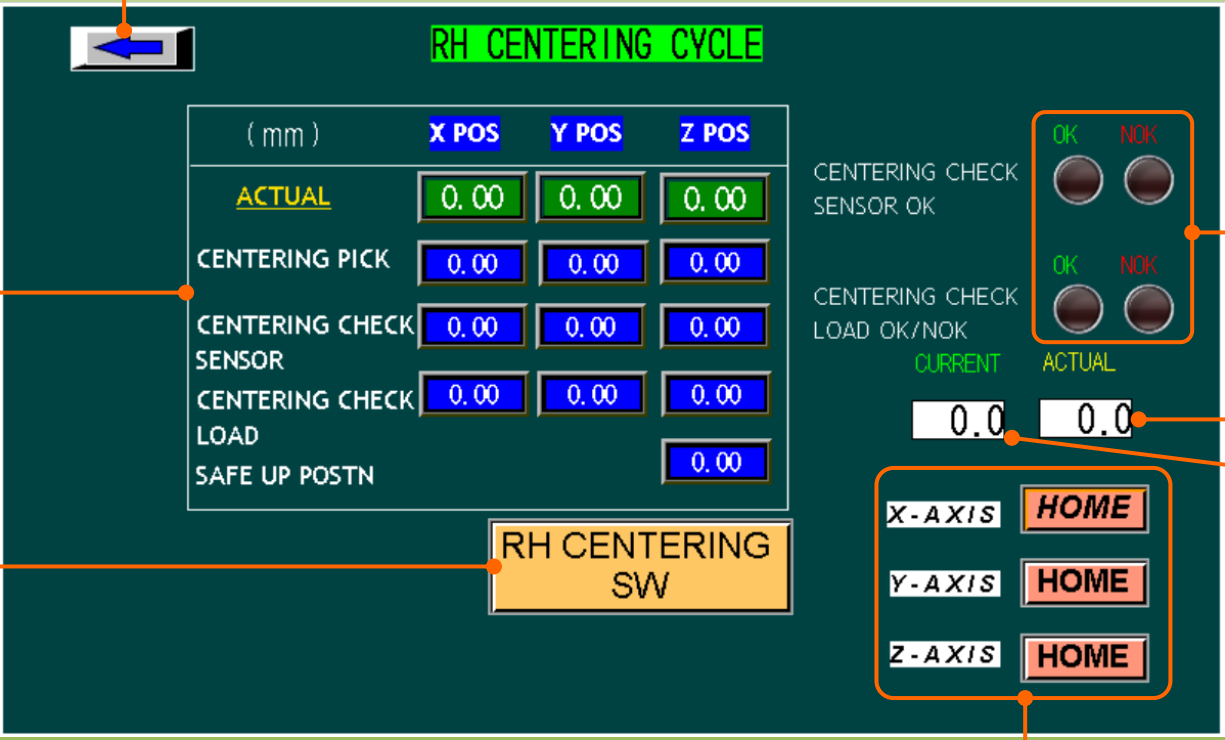


1. Next screen navigation button.
2. Previous screen navigation button.
3. LH Centering status.
4. LH Centering Start Switch.
5. RH Centering Start Switch.
6. RH Centering status.



1. Next page navigation button.
2. Previous page navigation button.
3. Centering Cycle teach positions.
4. Centering Start switch.
5. X , Y & Z Axis Homing Cmd.
6. Actual Load.
7. Vertical Cylinder actual MM Value.
8. Centering Status.

RH CENTERING



RH CENTERING CYCLE

(mm)	X POS	Y POS	Z POS
ACTUAL	0.00	0.00	0.00
CENTERING PICK	0.00	0.00	0.00
CENTERING CHECK SENSOR	0.00	0.00	0.00
CENTERING CHECK LOAD	0.00	0.00	0.00
SAFE UP POSTN			0.00

RH CENTERING SW

CENTERING CHECK SENSOR OK: OK (green), NOK (red)

CENTERING CHECK LOAD OK/NOK: OK (green), NOK (red)

CURRENT: 0.0, ACTUAL: 0.0

X-AXIS HOME, Y-AXIS HOME, Z-AXIS HOME

Numbered callouts: (1) Previous page navigation button, (2) Centering Cycle teach positions, (3) Centering Cycle Start Switch, (4) X, Y & Z Axis homing cmd., (5) Actual Load, (6) Vertical Cylinder actual MM Value, (7) Centering Status.

1. Previous page navigation button.
2. Centering Cycle teach positions.
3. Centering Cycle Start Switch.
4. X, Y & Z Axis homing cmd.
5. Actual Load.
6. Vertical Cylinder actual MM Value.
7. Centering Status.



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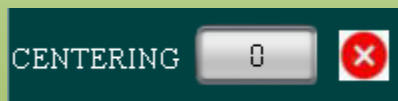
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Centering LH&RH: -

Overview-

- Centering used to check gripper displacement.
- Centering set point & Check bit provided in configuration screen i.e. After how many auto cycles centering will have carried out.



- Message will display when centering set point & actual auto cycles are equal.

RUN CENTERING FOR LH

Centering Check Point-

- Before start Centering teach or check position.

(mm)	X POS	Y POS	Z POS
ACTUAL	0.00	0.00	0.00
CENTERING PICK	0.00	0.00	0.00
CENTERING CHECK SENSOR	0.00	0.00	0.00
CENTERING CHECK LOAD	0.00	0.00	0.00
SAFE UP POSTN			0.00



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CENTERING CHECK SENSOR X & Y Positions & CENTERING CHECK LOAD (Not editable) X & Y Teach positions are same.

- SAFE UP POSTN: - Safe up position (Z-Axis) set point are used in all operation of Auto & Semiauto.

Every time positions Command to X, Y & Z-Axis, First Z-Axis move to safe up position then X & Y.

Centering Procedure-

Note: - Centering Speed not editable from HMI.

- When Centering set point and auto cycle pass count are equal & Centering check bit selected in configuration then message on HMI 'RUN CENTERING FOR LH'.
- Then go to centering screen & Press

LH CENTERING
SW

After centering start button status changes to green

LH CENTERING
SW



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- After Start centering X, Y & Z Axis go to CENTERING PICK POSITION. Close gripper to pick Dowell.
- Then go to CENTERING CHECK SENSOR POSITION. If both sensors are ok, then status on HMI shows OK.
- If centering NOK or Load alarm, then stop machine manually operate machine as per physical condition.

Sensor OK Status

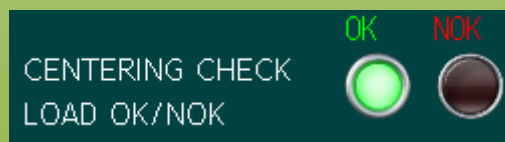


Sensors NOK Status



- If centering check sensor OK. Then Move to 'CENTERING CHECK LOAD', if load is greater than 5kg during load check machine will stop & operate machine as per physical condition.

Load OK Status



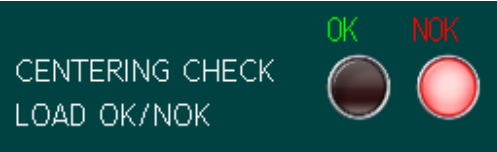


YFG TRUNK STUD INSERTION MACHINE.

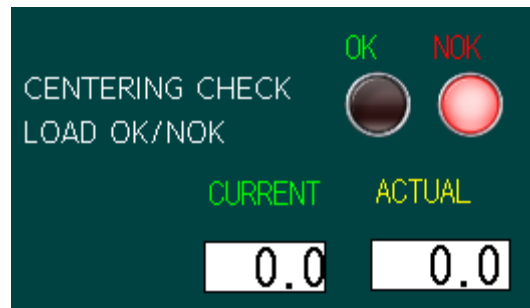
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Load NOK Status



- CURRENT: - Running Load.
- ACTUAL: - NOK Load during centering load check.



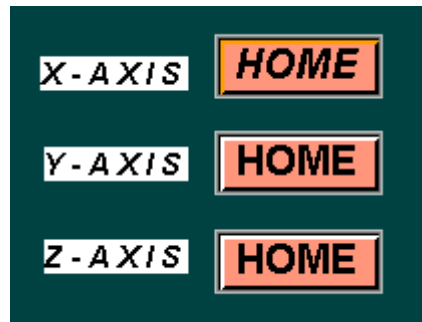
- Homing Button of X, Y & Z Axis.

Note: - If homing carried out then we need to teach all position points or manually/semiauto need to check all teach positions.



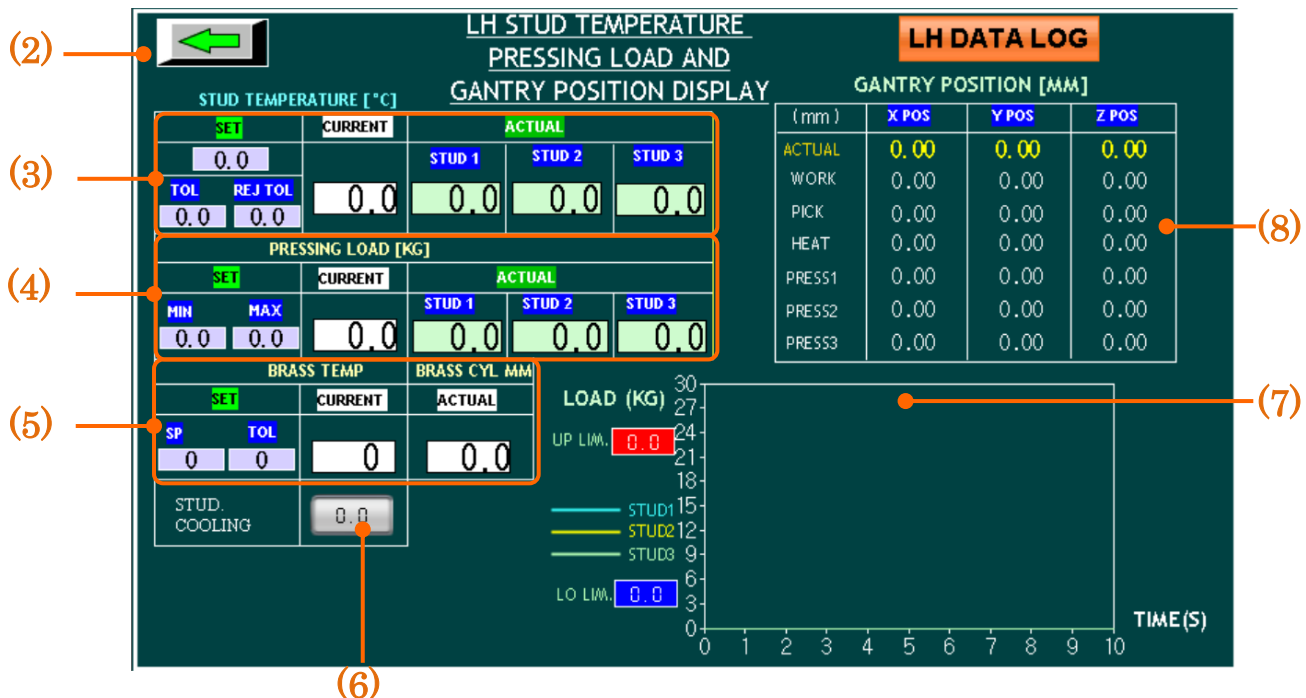
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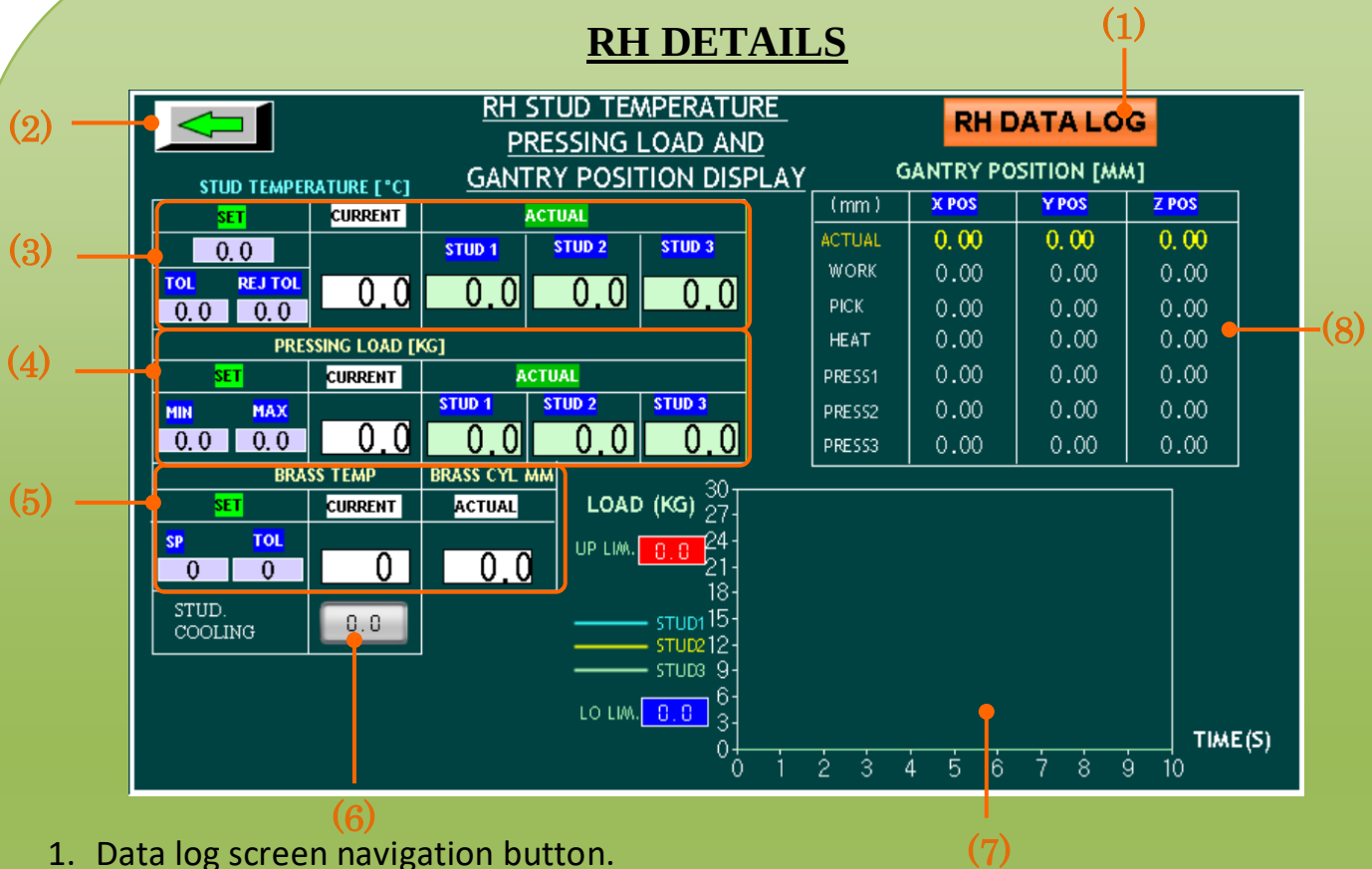
LH DETAILS

(1)
↓



1. Data log screen navigation button.
2. Previous page navigation button.
3. SET = Temp sp, Temp TOL & REJ TOL sp display set in configuration screen, CURRENT=Actual Temp, STUD1,2,3 = Actual temperature stud at temp check positon.
4. SET = LOAD sp, Load MIN & MAX sp display set in configuration screen, CURRENT=Actual Load, STUD1,2,3 = Actual Load during pressing.
5. SP= Brass Temp sp, Tol= Brass Temp Tolerance, CURRENT = Brass heater actual temp, BRASS CYL MM = Brass vertical cylinder actual value in MM.
6. SUD COOLING = Stud Cooling time set in configuration screen.
7. Load graph.
8. Actual Teach positions.

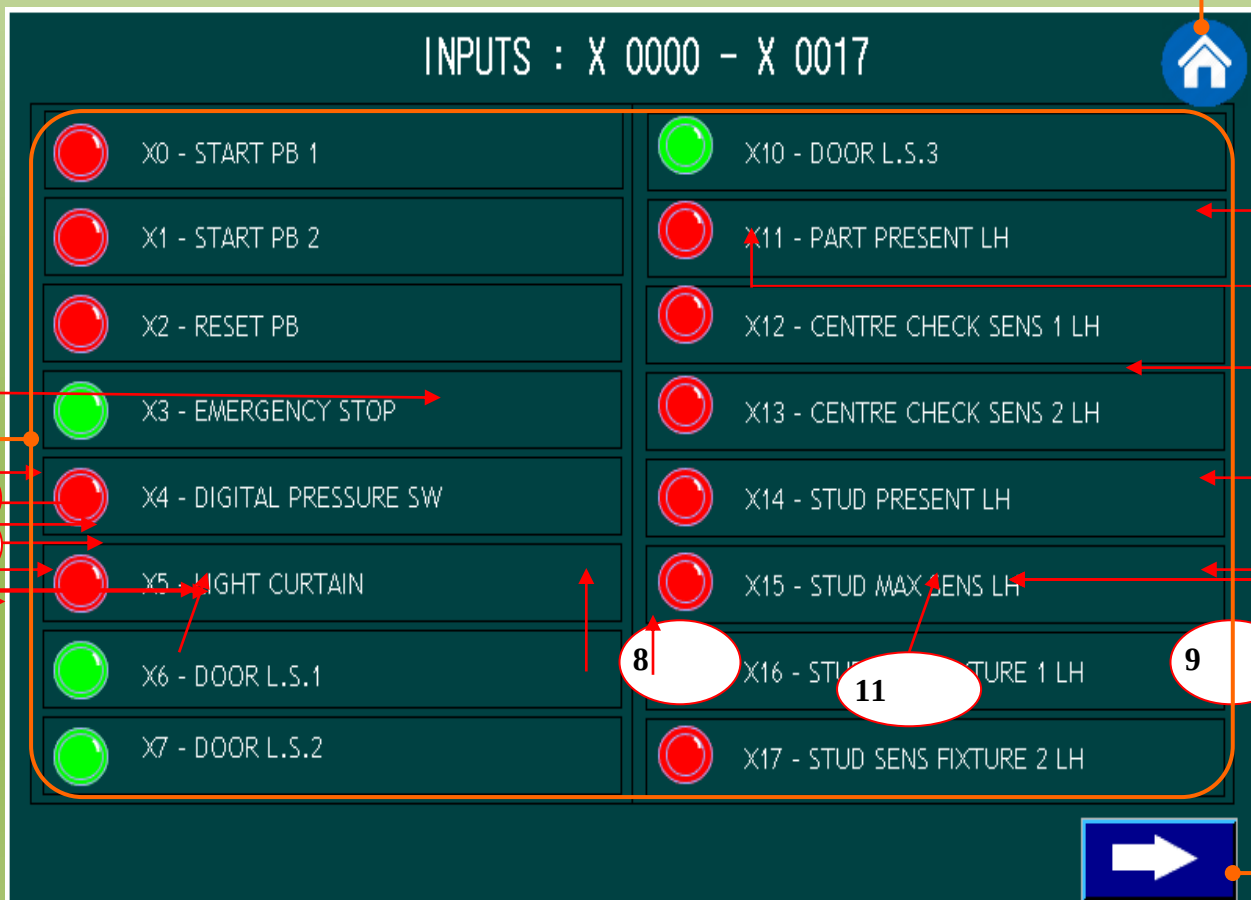
RH DETAILS



1. Data log screen navigation button.
2. Previous page navigation button.
3. SET = Temp sp, Temp TOL & REJ TOL sp display set in configuration screen, CURRENT=Actual Temp, STUD1,2,3 = Actual temperature stud at temp check positon.
4. SET = LOAD sp, Load MIN & MAX sp display set in configuration screen, CURRENT=Actual Load, STUD1,2,3 = Actual Load during pressing.
5. SP= Brass Temp sp, Tol= Brass Temp Tolerance, CURRENT = Brass heater actual temp, BRASS CYL MM = Brass vertical cylinder actual value in MM.
6. STUD COOLING = Stud Cooling time set in configuration screen.
7. Load graph.
8. Actual Teach positions.

INPUT SCREEN-1

INPUTS : X 0000 - X 0017



1. Home screen navigation button.
2. Plc Inputs.
3. Previous screen button.

INPUT SCREEN-2



INPUTS : X 0020 - X 0037

☐ X20 - STUD SENS FIXTURE 3 LH	☐ X30 - STUD GRIPPER RET RD LH
☐ X21 - BRASS PRESENT SENSOR LH	☐ X31 - DOT PUNCH FWD RD LH
☐ X22 - REJECTION BIN SENSOR LH	☐ X32 - DOT PUNCH RET RD LH
☐ X23 - PART CLAMP FWD RD LH	☐ X33-HOT COLD PRESS CYL-1 FWD RD SW LH
☐ X24 - PART CLAMP RET RD LH	☐ X34-HOT COLD PRESS CYL-1 RET RD SW LH
☐ X25 - PUSH CYL FWD RD LH	☐ X35-HOT COLD PRESS CYL-2 FWD RD SW LH
☐ X26 - PUSH CYL RET RD LH	☐ X36-HOT COLD PRESS CYL-2 RET RD SW LH
☐ X27 - STUD GRIPPER FWD RD LH	☐ X37 - SLIDE CYL FWD RD SW LH

(1) Home screen navigation button.

(2) Plc Inputs.

(3) Previous screen navigation button.

(4) Next screen navigation button.

1. Home screen navigation button.
2. Plc Inputs.



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3. Previous screen button.
4. Next screen button.

INPUT SCREEN-3

(1)

INPUTS : X 0040 - X 0057

☐ X40 - SLIDE CYL RET RD SW LH	☐ X50 - AXIS X END LS-1 LH
☐ X41 - FIX SELN IN-1 LH	☐ X51 - AXIS X END LS-2 LH
☐ X42 - FIX SELN IN-2 LH	☐ X52 - AXIS X HOME SENSOR LH
☐ X43 - FIX SELN IN-3 LH	☐ X53 - AYIS-Y END LS-1 LH
☐ X44 - FIX SELN IN-4 LH	☐ X54 - AYIS-Y END LS-2 LH
☐ X45 - STUD PRESENT SENSOR RH	☐ X55 - AYIS-Y HOME SENSOR LH
☐ X46 - BOWL FEEDER FULL SENSOR RH	☐ X56 - AZIS-Z END LS-1 LH
☐ X47 - INDUCTION HEATER FAULT LH	☐ X57 - AZIS-Z END LS-2 LH

(2)

(3) (4)

1. Home screen navigation button.

2. Plc Inputs.
3. Previous screen button.
4. Next screen button.

INPUT SCREEN-4

INPUTS : X 0060 - X 0077

(1)



(2)

 X60 - AYIS-Z HOME SENSOR LH	 X70 - PART PRESENT RH
 X61 - LOCK CYL FWD LH	 X71 - CENTRE CHECK SENSOR 1 RH
 X62 - LOCK CYL REV LH	 X72 - CENTRE CHECK SENSOR 2 RH
 X63 - INDUCTION HEATER FAULT RH	 X73 - STUD 1 PRESENT SENS RH
 X64 - FIX SEL IN-1 RH	 X74 - STUD 2 PRESENT SENS RH
 X65 - FIX SEL IN-2 RH	 X75 - STUD 3 PRESENT SENS RH
 X66 - FIX SEL IN-3 RH	 X76 - BRASS PRESENT SENSOR RH
 X67 - FIX SEL IN-4 RH	 X77 - PART CLAMP FWD RD RH



(3)



(4)



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1. Home screen navigation button.
2. Plc Inputs.
3. Previous screen button.
4. Next screen button.

INPUT SCREEN-5

(1)
↓

INPUTS : X 0100 – X 0117

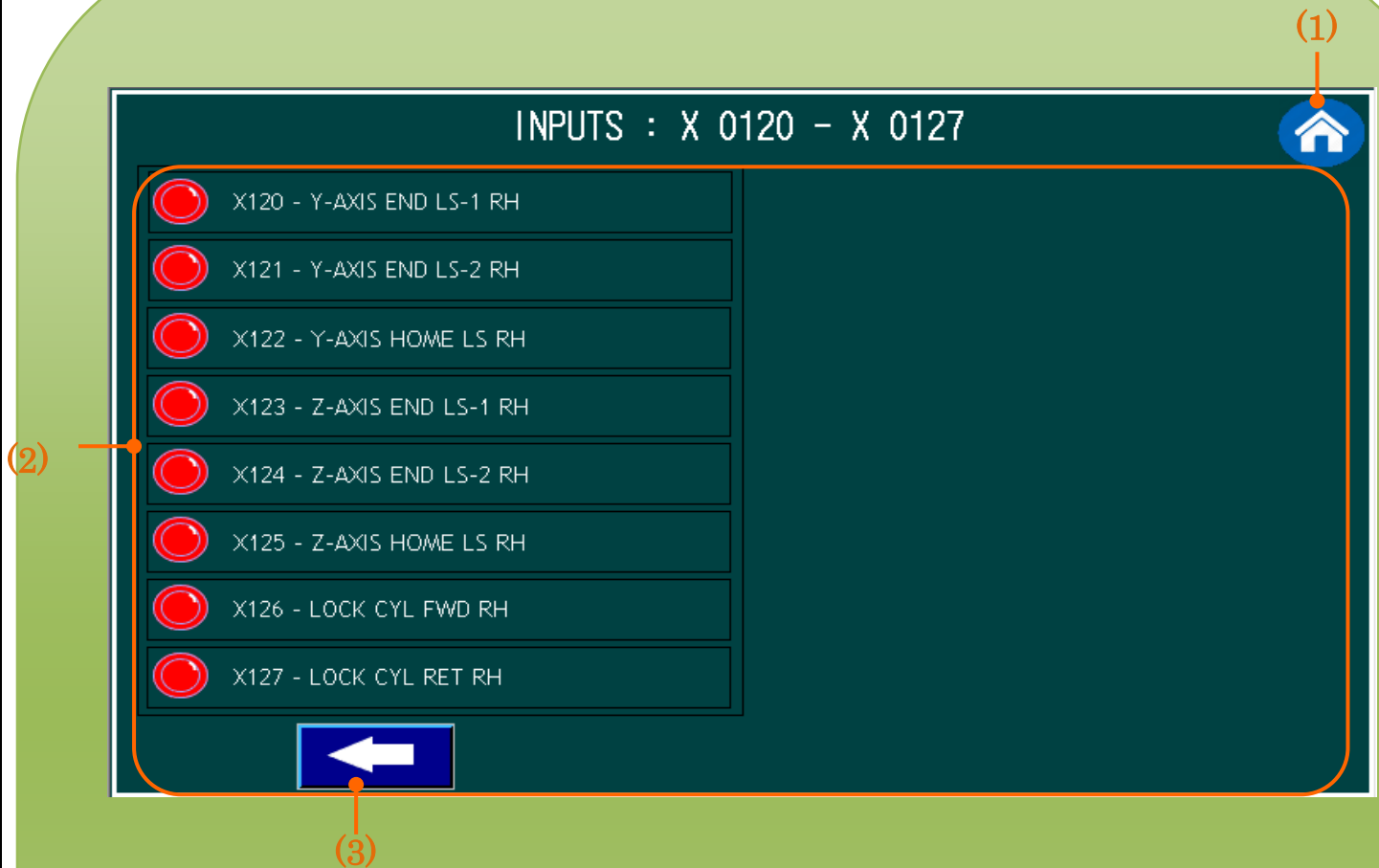


 X100 - PART CLAMP RET RD RH	 X110 -REJECTION BIN SENSOR RH
 X101 - PUSH CYL FWD RD RH	 X111 - HOT COLD CYL FWD RH
 X102 - PUSH CYL RET RD RH	 X112 - HOT COLD CYL RET RH
 X103 - STUD GRIPPER FWD RD RH	 X113 - SLIDE CYL FWD RD SW-RH
 X104 - STUD GRIPPER RET RD RH	 X114 - SLIDE CYL RET RD SW-RH
 X105 - DOT PUNCH FWD RD SW RH	 X115 - X-AXIS END LS-1 RH
 X106 - DOT PUNCH RET RD SW RH	 X116 - X-AXIS END LS-2 RH
 X107 -HOT COLD PRESS CYL-1 FWD RD SW-RH	 X117 - X-AXIS HOME LS RH



1. Home screen navigation button.
2. Plc Inputs.
3. Previous screen button.
4. Next screen button.

INPUT SCREEN-6



1. Home screen navigation button.
2. Plc Inputs.
3. Previous screen button.

OUTPUT SCREEN-1



1. Home screen navigation button.
2. Plc Outputs.
3. Next screen button.

OUTPUT SCREEN-2



1. Home screen navigation button.
2. Plc Outputs.

3. Previous screen button.

4. Next screen button.

OUTPUT SCREEN-3

OUTPUTS : Y 0040 - Y 0057



 Y40 - REV LIMIT SWITCH X-AXIS LH

 Y41 - FWD LIMIT SW Y-AXIS LH

 Y42 - REV LIMIT SWITCH Y-AXIS LH

 Y43 - FWD LIMIT SW Z-AXIS LH

 Y44 - REV LIMIT SWITCH Z-AXIS LH

 Y45 - EMERGENCY STOP DRIVE LH

 Y46 - EMERGENCY STOP DRIVE RH

 Y47 - SLIDE CYL FWD SOL RH

 Y50 - SLIDE CYL FWD SOL RH

 Y51- SLIDE CYL RET SOL RH

 Y52 - PART CLAMP FWD SOL RH

 Y53 - PART CLAMP RET SOL RH

 Y54 - STUD PUSHER CYL FWD SOL RH

 Y55 - STUD PUSHER CYL RET SOL

 Y56 - STUD GRIPPER FWD SOL RH

 Y57 - STUD GRIPPER RET SOL RH





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1. Home screen navigation button.
2. Plc Outputs.
3. Previous screen button.
4. Next screen button.

OUTPUT SCREEN-4

OUTPUTS : Y 0060 - Y 0077



	Y60 - DOT PUNCH SOL RH		Y70 - HOT COLD PRESS CYL-2 RET SOL RH
	Y61 - VERT CYL FWD RH		Y71 - AIR OFF COOL FIXTURE-1 RH
	Y62 - VERT CYL RET RH		Y72 - AIROFF COOL FIXTURE-2 RH
	Y63 - INDUCTION HEATER ON/OFF RH		Y73 - AIR OFF COOL FIXTURE-3 RH
	Y64 - LINEAR FEDDER START RH		Y74 - AIR OFF COOL GRIPPER RH
	Y65 - BOWL FEEDER START RH		Y75 - LOCK CYL FWD LH
	Y66 - AXIS-Z BREAK RH		Y76 - LOCK CYL RET LH
	Y67 - HOT COLD PRESS CYL-2 FWD SOL-RH		Y77 - FWD LIMIT SW X-AXIS RH





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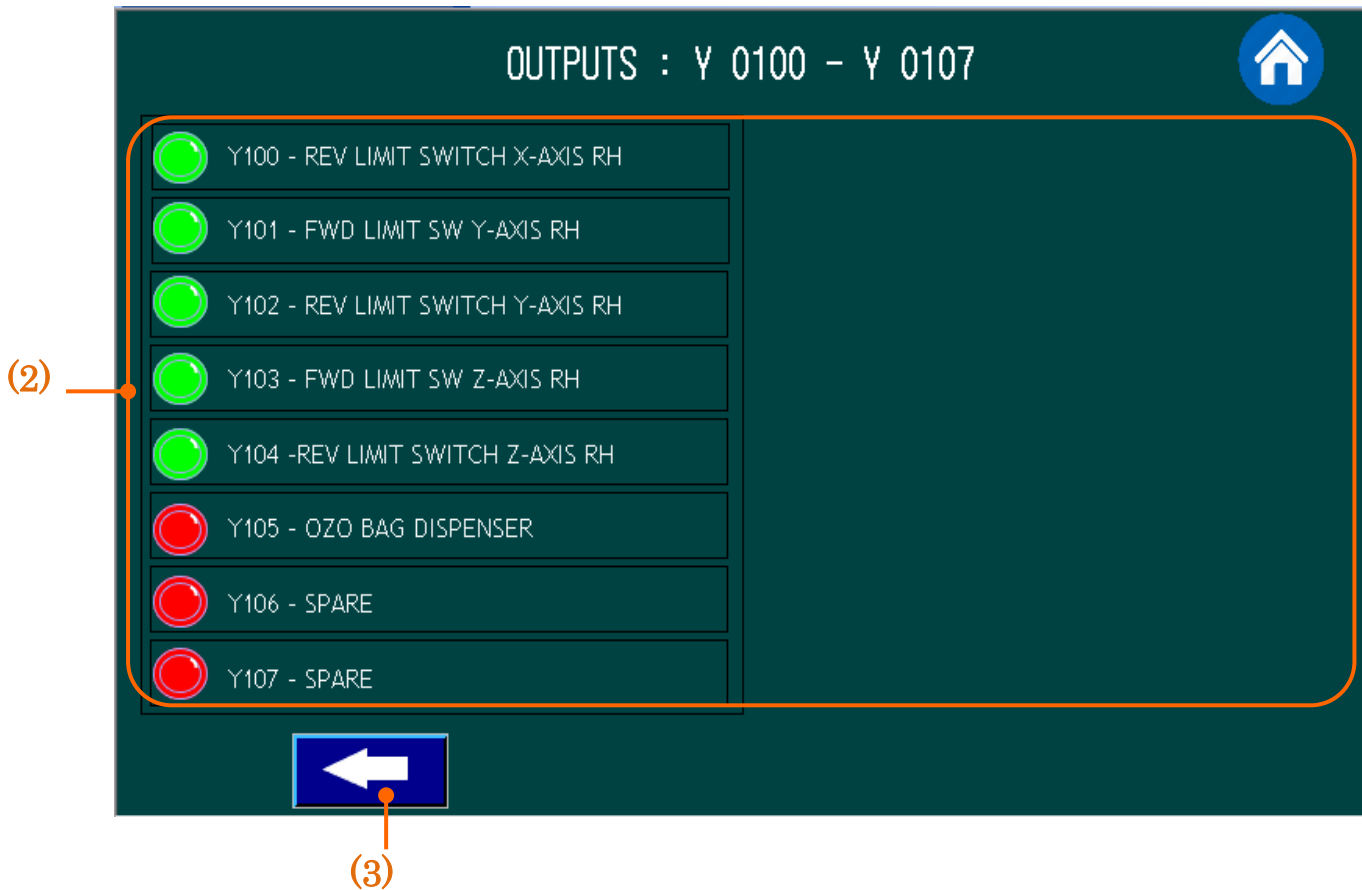
(3)

(4)

1. Home screen navigation button.
2. Plc Outputs.
3. Previous screen button.
4. Next screen button.

OUTPUT SCREEN-5

(1)
↓



1. Home screen navigation button.
2. Plc Outputs.
3. Previous screen button.

MANUAL SCREEN-1

MANUAL



LH CYLINDERS

RH CYLINDERS

DESCRIPTION	KEY	STS	F/B	DESCRIPTION	KEY	STS	F/B
STUD GRIPPER FWD	OFF	(Y16)	X27	STUD GRIPPER FWD	OFF	(Y56)	X103
STUD GRIPPER RET	OFF	(Y17)	X30	STUD GRIPPER RET	OFF	(Y57)	X104
PART CLAMP FWD	OFF	(Y12)	X23	PART CLAMP FWD	OFF	(Y52)	X77
PART CLAMP RET	OFF	(Y13)	X24	PART CLAMP RET	OFF	(Y53)	X100
STUD PUSH FWD	OFF	(Y14)	X25	STUD PUSH FWD	OFF	(Y54)	X101
STUD PUSH RET	OFF	(Y15)	X26	STUD PUSH RET	OFF	(Y55)	X102
CAMERA TRIG	OFF	LH LIGHT OFF	RH LIGHT OFF	CAMERA SCENE SW.	OFF		

1. Home screen navigation button.
2. Next screen button.
3. LH Cylinder manual operation.
4. Camera Trigger ON/OFF button, Lh light off button, Rh light off button, Camera scene switch OFF button.
5. RH Cylinder manual operation.

MANUAL SCREEN-2

(3)


MANUAL

(2)


(1)


LH CYLINDERS				RH CYLINDERS			
	KEY	STS	F/B	DESCRIPTION	KEY	STS	F/B
SLIDE CYL FWD	OFF	(Y10)	X37	SLIDE CYL FWD	OFF	(Y50)	X113
SLIDE CYL RET	OFF	(Y11)	X40	SLIDE CYL RET	OFF	(Y51)	X114
PRESS VERT CYL FWD	OFF	(Y27)	0.0	PRESS VERT CYL FWD	OFF	(Y61)	0.0
PRESS VERT CYL RET	OFF	(Y30)		PRESS VERT CYL RET	OFF	(Y62)	
HOT COLD CYL FWD	OFF	(Y21)	X35	HOT COLD CYL FWD	OFF	(Y67)	X111
HOT COLD CYL RET	OFF	(Y22)	X36	HOY COLD CYL RET	OFF	(Y70)	X112
DOT PUNCH	OFF	(Y20)	X31 X32	DOT PUNCH	OFF	(Y60)	X105 X106

1. Alarm screen Navigation button.
2. Next screen button.
3. Back screen button.
4. LH & RH Cylinder's manual operation.

MANUAL SCREEN-3

(3) 
(2) 
(1) 

MANUAL

LH CYLINDERS				RH CYLINDERS			
DESCRIPTION	KEY	STS	F/B	DESCRIPTION	KEY	STS	F/B
INDUCTION HEATER	OFF			INDUCTION HEATER	OFF		
LINEAR FEEDER STRT	ON			LINEAR FEEDER STRT	ON		
BOWL FEEDER STRT	OFF			BOWL FEEDER STRT	OFF		
STUD-1 COOL AIR	OFF			STUD-1 COOL AIR	OFF		
STUD-2 COOL AIR	OFF			STUD-2 COOL AIR	OFF		
STUD-3 COOL AIR	OFF			STUD-3 COOL AIR	OFF		
AIR OFF COOL GRIPPER	OFF			AIR OFF COOL GRIPPER	OFF		

(4) 

1. Alarm screen Navigation button.
2. Next screen button.
3. Back screen button.
4. LH & RH Cylinder's manual operation.

MANUAL SCREEN-4

(2)


(1)


MANUAL

LH CYLINDERS				RH CYLINDERS			
DESCRIPTION	KEY	STS	F/B	DESCRIPTION	KEY	STS	F/B
LH LOCK FWD	OFF		X61	RH LOCK FWD	OFF		X126
LH LOCK RET	OFF		X62	RH LOCK RET	OFF		X127
CAM LIGHT	OFF			CAM LIGHT	OFF		

(3)

1. Alarm screen Navigation button.
2. Back screen button.



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3. LH & RH Cylinder's manual operation.

LH ALARM SCREEN

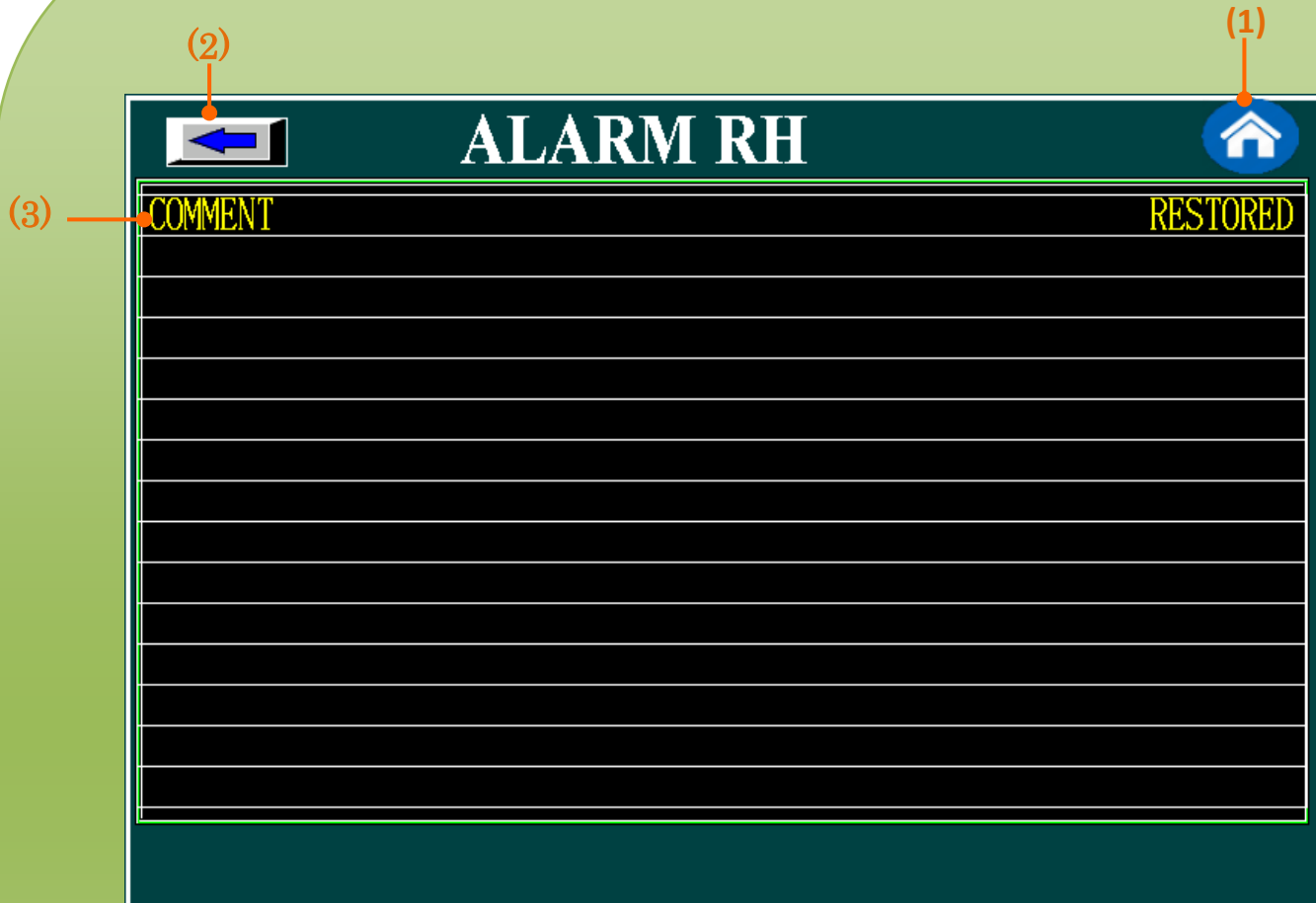
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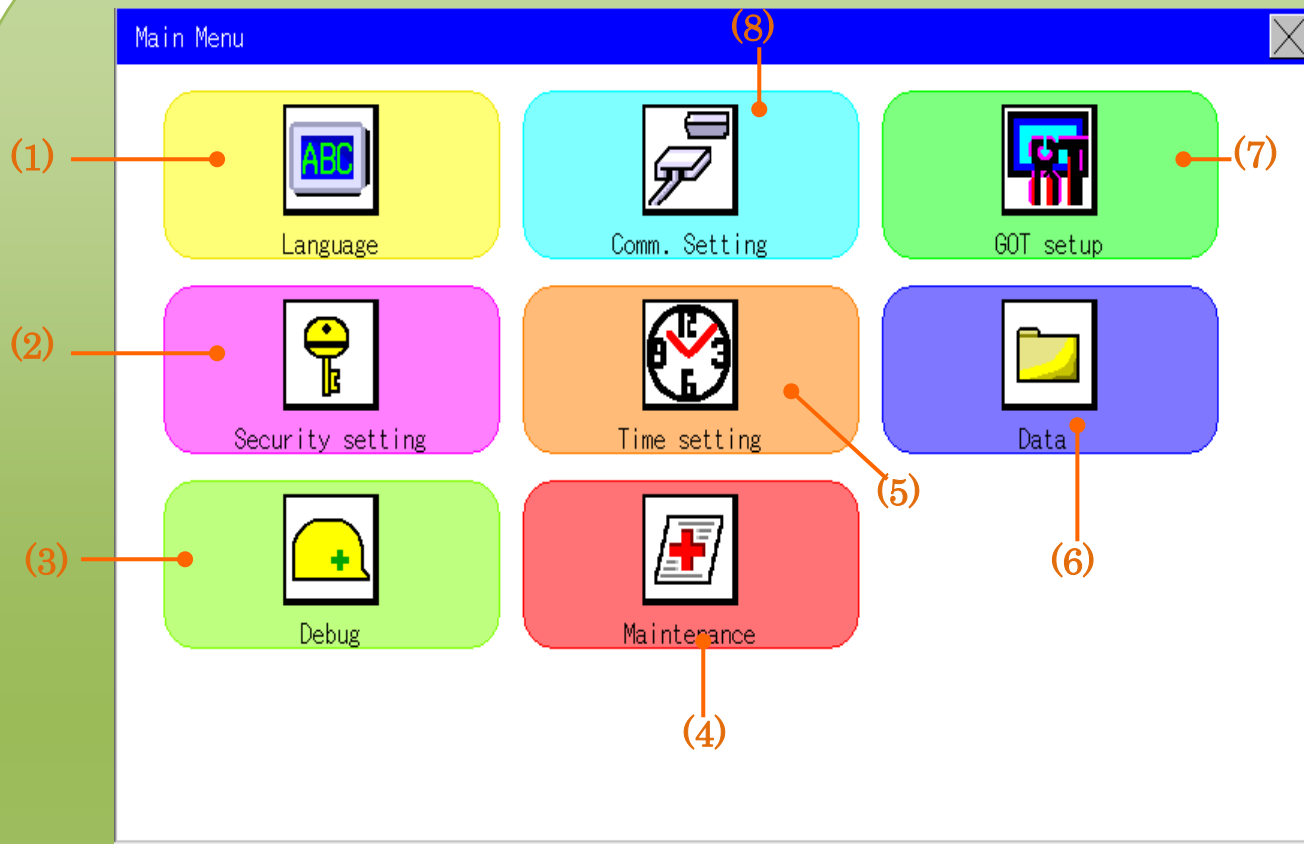
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RH ALARM SCREEN



1. Home screen navigation button.
2. Next screen button.
3. Alarm display.

UTILITY SCREEN



1. Language setting screen navigation button.
2. Security setting screen navigation button.
3. Debugging navigation button.
4. Maintenance setting screen navigation button.
5. Time setting navigation button.
6. Data navigation button.
7. Setup navigation button.
8. Communication setting button.

ALARM ON HMI CHECK POINTS

Alarm on HMI	Cause	Check point
GRIPPER RET ERROR SEMI AUTO LH	Gripper Return Reed switch error	Check gripper return reed switch
GRIPPER RET ERROR CENTERING LH		
GRIPPER FWD ERROR CENTERING LH	Gripper forward Reed switch error	Check gripper forward reed switch
LOAD ERROR LH	Load greater than min & max set point or greater than 50 kg	
LIGHT CURTAIN ERROR LH	Light Curtain Interrupted	Check light curtain
EMERGENCY PRESSED ERROR		
DRIVE NOT READY ERROR X- AXS LH	Drive error	Emergency released ,Ethernet communication, Check Axis end limit switch
DRIVE NOT READY ERROR Y- AXS LH		
DRIVE NOT READY ERROR Z- AXS LH		
DIGITAL PRESSURE SWITCH ERROR LH	Pressure not within range	Check incoming pressure of machine
DOOR LS 1 ERROR LH	Door not closed properly	Check door
DOOR LS 2 ERROR LH		
DOOR LS 3 ERROR LH		
STUD PUSH FWD ERROR PICK POS CMD LH	Pusher forward reed switch error	Check reed switch
STUD GRIPPER RET ERROR PICK POS CMD LH	Pusher return reed switch error	
GRIPPER CLOSE ERROR LH	Gripper return reed switch error	
STUD-1 PRSN ERROR LH	Auto cycle start and stud all ready present	Check stud presence or sensor
STUD-2 PRSN ERROR LH		
STUD-3 PRSN ERROR LH		
PART CLAMP ERROR LH	Part clamp reed switch error	Check part clamp cylinder reed switch
TEMP ERROR LH	Not Used	
SEMIAUTO SELECTION ERROR LH	If semi auto not selected & any position command then selection error	Reset error select semi auto then start semi auto position commands
GRIPPER RET ERROR CENTERING LH	Reed switch error	Check reed switch
GRIPPER RET HOMING ERROR LH		
STUD GRIPPER FORWARD ERROR IN CENTERING LH		

PART PRESENT ERROR AUTO LH	sensor error	check sensor
STUD PRESENT ERROR AUTO LH		
WORK HOME ERROR X-AXIS AUTO LH	Position error	Check teach points of work home or operate manually check physical position
WORK HOME ERROR Y-AXIS AUTO LH		
WORK HOME ERROR Z-AXIS AUTO LH		
PID TEMPERATURE ERROR AUTO LH	Brass temperature error	Analog input & SSR working, PID working with temperature deviation
HOT COLD CYLINDER RET ERROR AUTO LH	Reed switch error	Check reed switch
VERTICAL CYLINDER RET ERROR AUTO LH	Smat sensor error	Smat sensor wiring
LOCK CYLINDER ERROR AUTO LH	Reed switch error	
SLIDE CYLINDER RET ERROR AUTO LH		
PUSHER CYLINDER FWD ERROR AUTO LH		
CAMERA SCENE CHANGE NOT OK LH	Camera error	camera communication
CAMERA TRIGGER NOT OK LH	Camera error	stud size, stud present or not
PUSHER CYLINDER RET ERROR AUTO LH	Reed switch error	Check reed switch
CLAMP CYLINDER RET ERROR AUTO LH		
GRIPPER RET ERROR AUTO LH		
GRIPPER FWD ERROR AUTO LH		
PUSHER CYLINDER RET ERROR AUTO LH		
GRIPPER RET ERROR AUTO LH		
PART DOT PUNCH ERROR LH		
PART DECLAMP ERROR LH	sensor error	
TEMPERATURE RETRY ERROR LH	Stud heat & not within range temperature set point retry count	Temperature check position with respect to heater, Heater controller setting, Temperature check sensor position
CAMERA TRIGGER ERROR LH		
GRIPPER RETURN ERROR HOMING LH	Reed switch error	Check reed switch
PART DECLAMP ERROR HOMING LH		
PUSHER RETURN ERROR		



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HOMING LH		
RUN CENTERING FOR LH		
FIXTURE CODE ERROR LH	Wrong fixture connected with respect to fixture number from Hmi	Check connected fixture
REJECTION BIN	Load Error, Stud present sensor 1-2-3 error, Vertical cylinder hot press error, Brass Final cool press error	Reject part
CAMERA COMMUNICATION ERROR	Camera error	Check Communication cable, Camera is in run mode
CAMERA NOT IN RUN MODE		
CAMERA SCENE CHANGE ERROR		
CAMERA TRIGGER ERROR		
CAMERA NOK	Camera error	Check stud ok or mot or check stud size
SLIDE CYLINDER FWD START ERROR LH	Reed switch error	Check reed switch
LOCK CYLINDER FWD ERROR LH		
BRASS PRESENT ERROR LH	Brass error	Check brass present
VERT CYL FWD ERROR IN BRASS CHK LH	Brass distance error	Check smat sensor
VERT CYL RET ERROR AFTER INS CHK LH		
HOT COLD CYL FWD ERROR LH	Reed switch error	Check reed switch
VERT CYL FWD HOT PRESS CMD ERROR LH	Distance error	Check smat sensor, Do calibration, Overview distance set point
VERT CYL REV ERROR AFTER HOT PRESS LH		
HOT COLD REV ERROR LH	Reed switch error	
FNL COOL PRESS ERROR LH	Distance error	Check smat sensor, Do calibration, Overview distance set point
VERT CYL RET ERROR FNL PRESS LH		
LOCK CYL RET ERROR LH	Reed switch error	Check reed switch
SLIDE CYL RET ERROR LH		
VERT CYL RET ERROR IN HOMING LH	Distance error	Smat sensor wiring
HOT COLD CYL RET ERROR HOMING LH	Reed switch error	Check reed switch
LOCK CYL RET ERROR HOMING LH		
SLIDE CYL RET ERROR HOMING LH		
INDUCTIO HEATER ERROR LH	Heater error	Check heater setting or wiring
BRASS PRESENT SENSOR	Brass not place before cycle start	Check brass present



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ERROR LH		
GRIPPER RET ERROR IN HOMING LH	Reed switch error	Check reed switch
CLAMP FWD ERROR LH		
CLAMP RET ERROR LH		
CLAMP RET ERROR LH		

STUD INSERT AM PM CHECK POINTS

	Points to be check	Where to check	Daily	Monthly	Quarterly	Biannua l	Annua l
General points							
1	Check given power supply and earthing .ensure it is within range as per machine required	Control panel	√	√	√	√	√
2	Check Pneumatic pressure .ensure it is within range as per required	Pneumatic device panel	√	√	√	√	√
3	Check looseness of electric connectors. If the connectors are loosen, push it securely or tighten	External connector terminal ,Electric indo connector		√	√	√	√
4	Check loss ness of pneumatic connector, If the connectors are loosen, push it securely or tighten	On machine side & fixture side punmatic connector		√	√	√	√
5	Check X,Y,Z gantry sensor			√	√	√	√
6	Check X,Y,Z gantry position as per master part, if it is not in position tech points as per given process in him	X,Y,Z gantry	√	√	√	√	√
7	Check Brass insert heater mounting arrangement	On fixture	√	√	√	√	√
8	Check induction heater coil			√	√	√	√
9	Check temperature sensor calibration						√
10	Check PID controller						√



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	calibration						
11	Check communication cable of camera	Camera to PLC		✓	✓	✓	✓
12	Check looseness of camera mounting bolts, screws. Tighten them if necessary as per match marking.	Camera mounting				✓	✓
13	Check camera teaching using master parts if it is not working, teach camera under supplier guidance.	Camera monitor		✓			
14	Check linear fedder working, check proper stud moving ,if stud stuck check mechanical arrangement	Linner fedder		✓	✓	✓	✓
15	Check bowel fedder working, check proper stud moving ,if stud stuck check mechanical arrangement	Bowler fedder		✓	✓	✓	✓
16	Check stud gripper working	On fixture		✓	✓	✓	✓

DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.



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- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting into electrical connector.



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